

Development

Research Methods in Developmental Psychology

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What do infants know about the world in which they live – and how do they grow and change with age? These are the kinds of questions answered by developmental scientists. This module describes different research techniques that are used to study psychological phenomena in infants and children, research designs that are used to examine age-related changes in development, and unique challenges and special issues associated with conducting research with infants and children. Child development is a fascinating field of study, and many interesting questions remain to be examined by future generations of developmental scientists – maybe you will be among them!

Learning Objectives

- Describe different research methods used to study infant and child development
- Discuss different research designs, as well as their strengths and limitations
- Report on the unique challenges associated with conducting developmental research

Introduction

A group of children were playing hide-and-seek in the yard. Pilar raced to her hiding spot as her six-year-old cousin, Lucas, loudly counted, "... six, seven, eight, nine, ten! Ready or not, here I come!". Pilar let out a small giggle as Lucas ran over to find her – in the exact location

where he had found his sister a short time before. At first glance, this behavior is puzzling: why would Pilar hide in exactly the same location where someone else was just found? Whereas older children and adults realize that it is likely best to hide in locations that have not been searched previously, young children do not have the same cognitive sophistication. But why not... and when do these abilities first develop?



Children's games are more than just fun. They can be opportunities for researchers to learn about how children think, feel, and behave. [Image: gomattolson, <https://goo.gl/kajAEn>, CC BY-NC-SA 2.0, <https://goo.gl/i6LJ8m>]

Developmental psychologists investigate questions like these using research methods that are tailored to the particular capabilities of the infants and children being studied. Importantly, research in developmental psychology is more than simply examining how children behave during games of hide-and-seek – the results obtained from developmental research have been used to inform best practices in parenting, education, and policy.

This module describes different *research techniques* that are used to study psychological phenomena in infants and children, *research designs* that are used to examine age-related changes in developmental processes and changes over time, and *unique challenges and special issues*

associated with conducting research with infants and children.

Research Methods

Infants and children—especially younger children—cannot be studied using the same *research methods* used in studies with adults. Researchers, therefore, have developed many creative ways to collect information about infant and child development. In this section, we highlight some of the methods that have been used by researchers who study infants and older children, separating them into three distinct categories: involuntary or obligatory responses, voluntary responses, and psychophysiological responses. We will also discuss other methods such as the use of surveys and questionnaires. At the end of this section, we give an example of how interview techniques can be used to study the beliefs and perceptions of older children and adults – a method that cannot be used with infants or very young children.

Involuntary or obligatory responses

One of the primary challenges in studying very young infants is that they have limited motor control– they cannot hold their heads up for short amounts of time, much less grab an interesting toy, play the piano, or turn a door knob. As a result, infants cannot actively engage with the environment in the same way as older children and adults. For this reason, developmental scientists have designed research methods that assess involuntary or obligatory responses. These are behaviors in which people engage without much conscious thought or effort. For example, think about the last time you heard your name at a party – you likely turned your head to see who was talking without even thinking about it. Infants and young children also demonstrate involuntary responses to stimuli in the environment. When infants hear the voice of their mother, for instance, their heart rate increases – whereas if they hear the voice of a stranger, their heart rate decreases (Kisilevsky et al., 2003). Researchers study involuntary behaviors to better understand what infants know about the world around them.

One research method that capitalizes on involuntary or obligatory responses is a procedure known as habituation. In habituation studies, infants are presented with a stimulus such as a photograph of a face over and over again until they become bored with it. When infants become bored, they look away from the picture. If infants are then shown a new picture--such as a photograph of a different face-- their interest returns and they look at the new picture. This is a phenomenon known as dishabituation. Habituation procedures work because infants generally look longer at novel stimuli relative to items that are familiar to them. This research technique takes advantage of involuntary or obligatory responses because infants are constantly looking around and observing their environments; they do not have to be taught to engage with the world in this way.



Even when infants are so young that they have very little motor control, researchers can observe and record involuntary responses, such as an infant's eye movements, to get valuable insight into what they understand about the world around them. [Image: Greg Kintz, <https://goo.gl/mqD52S>, CC BY-NC-SA, <https://goo.gl/i6LJ8m>]

One classic habituation study was conducted by Baillargeon and colleagues (1985). These

researchers were interested in the concept of **object permanence**, or the understanding that objects exist even when they cannot be seen or heard. For example, you know your toothbrush exists even though you are probably not able to see it right this second. To investigate object permanence in 5-month-old infants, the researchers used a **violation of expectation paradigm**. The researchers first habituated infants to an opaque screen that moved back and forth like a drawbridge (using the same procedure you just learned about in the previous paragraph). Once the infants were bored with the moving screen, they were shown two different scenarios to test their understanding of physical events. In both of these test scenarios, an opaque box was placed behind the moving screen. What differed between these two scenarios, however, was whether they confirmed or violated the **solidity principle** – the idea that two solid objects cannot occupy the same space at the same time. In the *possible* scenario, infants watched as the moving drawbridge stopped when it hit the opaque box (as would be expected based on the solidity principle). In the *impossible* scenario, the drawbridge appeared to move right through the space that was occupied by the opaque box! This impossible scenario violates the solidity principle in the same way as if you got out of your chair and walked through a wall, reappearing on the other side.

The results of this study revealed that infants looked longer at the impossible test event than at the possible test event. The authors suggested that the infants reacted in this way because they were surprised – the demonstration went against their expectation that two solids cannot



Filling your basket at the supermarket is an example of a voluntary response behavior. The choices you make are under your control. [Image: Ikayama, <https://goo.gl/SvGDCz>, CC BY-NC-SA 2.0, <https://goo.gl/i6LJ8m>]

move through one another. The findings indicated that 5-month-old infants understood that the box continued to exist even when they could not see it. Subsequent studies indicated that 3½- and 4½-month-old infants also demonstrate object permanence under similar test conditions (Baillargeon, 1987). These findings are notable because they suggest that infants understand object permanence much earlier than had been reported previously in research examining **voluntary responses** (although see more recent research by Cashon & Cohen, 2000).

Voluntary responses

As infants and children age, researchers are increasingly able to study their understanding

of the world through their voluntary responses. Voluntary responses are behaviors that a person completes by choice. For example, think about how you act when you go to the grocery store: you select whether to use a shopping cart or a basket, you decide which sections of the store to walk through, and you choose whether to stick to your grocery list or splurge on a treat. Importantly, these behaviors are completely up to you (and are under your control). Although they do not do a lot of grocery shopping, infants and children also have voluntary control over their actions. Children, for instance, choose which toys to play with.

Researchers study the voluntary responses of infants and young children in many ways. For example, developmental scientists study recall memory in infants and young children by looking at voluntary responses. Recall memory is memory of past events or episodes, such as what you did yesterday afternoon or on your last birthday. Whereas older children and adults are simply asked to talk about their past experiences, recall memory has to be studied in a different way in infants and very young children who cannot discuss the past using language. To study memory in these subjects researchers use a behavioral method known as elicited imitation (Lukowski & Milojevich, in press).

In the elicited imitation procedure, infants play with toys that are designed in the lab to be unlike the kinds of things infants usually have at home. These toys (or event sequences, as researchers call them) can be put together in a certain way to produce an outcome that infants commonly enjoy. One of these events is called Find the Surprise. As shown in Figure 1, this toy has a door on the front that is held in place by a latch – and a small plastic figure is hidden on the inside. During the first part of the study, infants play with the toy in whichever way they want for a few minutes. The researcher then shows the infant how to make the toy work by (1) flipping the latch out of the way and (2) opening the door, revealing the plastic toy inside. The infant is allowed to play with the toy again either immediately after the demonstration or after a longer delay. As the infant plays, the researcher records whether the infant finds the surprise using the same procedure that was demonstrated.



Figure 1. The two-step event sequence Find the Surprise. The picture on the left shows all of the toys needed to complete the event. The picture in the middle shows a hand flipping the latch out of the way so the door can be opened (step 1). The picture on the right shows a hand opening the door, ultimately revealing a plastic figurine hidden inside (step 2). These images are from a larger figure initially published in Lukowski and Milojevich (2013) (reprinted with permission).

Use of the elicited imitation procedure has taught developmental scientists a lot about how recall memory develops. For example, we now know that 6-month-old infants remember one step of a 3-step sequence for 24 hours (Barr, Dowden, & Hayne, 1996; Collie & Hayne, 1999). Nine-month-olds remember the individual steps that make up a 2-step event sequence for 1 month, but only 50% of infants remember to do the first step of the sequence before the second (Bauer, Wiebe, Carver, Waters, & Nelson, 2003; Bauer, Wiebe, Waters, & Bangston, 2001; Carver & Bauer, 1999). When children are 20 months old, they remember the individual steps and temporal order of 4-step events for at least 12 months – the longest delay that has been tested to date (Bauer, Wenner, Dropik, & Wewerka, 2000).

Psychophysiology

Behavioral studies have taught us important information about what infants and children know about the world. Research on behavior alone, however, cannot tell scientists how brain development or biological changes impact (or are impacted by) behavior. For this reason, researchers may also record psychophysiological data, such as measures of heart rate, hormone levels, or brain activity. These measures may be recorded by themselves or in combination with behavioral data to better understand the bidirectional relations between biology and behavior.

One manner of understanding associations between brain development and behavioral advances is through the recording of event-related potentials, or ERPs. ERPs are recorded by fitting a research participant with a stretchy cap that contains many small sensors or electrodes. These electrodes record tiny electrical currents on the scalp of the participant in response to the presentation of particular stimuli, such as a picture or a sound (for additional information on recording ERPs from infants and children, see DeBoer, Scott, & Nelson, 2005). The recorded responses are then amplified thousands of times using specialized equipment so that they look like squiggly lines with peaks and valleys. Some of these brain responses have been linked



Special equipment has been developed to allow researchers to record the brain activity of very young and very small research subjects. [Image: Matt Cain, <https://goo.gl/XIAIOe>, CC BY-NC-SA 2.0, <https://goo.gl/i6LJ8m>]

to psychological phenomena. For example, researchers have identified a negative peak in the recorded waveform that they have called the N170 (Bentin, Allison, Puce, Perez, & McCarthy, 2010). The peak is named in this way because it is negative (hence the N) and because it occurs about 140ms to 170ms after a stimulus is presented (hence the 170). This peak is particularly sensitive to the presentation of faces, as it is commonly more negative when participants are presented with photographs of faces rather than with photographs of objects. In this way, researchers are able to identify brain activity associated with real world thinking and behavior.

The use of ERPs has provided important insight as to how infants and children understand the world around them. In one study (Webb, Dawson, Bernier, & Panagiotides, 2006), researchers examined face and object processing in children with autism spectrum disorders, those with developmental delays, and those who were typically developing. The children wore electrode caps and had their brain activity recorded as they watched still photographs of faces (of their mother or of a stranger) and objects (including those that were familiar or unfamiliar to them). The researchers examined differences in face and object processing by group by observing a component of the brainwave they called the prN170 (because it was believed to be a precursor to the adult N170). Their results showed that the height of the prN170 peak (commonly called the *amplitude*) did not differ when faces or objects were presented to typically developing children. When considering children with autism, however, the peaks were higher when objects were presented relative to when faces were shown. Differences were also found in how long it took the brain to reach the negative peak (commonly called the *latency* of the response). Whereas the peak was reached more quickly when typically developing children were presented with faces relative to objects, the opposite was true for children with autism. These findings suggest that children with autism are in some way processing faces differently than typically developing children (and, as reported in the manuscript, children with more general developmental delays).

Parent-report questionnaires

Developmental science has come a long way in assessing various aspects of infant and child development through behavior and psychophysiology – and new advances are happening every day. In many ways, however, the very youngest of research participants are still quite limited in the information they can provide about their own development. As such, researchers often ask the people who know infants and children best – commonly, their parents or guardians – to complete surveys or questionnaires about various aspects of their lives. These parent-report data can be analyzed by themselves or in combination with any collected behavioral or psychophysiological data.



Parents spend countless hours together with children observing their behavior. Developmental psychologists sometimes use surveys to collect information from parents that can be used to answer important research questions. [Image: JerryLai0208, <https://goo.gl/ICw3Tb>, CC BY-NC-SA 2.0, <https://goo.gl/i6LJ8m>]

One commonly used parent-report questionnaire is the Child Behavior Checklist (CBCL; Achenbach & Rescorla, 2000). Parents complete the preschooler version of this questionnaire by answering questions about child strengths, behavior problems, and disabilities, among other things. The responses provided by parents are used to identify whether the child has any behavioral issues, such as sleep difficulties, aggressive behaviors, depression, or attention deficit/hyperactivity problems.

A recent study used the CBCL-Preschool questionnaire (Achenbach & Rescorla, 2000) to examine preschooler functioning in relation to levels of stress experienced by their mothers while they were pregnant (Ronald, Pennell, & Whitehouse,

2011). Almost 3,000 pregnant women were recruited into the study during their pregnancy and were interviewed about their stressful life experiences. Later, when their children were 2 years old, mothers completed the CBCL-Preschool questionnaire. The results of the study showed that higher levels of maternal stress during pregnancy (such as a divorce or moving to a new house) were associated with increased attention deficit/hyperactivity problems in children over 2 years later. These findings suggest that stressful events experienced during prenatal development may be associated with problematic child behavioral functioning years later – although additional research is needed.

Interview techniques

Whereas infants and very young children are unable to talk about their own thoughts and behaviors, older children and adults are commonly asked to use language to discuss their thoughts and knowledge about the world. In fact, these verbal report paradigms are among the most widely used in psychological research. For instance, a researcher might present a child with a vignette or short story describing a moral dilemma, and the child would be asked to give their own thoughts and beliefs (Walrath, 2011). For example, children might react to the following:

"Mr. Kohut's wife is sick and only one medication can save her life. The medicine is extremely expensive and Mr. Kohut cannot afford it. The druggist will not lower the price. What should Mr. Kohut do, and why?"

Children can provide written or verbal answers to these types of scenarios. They can also offer their perspectives on issues ranging from attitudes towards drug use to the experience of fear while falling asleep to their memories of getting lost in public places – the possibilities are endless. Verbal reports such as interviews and surveys allow children to describe their own experience of the world.

Research Design

Now you know about some tools used to conduct research with infants and young children. Remember, *research methods* are the tools that are used to collect information. But it is easy to confuse research methods and research design. Research design is the strategy or blueprint for deciding how to collect and analyze information. Research design dictates which methods are used and how.

Researchers typically focus on two distinct types of comparisons when conducting research with infants and children. The first kind of comparison examines *change within individuals*. As the name suggests, this type of analysis measures the ways in which a specific person changes (or remains the same) over time. For example, a developmental scientist might be interested in studying the same group of infants at 12 months, 18 months, and 24 months to examine how vocabulary and grammar change over time. This kind of question would be best answered using a longitudinal research design. Another sort of comparison focuses on *changes between groups*. In this type of analysis, researchers study average changes in behavior between groups of different ages. Returning to the language example, a scientist might study the vocabulary and grammar used by 12-month-olds, 18-month-olds, and 24-month-olds to examine how language abilities change with age. This kind of question would be best answered using a cross-sectional research design.

Longitudinal research designs

Longitudinal research designs are used to examine behavior in the same infants and children over time. For example, when considering our example of hide-and-seek behaviors in preschoolers, a researcher might conduct a longitudinal study to examine whether 2-year-olds develop into better hidiers over time. To this end, a researcher might observe a group of 2-year-old children playing hide-and-seek with plans to observe them again when they are 4

years old – and again when they are 6 years old. This study is longitudinal in nature because the researcher plans to study the same children as they age. Based on her data, the researcher might conclude that 2-year-olds develop more mature hiding abilities with age. Remember, researchers examine games such as hide-and-seek not because they are interested in the games themselves, but because they offer clues to how children think, feel and behave at various ages.



Example of longitudinal research design

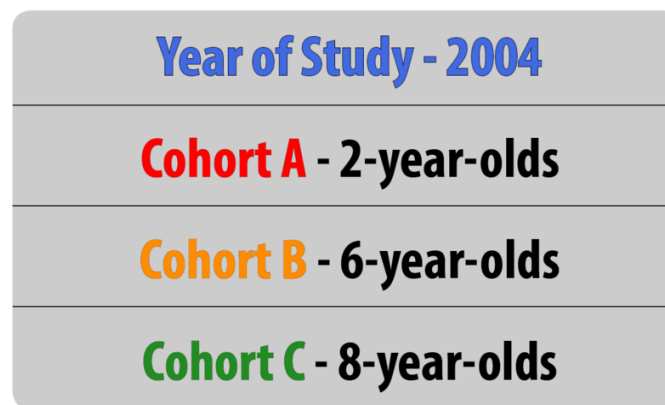
Longitudinal studies may be conducted over the short term (over a span of months, as in Wiebe, Lukowski, & Bauer, 2010) or over much longer durations (years or decades, as in Lukowski et al., 2010). For these reasons, longitudinal research designs are optimal for studying stability and change over time. Longitudinal research also has limitations, however. For one, longitudinal studies are expensive: they require that researchers maintain continued contact with participants over time, and they necessitate that scientists have funding to conduct their work over extended durations (from infancy to when participants were 19 years old in Lukowski et al., 2010). An additional risk is attrition. Attrition occurs when participants fail to complete all portions of a study. Participants may move, change their phone numbers, or simply become disinterested in participating over time. Researchers should account for the possibility of attrition by enrolling a larger sample into their study initially, as some participants will likely drop out over time.

The results from longitudinal studies may also be impacted by repeated assessments. Consider how well you would do on a math test if you were given the exact same exam every day for a week. Your performance would likely improve over time not necessarily because you developed better math abilities, but because you were continuously practicing the same math problems. This phenomenon is known as a practice effect. Practice effects occur when participants become better at a task over time because they have done it again and again; not due to natural psychological development. A final limitation of longitudinal research is that the results may be impacted by cohort effects. Cohort effects occur when the results of the study are affected by the particular point in historical time during which participants are tested. As an example, think about how peer relationships in childhood have likely changed

since February 2004 – the month and year Facebook was founded. Cohort effects can be problematic in longitudinal research because only one group of participants are tested at one point in time – different findings might be expected if participants of the same ages were tested at different points in historical time.

Cross-sectional designs

Cross-sectional research designs are used to examine behavior in participants of different ages who are tested at the same point in time. When considering our example of hide-and-seek behaviors in children, for example, a researcher might want to examine whether older children more often hide in novel locations (those in which another child in the same game has never hidden before) when compared to younger children. In this case, the researcher might observe 2-, 4-, and 6-year-old children as they play the game (the various age groups represent the “cross sections”). This research is cross-sectional in nature because the researcher plans to examine the behavior of children of different ages within the same study at the same time. Based on her data, the researcher might conclude that 2-year-olds more commonly hide in previously-searched locations relative to 6-year-olds.



Example of cross-sectional research design

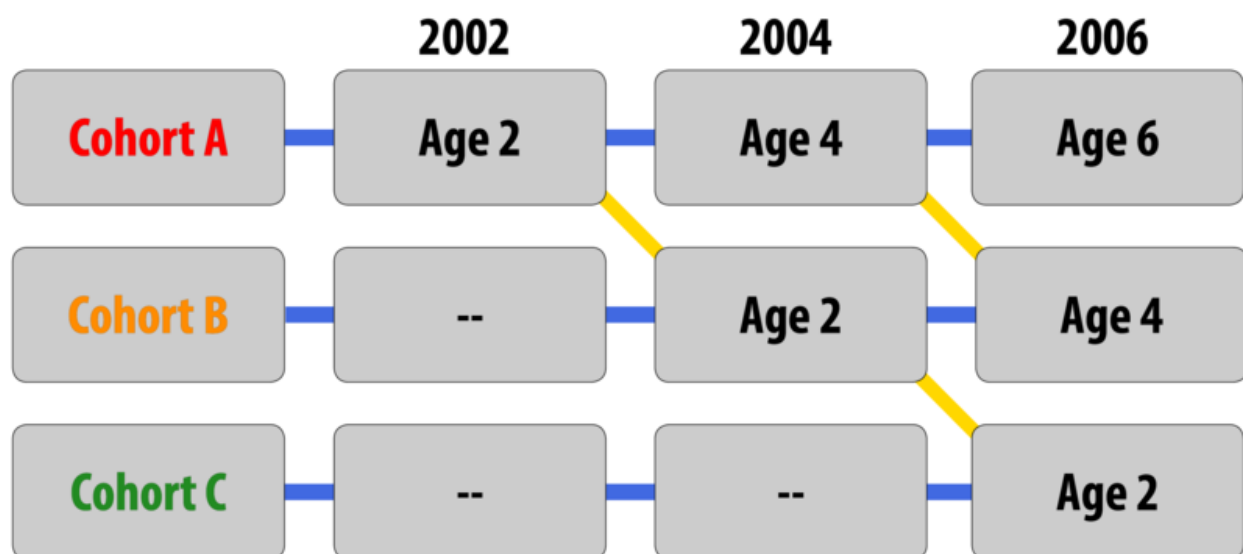
Cross-sectional designs are useful for many reasons. Because participants of different ages are tested at the same point in time, data collection can proceed at a rapid pace. In addition, because participants are only tested at one point in time, practice effects are not an issue – children do not have the opportunity to become better at the task over time. Cross-sectional designs are also more cost-effective than longitudinal research designs because there is no need to maintain contact with and follow-up on participants over time.

One of the primary limitations of cross-sectional research, however, is that the results yield information on *age-related change, not development per se*. That is, although the study described above can show that 6-year-olds are more advanced in their hiding behavior than 2-year-olds, the data used to come up with this conclusion were collected from different children. It could be, for instance, that this specific sample of 6-year-olds just happened to be particularly clever at hide-and-seek. As such, the researcher cannot conclude that 2-year-olds develop into better hidiers with age; she can only state that 6-year-olds, on average, are more sophisticated hidiers relative to children 4 years younger.

Sequential research designs

Sequential research designs include elements of both longitudinal and cross-sectional research designs. Similar to longitudinal designs, sequential research features participants who are followed over time; similar to cross-sectional designs, sequential work includes participants of different ages. This research design is also distinct from those that have been discussed previously in that children of different ages are enrolled into a study at various points in time to examine age-related changes, development within the same individuals as they age, and account for the possibility of cohort effects.

Consider, once again, our example of hide-and-seek behaviors. In a study with a sequential design, a researcher might enroll three separate groups of children (Groups A, B, and C). Children in Group A would be enrolled when they are 2 years old and would be tested again when they are 4 and 6 years old (similar in design to the longitudinal study described



Example of sequential research design

previously). Children in Group B would be enrolled when they are 4 years old and would be tested again when they are 6 and 8 years old. Finally, children in Group C would be enrolled when they are 6 years old and would be tested again when they are 8 and 10 years old.

Studies with sequential designs are powerful because they allow for both longitudinal and cross-sectional comparisons. This research design also allows for the examination of cohort effects. For example, the researcher could examine the hide-and-seek behavior of 6-year-olds in Groups A, B, and C to determine whether performance differed by group when participants were the same age. If performance differences were found, there would be evidence for a cohort effect. In the hide-and-seek example, this might mean that children from different time periods varied in the amount they giggled or how patient they are when waiting to be found. Sequential designs are also appealing because they allow researchers to learn a lot about development in a relatively short amount of time. In the previous example, a four-year research study would provide information about 8 years of developmental time by enrolling children ranging in age from two to ten years old.

Because they include elements of longitudinal and cross-sectional designs, sequential research has many of the same strengths and limitations as these other approaches. For example, sequential work may require less time and effort than longitudinal research, but more time and effort than cross-sectional research. Although practice effects may be an issue if participants are asked to complete the same tasks or assessments over time, attrition may

	Advantages	Disadvantages
Longitudinal	• Examines changes within individuals over time • Provides a developmental analysis	• Expensive • Takes a long time • Participant attrition • Possibility of practice effects • Cannot examine cohort effects
Cross-sectional	• Examines changes between participants of different ages at the same point in time • Provide information on age-related change	• Cannot examine change over time • Cannot examine cohort effects
Sequential	• Examines changes within individuals over time • Examines changes between participants of different ages at the same point in time • Can be used to examine cohort effects	• May be expensive • Possibility of practice effects

Advantages and disadvantages of different research designs

be less problematic than what is commonly experienced in longitudinal research since participants may not have to remain involved in the study for such a long period of time.

When considering the best research design to use in their research, scientists think about their main research question and the best way to come up with an answer. A table of advantages and disadvantages for each of the described research designs is provided here to help you as you consider what sorts of studies would be best conducted using each of these different approaches.

Challenges Associated with Conducting Developmental Research

The previous sections describe research tools to assess development in infancy and early childhood, as well as the ways that research designs can be used to track age-related changes and development over time. Before you begin conducting developmental research, however, you must also be aware that testing infants and children comes with its own unique set of challenges. In the final section of this module, we review some of the main issues that are encountered when conducting research with the youngest of human participants. In particular, we focus our discussion on ethical concerns, recruitment issues, and participant attrition.

Ethical concerns

As a student of psychological science, you may already know that **Institutional Review Boards (IRBs)** review and approve of all research projects that are conducted at universities, hospitals, and other institutions. An IRB is typically a panel of experts who read and evaluate proposals for research. IRB members want to ensure that the proposed research will be carried out ethically and that the potential benefits of the research outweigh the risks and harm for participants. What you may not know though, is that the IRB considers some groups of participants to be more vulnerable or at-risk than others. Whereas university students are generally not viewed as vulnerable or at-risk, infants and young children commonly fall into this category. What makes infants and young children more vulnerable during research than young adults? One reason infants and young children are perceived as being at increased risk is due to their limited cognitive capabilities, which makes them unable to state their willingness to participate in research or tell researchers when they would like to drop out of a study. For these reasons, infants and young children require special accommodations as they participate in the research process.

When thinking about special accommodations in developmental research, consider the

informed consent process. If you have ever participated in psychological research, you may know through your own experience that adults commonly sign an informed consent statement (a contract stating that they agree to participate in research) after learning about a study. As part of this process, participants are informed of the procedures to be used in the research, along with any expected risks or benefits. Infants and young children cannot verbally indicate their willingness to participate, much less understand the balance of potential risks and benefits. As such, researchers are oftentimes required to obtain written informed consent from the parent or legal guardian of the child participant, an adult who is almost always present as the study is conducted. In fact, children are not asked to indicate whether they would like to be involved in a study at all (a process known as **assent**) until they are approximately seven years old. Because infants and young children also cannot easily indicate if they would like to discontinue their participation in a study, researchers must be sensitive to changes in the state of the participant (determining whether a child is too tired or upset to continue) as well as to parent desires (in some cases, parents might want to discontinue their involvement in the research). As in adult studies, researchers must always strive to protect the rights and well-being of the minor participants and their parents when conducting developmental science.

Recruitment

An additional challenge in developmental science is participant recruitment. Recruiting university students to participate in adult studies is typically easy. Many colleges and universities offer extra credit for participation in research and have locations such as bulletin boards and school newspapers where research can be advertised. Unfortunately, young children cannot be recruited by making announcements in Introduction to Psychology courses, by posting ads on campuses, or through online platforms such as **Amazon Mechanical Turk**. Given these limitations, how do researchers go about finding infants and young children to be in their studies?

The answer to this question varies along multiple dimensions. Researchers must consider the number of participants they need and the financial resources available to them, among other things. Location may also be an important consideration. Researchers who need large numbers of infants and children may attempt to do so by obtaining infant birth records from the state, county, or province in which they reside. Some areas make this information publicly available for free, whereas birth records must be purchased in other areas (and in some locations birth records may be entirely unavailable as a recruitment tool). If birth records are available, researchers can use the obtained information to call families by phone or mail them letters describing possible research opportunities. All is not lost if this recruitment strategy

is unavailable, however. Researchers can choose to pay a recruitment agency to contact and recruit families for them. Although these methods tend to be quick and effective, they can also be quite expensive. More economical recruitment options include posting advertisements and fliers in locations frequented by families, such as mommy-and-me classes, local malls, and preschools or day care centers. Researchers can also utilize online social media outlets like Facebook, which allows users to post recruitment advertisements for a small fee. Of course, each of these different recruitment techniques requires IRB approval.

Attrition



Participating in developmental research can sometimes be difficult for both children and their parents. This can contribute to a higher attrition rate than is typical in other types of research. [Image: Tina Franklin, <https://goo.gl/bN19Gm>, CC BY 2.0, <https://goo.gl/5YbMw6>]

Another important consideration when conducting research with infants and young children is **attrition**. Although attrition is quite common in longitudinal research in particular, it is also problematic in developmental science more generally, as studies with infants and young children tend to have higher attrition rates than studies with adults. For example, high attrition rates in ERP studies oftentimes result from the demands of the task: infants are required to sit still and have a tight, wet cap placed on their heads before watching still photographs on a computer screen in a dark, quiet room. In other cases, attrition may be due to motivation (or a lack thereof). Whereas adults may be motivated to participate in research in order to receive money or extra course credit, infants and young children are not as easily enticed. In addition, infants and young children are more likely to tire

easily, become fussy, and lose interest in the study procedures than are adults. For these reasons, research studies should be designed to be as short as possible – it is likely better to break up a large study into multiple short sessions rather than cram all of the tasks into one long visit to the lab. Researchers should also allow time for breaks in their study protocols so that infants can rest or have snacks as needed. Happy, comfortable participants provide the best data.

Conclusions

Child development is a fascinating field of study – but care must be taken to ensure that researchers use appropriate methods to examine infant and child behavior, use the correct experimental design to answer their questions, and be aware of the special challenges that are part-and-parcel of developmental research. After reading this module, you should have a solid understanding of these various issues and be ready to think more critically about research questions that interest you. For example, when considering our initial example of hide-and-seek behaviors in preschoolers, you might ask questions about what other factors might contribute to hiding behaviors in children. Do children with older siblings hide in locations that were previously searched less often than children without siblings? What other abilities are associated with the development of hiding skills? Do children who use more sophisticated hiding strategies as preschoolers do better on other tests of cognitive functioning in high school? Many interesting questions remain to be examined by future generations of developmental scientists – maybe you will make one of the next big discoveries!

Outside Resources

Sample Child Behavior Checklist for Ages 6-18

<https://aseba.org/wp-content/uploads/2019/02/schoolagecbcl.pdf>

Video: A 3 and a half minute video depicting the violation of expectation paradigm.

<https://www.youtube.com/watch?v=l1VK2iawS34>

Video: A popular TED talk by Dr. Laura Schulz on the topic of how babies make decisions.

<https://www.youtube.com/watch?v=y1KIVZw7Jxk>

Video: A popular TED talk by Dr. Patricia Kuhl on the topic of how babies learn language.

<https://www.youtube.com/watch?v=M-ymanHajN8>

Video: A two and a half minute video showing how ERP (brain activity) can be measured with children in the laboratory.

<https://www.youtube.com/watch?v=gMBYs-pP-9o>

Web: A link to Angela Lukowski's research laboratory. The site includes descriptions of the research and researchers as well as a list of publications.

<http://memorydevelopment.sococo.uci.edu/>

Web: The International Congress on Infant Studies - a professional society focused on infant research

<http://www.infantstudies.org/>

Web: The Society for Research on Adolescence - a professional society focused on research on adolescence

<http://www.s-r-a.org/>

Web: The Society for Research on Child Development - a professional society focused on child development research

<http://www.srcd.org/>

Discussion Questions

1. Why is it important to conduct research on infants and children?

2. What are some possible benefits and limitations of the various research methods discussed in this module?
3. Why is it important to examine cohort effects in developmental research?
4. Think about additional challenges or unique issues that might be experienced by developmental scientists. How would they handle the challenges and issues you've addressed?
5. Work with your peers to design a study to identify whether children who were good hiders as preschoolers are more cognitively advanced in high school. What research design would you use and why? What are the advantages and limitations of the design you selected?

Vocabulary

Assent

When minor participants are asked to indicate their willingness to participate in a study. This is usually obtained from participants who are at least 7 years old, in addition to parent or guardian consent.

Attrition

When a participant drops out, or fails to complete, all parts of a study.

Bidirectional relations

When one variable is likely both cause and consequence of another variable.

Cohort effects

When research findings differ for participants of the same age tested at different points in historical time.

Cross-sectional research

A research design used to examine behavior in participants of different ages who are tested at the same point in time.

Dishabituation

When participants demonstrated increased attention (through looking or listening behavior) to a new stimulus after having been habituated to a different stimulus.

Elicited imitation

A behavioral method used to examine recall memory in infants and young children.

Event-related potentials (ERP)

The recording of participant brain activity using a stretchy cap with small electrodes or sensors as participants engage in a particular task (commonly viewing photographs or listening to auditory stimuli).

Habituation

When participants demonstrated decreased attention (through looking or listening behavior) to repeatedly-presented stimuli.

Informed consent

The process of getting permission from adults for themselves and their children to take part in research.

Institutional Review Boards (IRBs)

A committee that reviews and approves research procedures involving human participants and animal subjects to ensure that the research is conducted in accordance with federal, institutional, and ethical guidelines.

Interview techniques

A research method in which participants are asked to report on their experiences using language, commonly by engaging in conversation with a researcher (participants may also be asked to record their responses in writing).

Involuntary or obligatory responses

Behaviors in which individuals engage that do not require much conscious thought or effort.

Longitudinal research

A research design used to examine behavior in the same participants over short (months) or long (decades) periods of time.

Motor control

The use of thinking to direct muscles and limbs to perform a desired action.

Object permanence

The understanding that objects continue to exist even when they cannot be directly observed (e.g., that a pen continues to exist even when it is hidden under a piece of paper).

Practice effect

When participants get better at a task over time by “practicing” it through repeated assessments instead of due to actual developmental change (practice effects can be particularly problematic in longitudinal and sequential research designs).

Psychophysiological responses

Recording of biological measures (such as heart rate and hormone levels) and neurological responses (such as brain activity) that may be associated with observable behaviors.

Recall memory

The process of remembering discrete episodes or events from the past, including encoding, consolidation and storage, and retrieval.

Research design

The strategy (or “blueprint”) for deciding how to collect and analyze research information.

Research methods

The specific tools and techniques used by researchers to collect information.

Sequential research designs

A research design that includes elements of cross-sectional and longitudinal research designs. Similar to cross-sectional designs, sequential research designs include participants of different ages within one study; similar to longitudinal designs, participants of different ages are followed over time.

Solidity principle

The idea that two solid masses should not be able to move through one another.

Verbal report paradigms

Research methods that require participants to report on their experiences, thoughts, feelings, etc., using language.

Vignette

A short story that presents a situation that participants are asked to respond to.

Violation of expectation paradigm

A research method in which infants are expected to respond in a particular way because one of two conditions violates or goes against what they should expect based on their everyday experiences (e.g., it violates our expectations that Wile E. Coyote runs off a cliff but does not immediately fall to the ground below).

Voluntary responses

Behaviors that a person has control over and completes by choice.

References

- Achenbach, T. M., & Rescorla, L. A. (2000). *Manual for the ASEBA preschool forms and profiles: An integrated system of multi-informant assessment*. Burlington, VT: University of Vermont Department of Psychiatry.
- Baillargeon, R. (1987). Object permanence in 3½- and 4½-month-old infants. *Developmental Psychology*, 23, 655-664. doi: 10.1037/0012-1649.23.5.655
- Baillargeon, R., Spelke, E., & Wasserman, S. (1985). Object permanence in five-month-old infants. *Cognition*, 20, 191-208. doi: 10.1016/0010-0277(85)90008-3
- Barr, R., Dowden, A., & Hayne, H. (1996). Developmental changes in deferred imitation by 6- to 24-month-old infants. *Infant Behavior and Development*, 19, 159-170. doi: 10.1016/s0163-6383(96)90015-6
- Bauer, P. J., Wenner, J. A., Dropik, P. L., & Wewerka, S. S. (2000). Parameters of remembering and forgetting in the transition from infancy to early childhood. *Monographs of the Society for Research in Child Development*, 65, 1-204. doi: 10.1016/j.imlet.2014.04.001
- Bauer, P. J., Wiebe, S. A., Carver, L. J., Waters, J. M., & Nelson, C. A. (2003). Developments in long-term explicit memory late in the first year of life: Behavioral and electrophysiological indices. *Psychological Science*, 14, 629-635. doi: 10.1046/j.0956-7976.2003.psci_1476.x
- Bauer, P. J., Wiebe, S. A., Waters, J. M., & Bangston, S. K. (2001). Reexposure breeds recall: Effects of experience on 9-month-olds' ordered recall. *Journal of Experimental Child Psychology*, 80, 174-200. doi: 10.1006/jecp.2000.2628
- Bentin, S., Allison, T., Puce, A., Perez, E., & McCarthy, G. (2010). Electrophysiological studies of face perception in humans. *Journal of Cognitive Neuroscience*, 8, 551-565. doi: 10.1162/jocn.1996.8.6.551
- Carver, L. J., & Bauer, P. J. (1999). When the event is more than the sum of its parts: 9-month-olds' long-term ordered recall. *Memory*, 7, 147-174. doi: 10.1080/741944070
- Cashon, C. H., & Cohen, L. B. (2000). Eight-month-old infants' perception of possible and impossible events. *Infancy*, 1, 429-446. doi: 10.1016/s0163-6383(98)91561-2
- Collie, R., & Hayne, H. (1999). Deferred imitation by 6- and 9-month-old infants: More evidence for declarative memory. *Developmental Psychobiology*, 35, 83-90. doi: 10.1002/(sici)1098-2302(199909)35:23.0.co;2-s
- DeBoer, T., Scott, L. S., & Nelson, C. A. (2005). ERPs in developmental populations. In T. C. Handy (Ed.), *Event-related potentials: A methods handbook* (pp. 263-297). Cambridge, MA: The MIT Press.

- Lukowski, A. F., & Milojevich, H. M. (2016). Examining recall memory in infancy and early childhood using the elicited imitation paradigm. *Journal of Visualized Experiments*, 110, e53347.
- Lukowski, A. F., Koss, M., Burden, M. J., Jonides, J., Nelson, C. A., Kaciroti, N., ... Lozoff, B. (2010). Iron deficiency in infancy and neurocognitive functioning at 19 years: Evidence of long-term deficits in executive function and recognition memory. *Nutritional Neuroscience*, 13, 54-70. doi: 10.1179/147683010x12611460763689
- Ronald, A., Pennell, C. E., & Whitehouse, A. J. O. (2011). Prenatal maternal stress associated with ADHD and autistic traits in early childhood. *Frontiers in Psychology*, 1, 1-8. doi: 10.3389/fpsyg.2010.00223
- Walrath, R. (2011). Kohlberg's theory of moral development. In *Encyclopedia of Child Behavior and Development* (pp. 859-860).
- Webb, S. J., Dawson, G., Bernier, R., & Panagiotides, H. (2006). ERP evidence of atypical face processing in young children with autism. *Journal of Autism and Developmental Disorders*, 36, 884-890. doi: 10.1007/s10803-006-0126-x
- Wiebe, S. A., Lukowski, A. F., & Bauer, P. J. (2010). Sequence imitation and reaching measures of executive control: A longitudinal examination in the second year of life. *Developmental Neuropsychology*, 35, 522-538. doi: 10.1080/87565641.2010.494751

Cognitive Development in Childhood

Robert Siegler

This module examines what cognitive development is, major theories about how it occurs, the roles of nature and nurture, whether it is continuous or discontinuous, and how research in the area is being used to improve education.

Learning Objectives

- Be able to identify and describe the main areas of cognitive development.
- Be able to describe major theories of cognitive development and what distinguishes them.
- Understand how nature and nurture work together to produce cognitive development.
- Understand why cognitive development is sometimes viewed as discontinuous and sometimes as continuous.
- Know some ways in which research on cognitive development is being used to improve education.

Introduction

By the time you reach adulthood you have learned a few things about how the world works. You know, for instance, that you can't walk through walls or leap into the tops of trees. You know that although you cannot see your car keys they've got to be around here someplace. What's more, you know that if you want to communicate complex ideas like ordering a triple-shot soy vanilla latte with chocolate sprinkles it's better to use words with meanings attached to them rather than simply gesturing and grunting. People accumulate all this useful

knowledge through the process of cognitive development, which involves a multitude of factors, both inherent and learned.

Cognitive development refers to the development of thinking across the lifespan. Defining thinking can be problematic, because no clear boundaries separate thinking from other mental activities. Thinking obviously involves the higher mental processes: problem solving, reasoning, creating, conceptualizing, categorizing, remembering, planning, and so on. However, thinking also involves other mental processes that seem more basic and at which even toddlers are skilled—such as perceiving objects and events in the environment, acting skillfully on objects to obtain goals, and understanding and producing language. Yet other areas of human development that involve thinking are not usually associated with cognitive development, because thinking isn't a prominent feature of them—such as personality and temperament.



Cognitive development in childhood is about change. From birth to adolescence a young person's mind changes dramatically in many important ways. [Image: One Laptop per Child, <https://goo.gl/L1eAsO>, CC BY 2.0, <https://goo.gl/9uSnqN>]

As the name suggests, cognitive development is about change. Children's thinking changes in dramatic and surprising ways. Consider DeVries's (1969) study of whether young children understand the difference between appearance and reality. To find out, she brought an unusually even-tempered cat named Maynard to a psychology laboratory and allowed the 3- to 6-year-old participants in the study to pet and play with him. DeVries then put a mask of a fierce dog on Maynard's head, and asked the children what Maynard was. Despite all of the children having identified Maynard previously as a cat, now most 3-year-olds said that he was a dog and claimed that he had a dog's bones and a dog's stomach. In contrast, the 6-year-olds weren't fooled; they had no doubt that Maynard remained a cat. Understanding how children's thinking changes so dramatically in just a few years is one of the fascinating challenges in studying cognitive development.

There are several main types of theories of child development. Stage theories, such as **Piaget's stage theory**, focus on whether children progress through qualitatively different stages of

development. Sociocultural theories, such as that of Lev Vygotsky, emphasize how other people and the attitudes, values, and beliefs of the surrounding culture, influence children's development. Information processing theories, such as that of David Klahr, examine the mental processes that produce thinking at any one time and the transition processes that lead to growth in that thinking.

At the heart of all of these theories, and indeed of all research on cognitive development, are two main questions: (1) How do nature and nurture interact to produce cognitive development? (2) Does cognitive development progress through qualitatively distinct stages? In the remainder of this module, we examine the answers that are emerging regarding these questions, as well as ways in which cognitive developmental research is being used to improve education.

Nature and Nurture

The most basic question about child development is how nature and nurture together shape development. Nature refers to our biological endowment, the genes we receive from our parents. Nurture refers to the environments, social as well as physical, that influence our development, everything from the womb in which we develop before birth to the homes in which we grow up, the schools we attend, and the many people with whom we interact.

The nature-nurture issue is often presented as an either-or question: Is our intelligence (for example) due to our genes or to the environments in which we live? In fact, however, every aspect of development is produced by the interaction of genes and environment. At the most basic level, without genes, there would be no child, and without an environment to provide nurture, there also would be no child.

The way in which nature and nurture work together can be seen in findings on visual development. Many people view vision as something that people either are born with or that is purely a matter of biological maturation, but it also depends on the right kind of experience at the right time. For example, development of depth perception, the ability to actively perceive the distance from oneself to objects in the environment, depends on seeing patterned light and having normal brain activity in response to the patterned light, in infancy (Held, 1993). If no patterned light is received, for example when a baby has severe cataracts or blindness that is not surgically corrected until later in development, depth perception remains abnormal even after the surgery.

Adding to the complexity of the nature-nurture interaction, children's genes lead to their



A child that is perceived to be attractive and calm may receive a different sort of care and attention from adults and as a result enjoy a developmental advantage. [Image: Cairn 111, <https://goo.gl/6RpBVt>, CC BY-NC-SA 2.0, <https://goo.gl/HEXbAA>]

eliciting different treatment from other people, which influences their cognitive development. For example, infants' physical attractiveness and temperament are influenced considerably by their genetic inheritance, but it is also the case that parents provide more sensitive and affectionate care to easygoing and attractive infants than to difficult and less attractive ones, which can contribute to the infants' later cognitive development (Langlois et al., 1995; van den Boom & Hoeksma, 1994).

Also contributing to the complex interplay of nature and nurture is the role of children in shaping their own cognitive development. From the first days out of the womb, children actively choose to

attend more to some things and less to others. For example, even 1-month-olds choose to look at their mother's face more than at the faces of other women of the same age and general level of attractiveness (Bartrip, Morton, & de Schonen, 2001). Children's contributions to their own cognitive development grow larger as they grow older (Scarr & McCartney, 1983). When children are young, their parents largely determine their experiences: whether they will attend day care, the children with whom they will have play dates, the books to which they have access, and so on. In contrast, older children and adolescents choose their environments to a larger degree. Their parents' preferences largely determine how 5-year-olds spend time, but 15-year-olds' own preferences largely determine when, if ever, they set foot in a library. Children's choices often have large consequences. To cite one example, the more that children choose to read, the more that their reading improves in future years (Baker, Dreher, & Guthrie, 2000). Thus, the issue is not whether cognitive development is a product of nature or nurture; rather, the issue is how nature and nurture work together to produce cognitive development.

Does Cognitive Development Progress Through Distinct Stages?

Some aspects of the development of living organisms, such as the growth of the width of a pine tree, involve **quantitative changes**, with the tree getting a little wider each year. Other changes, such as the life cycle of a ladybug, involve **qualitative changes**, with the creature

becoming a totally different type of entity after a transition than before (Figure 1). The existence of both gradual, quantitative changes and relatively sudden, qualitative changes in the world has led researchers who study cognitive development to ask whether changes in children's thinking are gradual and continuous or sudden and discontinuous.

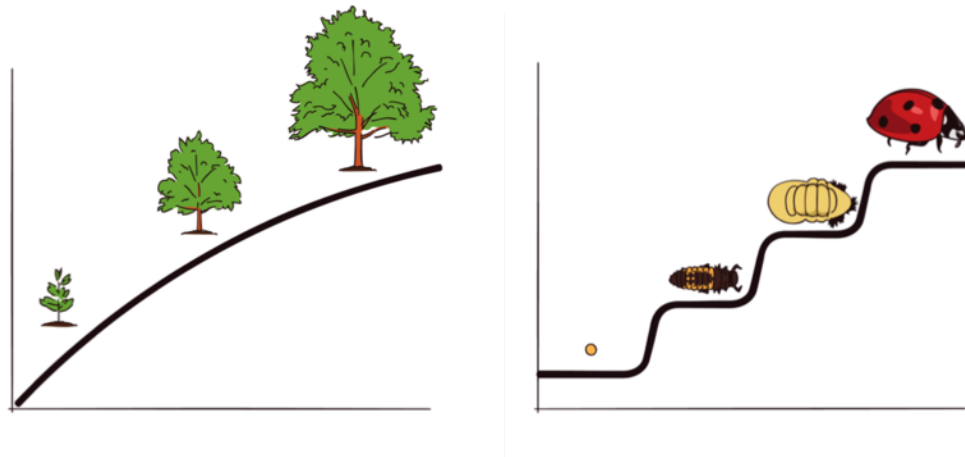


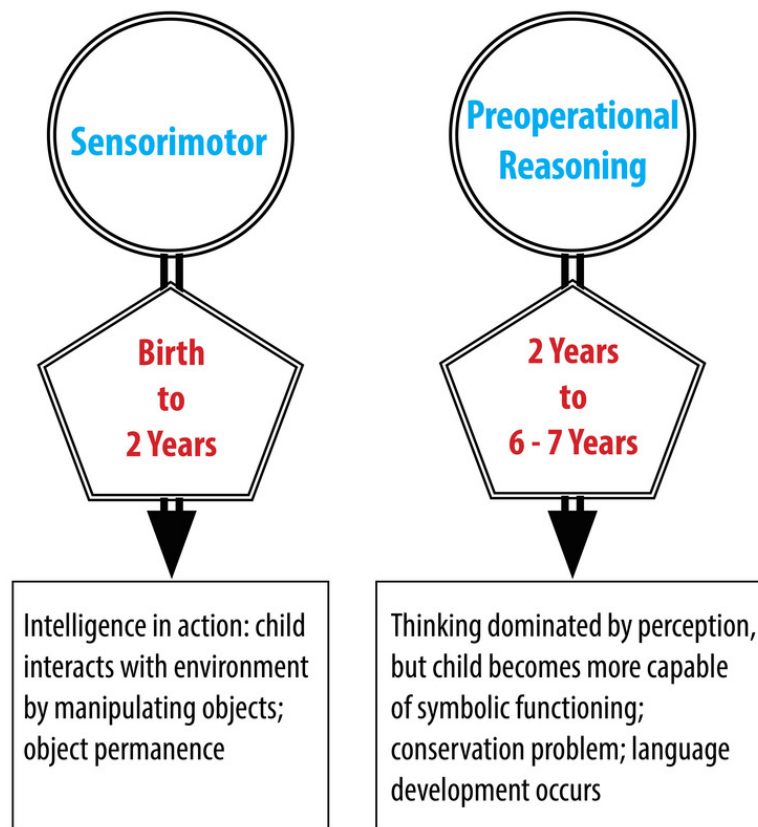
Figure 1: Continuous and discontinuous development. Some researchers see development as a continuous gradual process, much like a maple tree growing steadily in height and cross-sectional area. Other researchers see development as a progression of discontinuous stages, involving rapid discontinuous changes, such as those in the life cycle of a ladybug, separated by longer periods of slow, gradual change.

The great Swiss psychologist Jean Piaget proposed that children's thinking progresses through a series of four discrete stages. By "stages," he meant periods during which children reasoned similarly about many superficially different problems, with the stages occurring in a fixed order and the thinking within different stages differing in fundamental ways. The four stages that Piaget hypothesized were the sensorimotor stage (birth to 2 years), the preoperational reasoning stage (2 to 6 or 7 years), the concrete operational reasoning stage (6 or 7 to 11 or 12 years), and the formal operational reasoning stage (11 or 12 years and throughout the rest of life).

During the sensorimotor stage, children's thinking is largely realized through their perceptions of the world and their physical interactions with it. Their mental representations are very limited. Consider Piaget's object permanence task, which is one of his most famous problems. If an infant younger than 9 months of age is playing with a favorite toy, and another person removes the toy from view, for example by putting it under an opaque cover and not letting the infant immediately reach for it, the infant is very likely to make no effort to retrieve it and to show no emotional distress (Piaget, 1954). This is not due to their being uninterested in the toy or unable to reach for it; if the same toy is put under a clear cover, infants below 9

months readily retrieve it (Munakata, McClelland, Johnson, & Siegler, 1997). Instead, Piaget claimed that infants less than 9 months do not understand that objects continue to exist even when out of sight.

During the preoperational stage, according to Piaget, children can solve not only this simple problem (which they actually can solve after 9 months) but show a wide variety of other symbolic-representation capabilities, such as those involved in drawing and using language. However, such 2- to 7-year-olds tend to focus on a single dimension, even when solving problems would require them to consider multiple dimensions. This is evident in Piaget's (1952) conservation problems. For example, if a glass of water is poured into a taller, thinner glass, children below age 7 generally say that there now is more water than before. Similarly, if a clay ball is reshaped into a long, thin sausage, they claim that there is now more clay, and if a row of coins is spread out, they claim that there are now more coins. In all cases, the children are focusing on one dimension, while ignoring the changes in other dimensions (for example, the greater width of the glass and the clay ball).

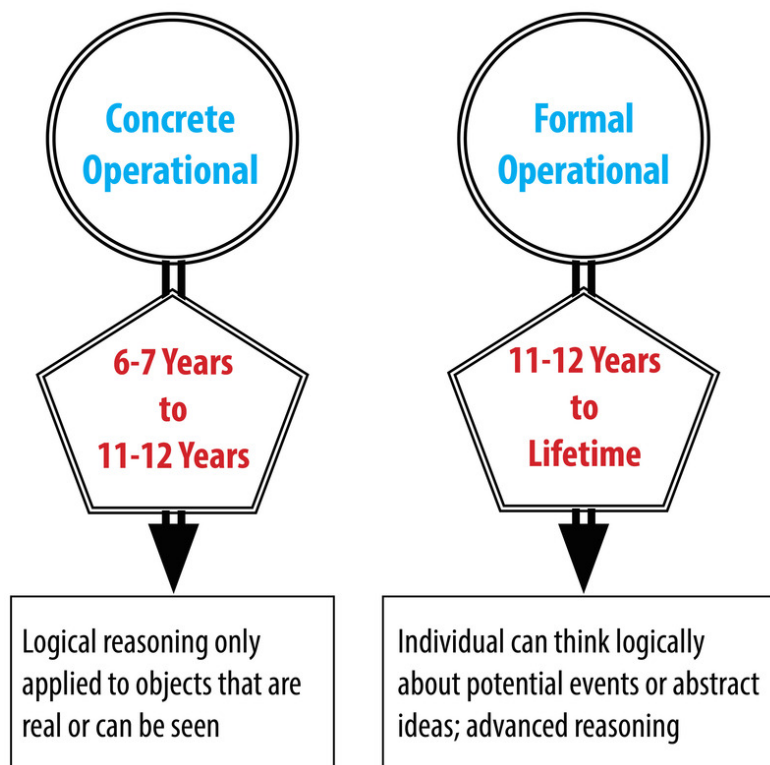


Piaget's Sensorimotor and Pre-operational Reasoning stages

Children overcome this tendency to focus on a single dimension during the **concrete operations stage**, and think logically in most situations. However, according to Piaget, they still cannot think in systematic scientific ways, even when such thinking would be useful. Thus, if asked to find out which variables influence the period that a pendulum takes to complete its arc, and given weights that they can attach to strings in order to do experiments with the pendulum to find out, most children younger than age 12, perform biased experiments from which no conclusion can be drawn, and then conclude that whatever they originally believed is correct. For example, if a boy believed that weight was the only variable that mattered, he might put the heaviest weight on the shortest string and push it the hardest, and then conclude that just as he thought, weight is the only variable that matters (Inhelder & Piaget, 1958).

Finally, in the formal operations period, children attain the reasoning power of mature adults, which allows them to solve the pendulum problem and a wide range of other problems. However, this **formal operations stage** tends not to occur without exposure to formal education in scientific reasoning, and appears to be largely or completely absent from some societies that do not provide this type of education.

Although Piaget's theory has been very influential, it has not gone unchallenged. Many more recent researchers have obtained findings indicating that cognitive development is



Piaget's Concrete and Formal Operations stages

considerably more continuous than Piaget claimed. For example, Diamond (1985) found that on the object permanence task described above, infants show earlier knowledge if the waiting period is shorter. At age 6 months, they retrieve the hidden object if the wait is no longer than 2 seconds; at 7 months, they retrieve it if the wait is no longer than 4 seconds; and so on. Even earlier, at 3 or 4 months, infants show surprise in the form of longer looking times if objects suddenly appear to vanish with no obvious cause (Baillargeon, 1987). Similarly, children's specific experiences can greatly influence when developmental changes occur. Children of pottery makers in Mexican villages, for example, know that reshaping clay does not change the amount of clay at much younger ages than children who do not have similar experiences (Price-Williams, Gordon, & Ramirez, 1969).

So, is cognitive development fundamentally continuous or fundamentally discontinuous? A reasonable answer seems to be, "It depends on how you look at it and how often you look." For example, under relatively facilitative circumstances, infants show early forms of object permanence by 3 or 4 months, and they gradually extend the range of times for which they can remember hidden objects as they grow older. However, on Piaget's original object permanence task, infants do quite quickly change toward the end of their first year from not reaching for hidden toys to reaching for them, even after they've experienced a substantial delay before being allowed to reach. Thus, the debate between those who emphasize discontinuous, stage-like changes in cognitive development and those who emphasize gradual continuous changes remains a lively one.

Applications to Education

Understanding how children think and learn has proven useful for improving education. One example comes from the area of reading. Cognitive developmental research has shown that **phonemic awareness**—that is, awareness of the component sounds within words—is a crucial skill in learning to read. To measure awareness of the component sounds within words, researchers ask children to decide whether two words rhyme, to decide whether the words start with the same sound, to identify the component sounds within words, and to indicate what would be left if a given sound were removed from a word. Kindergartners' performance on these tasks is the strongest predictor of reading achievement in third and fourth grade, even stronger than IQ or social class background (Nation, 2008). Moreover, teaching these skills to randomly chosen 4- and 5-year-olds results in their being better readers years later (National Reading Panel, 2000).

Another educational application of cognitive developmental research involves the area of mathematics. Even before they enter kindergarten, the mathematical knowledge of children



Activities like playing games that involve working with numbers and spatial relationships can give young children a developmental advantage over peers who have less exposure to the same concepts. [Image: Ben Husmann, <https://goo.gl/awOXSw>, CC BY 2.0, <https://goo.gl/9uSnqN>]

from low-income backgrounds lags far behind that of children from more affluent backgrounds. Ramani and Siegler (2008) hypothesized that this difference is due to the children in middle- and upper-income families engaging more frequently in numerical activities, for example playing numerical board games such as Chutes and Ladders. Chutes and Ladders is a game with a number in each square; children start at the number one and spin a spinner or throw a dice to determine how far to move their token. Playing this game seemed likely to teach children about numbers, because in it, larger numbers are associated with greater values on a variety of dimensions. In particular, the higher the number that a child's token reaches, the greater the distance the

token will have traveled from the starting point, the greater the number of physical movements the child will have made in moving the token from one square to another, the greater the number of number-words the child will have said and heard, and the more time will have passed since the beginning of the game. These spatial, kinesthetic, verbal, and time-based cues provide a broad-based, multisensory foundation for knowledge of numerical magnitudes (the sizes of numbers), a type of knowledge that is closely related to mathematics achievement test scores (Booth & Siegler, 2006).

Playing this numerical board game for roughly 1 hour, distributed over a 2-week period, improved low-income children's knowledge of numerical magnitudes, ability to read printed numbers, and skill at learning novel arithmetic problems. The gains lasted for months after the game-playing experience (Ramani & Siegler, 2008; Siegler & Ramani, 2009). An advantage of this type of educational intervention is that it has minimal if any cost—a parent could just draw a game on a piece of paper.

Understanding of cognitive development is advancing on many different fronts. One exciting area is linking changes in brain activity to changes in children's thinking (Nelson et al., 2006). Although many people believe that brain maturation is something that occurs before birth, the brain actually continues to change in large ways for many years thereafter. For example,

a part of the brain called the prefrontal cortex, which is located at the front of the brain and is particularly involved with planning and flexible problem solving, continues to develop throughout adolescence (Blakemore & Choudhury, 2006). Such new research domains, as well as enduring issues such as nature and nurture, continuity and discontinuity, and how to apply cognitive development research to education, insure that cognitive development will continue to be an exciting area of research in the coming years.

Conclusion

Research into cognitive development has shown us that minds don't just form according to a uniform blueprint or innate intellect, but through a combination of influencing factors. For instance, if we want our kids to have a strong grasp of language we could concentrate on phonemic awareness early on. If we want them to be good at math and science we could engage them in numerical games and activities early on. Perhaps most importantly, we no longer think of brains as empty vessels waiting to be filled up with knowledge but as adaptable organs that develop all the way through early adulthood.

Outside Resources

Book: Frye, D., Baroody, A., Burchinal, M., Carver, S. M., Jordan, N. C., & McDowell, J. (2013). Teaching math to young children: A practice guide. Washington, DC: National Center for Education Evaluation and Regional Assistance (NCEE), Institute of Education Sciences, U.S. Department of Education.

Book: Goswami, U. G. (2010). The Blackwell Handbook of Childhood Cognitive Development. New York: John Wiley and Sons.

Book: Kuhn, D., & Siegler, R. S. (Vol. Eds.). (2006). Volume 2: Cognition, perception, and language. In W. Damon & R. M. Lerner (Series Eds.), Handbook of child psychology (6th ed.). Hoboken, NJ: Wiley.

Book: Miller, P. H. (2011). Theories of developmental psychology (5th ed.). New York: Worth.

Book: Siegler, R. S., & Alibali, M. W. (2004). Children's thinking (4th ed.). Upper Saddle River, NJ: Prentice-Hall.

Discussion Questions

1. Why are there different theories of cognitive development? Why don't researchers agree on which theory is the right one?
2. Do children's natures differ, or do differences among children only reflect differences in their experiences?
3. Do you see development as more continuous or more discontinuous?
4. Can you think of ways other than those described in the module in which research on cognitive development could be used to improve education?

Vocabulary

Chutes and Ladders

A numerical board game that seems to be useful for building numerical knowledge.

Concrete operations stage

Piagetian stage between ages 7 and 12 when children can think logically about concrete situations but not engage in systematic scientific reasoning.

Conservation problems

Problems pioneered by Piaget in which physical transformation of an object or set of objects changes a perceptually salient dimension but not the quantity that is being asked about.

Continuous development

Ways in which development occurs in a gradual incremental manner, rather than through sudden jumps.

Depth perception

The ability to actively perceive the distance from oneself of objects in the environment.

Discontinuous development

Discontinuous development

Formal operations stage

Piagetian stage starting at age 12 years and continuing for the rest of life, in which adolescents may gain the reasoning powers of educated adults.

Information processing theories

Theories that focus on describing the cognitive processes that underlie thinking at any one age and cognitive growth over time.

Nature

The genes that children bring with them to life and that influence all aspects of their development.

Numerical magnitudes

The sizes of numbers.

Nurture

The environments, starting with the womb, that influence all aspects of children's development.

Object permanence task

The Piagetian task in which infants below about 9 months of age fail to search for an object that is removed from their sight and, if not allowed to search immediately for the object, act as if they do not know that it continues to exist.

Phonemic awareness

Awareness of the component sounds within words.

Piaget's theory

Theory that development occurs through a sequence of discontinuous stages: the sensorimotor, preoperational, concrete operational, and formal operational stages.

Preoperational reasoning stage

Period within Piagetian theory from age 2 to 7 years, in which children can represent objects through drawing and language but cannot solve logical reasoning problems, such as the conservation problems.

Qualitative changes

Large, fundamental change, as when a caterpillar changes into a butterfly; stage theories such as Piaget's posit that each stage reflects qualitative change relative to previous stages.

Quantitative changes

Gradual, incremental change, as in the growth of a pine tree's girth.

Sensorimotor stage

Period within Piagetian theory from birth to age 2 years, during which children come to represent the enduring reality of objects.

Sociocultural theories

Theory founded in large part by Lev Vygotsky that emphasizes how other people and the attitudes, values, and beliefs of the surrounding culture influence children's development.

References

- Baillargeon, R. (1987). Object permanence in 3 1/2- and 4 1/2-month-old infants. *Developmental Psychology*, 23, 655–664.
- Baker, L., Dreher, M. J., & Guthrie, J. T., (Eds.). (2000). *Engaging young readers: Promoting achievement and motivation*. New York: Guilford.
- Bartrip, J., Morton, J., & De Schonen, S. (2001). Responses to mother's face in 3-week to 5-month old infants. *British Journal of Developmental Psychology*, 19, 219–232
- Blakemore, S.-J., & Choudhury, S. (2006). Development of the adolescent brain: Implications for executive function and social cognition. *Journal of Child Psychiatry and Psychology*, 47, 296–312.
- Booth, J. L., & Siegler, R. S. (2006). Developmental and individual differences in pure numerical estimation. *Developmental Psychology*, 41, 189–201.
- DeVries, R. (1969). Constancy of genetic identity in the years three to six. Monographs of the Society for Research in Child Development, 34, 127.
- Diamond, A. (1985). Development of the ability to use recall to guide action, as indicated by infants' performance on AB. *Child Development*, 56, 868–883.
- Held, R. (1993). What can rates of development tell us about underlying mechanisms? In C. E. Granrud (Ed.), *Visual perception and cognition in infancy* (pp. 75–90). Hillsdale, NJ: Erlbaum.
- Inhelder, B., & Piaget, J. (1958). *The growth of logical thinking from childhood to adolescence*. New York: Basic Books.
- Langlois, J. H., Ritter, J. M., Casey, R. J., & Sawin, D. B. (1995). Infant attractiveness predicts maternal behaviors and attitudes. *Developmental Psychology*, 31, 464–472.
- Munakata, Y., McClelland, J. L., Johnson, M. H., & Siegler, R. S. (1997). Rethinking infant knowledge: Toward an adaptive process account of successes and failures in object permanence tasks. *Psychological Review*, 104, 686–713.
- Nation, K. (2008). Learning to read words. *The Quarterly Journal of Experimental Psychology*, 61, 1121–1133.
- National Reading Panel (2000). *Teaching children to read: An evidence-based assessment of the scientific research literature on reading and its implications for reading instruction*. Washington, DC: National Institute of Child Health and Human Development.
- Nelson, C. A., Thomas, K. M., & de Haan, M. (2006). Neural bases of cognitive development. In W. Damon & R. M. Lerner (Series Eds.) & D. Kuhn & R. S. Siegler (Vol. Eds.), *Handbook of child psychology: Volume 2: Cognition, perception, and language* (6th ed., pp. 3–57). Hoboken,

NJ: Wiley.

Piaget, J. (1954). *The construction of reality in the child*. New York: BasicBooks.

Piaget, J. (1952). *The child's concept of number*. New York: W. W. Norton.

Price-Williams, D. R., Gordon, W., & Ramirez, M. (1969). Skill and conservation: A study of pottery making children. *Developmental Psychology*, 1, 769.

Ramani, G. B., & Siegler, R. S. (2008). Promoting broad and stable improvements in low-income children's numerical knowledge through playing number board games. *Child Development*, 79, 375–394.

Scarr, S., & McCartney, K. (1983). How people make their own environments: A theory of genotype-environment effects. *Child Development*, 54, 424–435.

Siegler, R. S., & Ramani, G. B. (2009). Playing linear number board games—but not circular ones—improves low-income preschoolers' numerical understanding. *Journal of Educational Psychology*, 101, 545–560.

van den Boom, D. C., & Hoeksma, J. B. (1994). The effect of infant irritability on mother-infant interaction: A growth curve analysis. *Developmental Psychology*, 30, 581–590.

Social and Personality Development in Childhood

Ross Thompson

Childhood social and personality development emerges through the interaction of social influences, biological maturation, and the child's representations of the social world and the self. This interaction is illustrated in a discussion of the influence of significant relationships, the development of social understanding, the growth of personality, and the development of social and emotional competence in childhood.

Learning Objectives

- Provide specific examples of how the interaction of social experience, biological maturation, and the child's representations of experience and the self provide the basis for growth in social and personality development.
- Describe the significant contributions of parent-child and peer relationships to the development of social skills and personality in childhood.
- Explain how achievements in social understanding occur in childhood. Moreover, do scientists believe that infants and young children are egocentric?
- Describe the association of temperament with personality development.
- Explain what is "social and emotional competence" and provide some examples of how it develops in childhood.

Introduction

"How have I become the kind of person I am today?" Every adult ponders this question from time to time. The answers that readily come to mind include the influences of parents, peers, temperament, a moral compass, a strong sense of self, and sometimes critical life experiences such as parental divorce. Social and personality development encompasses these and many other influences on the growth of the person. In addition, it addresses questions that are at the heart of understanding how we develop as unique people. How much are we products of nature or nurture? How enduring are the influences of early experiences? The study of social and personality development offers perspective on these and other issues, often by showing how complex and multifaceted are the influences on developing children, and thus the intricate processes that have made you the person you are today (Thompson, 2006a).

Understanding social and personality development requires looking at children from three perspectives that interact to shape development. The first is the social context in which each child lives, especially the relationships that provide security, guidance, and knowledge. The second is biological maturation that supports developing social and emotional competencies and underlies temperamental individuality. The third is children's developing representations of themselves and the social world. Social and personality development is best understood as the continuous interaction between these social, biological, and representational aspects of psychological development.



Humans are inherently social creatures. Mostly, we work, play, and live together in groups. [Image: The Daring Librarian, <https://goo.gl/LmA2pS>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Relationships

This interaction can be observed in the development of the earliest relationships between infants and their parents in the first year. Virtually all infants living in normal circumstances develop strong emotional attachments to those who care for them. Psychologists believe that the development of these attachments is as biologically natural as learning to walk and not simply a byproduct of the parents' provision of food or warmth. Rather, attachments have evolved in humans because they promote children's motivation to stay close to those who care for them and, as a consequence, to benefit from the learning, security, guidance, warmth,

and affirmation that close relationships provide (Cassidy, 2008).



One of the first and most important relationships is between mothers and infants. The quality of this relationship has an effect on later psychological and social development. [Image: Premnath Thirumalaisamy, <https://goo.gl/66BROf>, CC BY-NC 2.0, <https://goo.gl/Fllc2e>]

parents are simply poorly emotionally equipped to take on the responsibility of caring for a child.

Although nearly all infants develop emotional attachments to their caregivers-- parents, relatives, nannies-- their sense of security in those attachments varies. Infants become *securely* attached when their parents respond sensitively to them, reinforcing the infants' confidence that their parents will provide support when needed. Infants become *insecurely* attached when care is inconsistent or neglectful; these infants tend to respond avoidantly, resistantly, or in a disorganized manner (Belsky & Pasco Fearon, 2008). Such insecure attachments are not necessarily the result of deliberately bad parenting but are often a byproduct of circumstances. For example, an overworked single mother may find herself overstressed and fatigued at the end of the day, making fully-involved childcare very difficult. In other cases, some

The different behaviors of securely- and insecurely-attached infants can be observed especially when the infant needs the caregiver's support. To assess the nature of attachment, researchers use a standard laboratory procedure called the "Strange Situation," which involves brief separations from the caregiver (e.g., mother) (Solomon & George, 2008). In the Strange Situation, the caregiver is instructed to leave the child to play alone in a room for a short time, then return and greet the child while researchers observe the child's response. Depending on the child's level of attachment, he or she may reject the parent, cling to the parent, or simply welcome the parent—or, in some instances, react with an agitated combination of responses.

Infants can be securely or insecurely attached with mothers, fathers, and other regular caregivers, and they can differ in their security with different people. The **security of attachment** is an important cornerstone of social and personality development, because infants and young children who are securely attached have been found to develop stronger friendships with peers, more advanced emotional understanding and early conscience

development, and more positive self-concepts, compared with insecurely attached children (Thompson, 2008). This is consistent with attachment theory's premise that experiences of care, resulting in secure or insecure attachments, shape young children's developing concepts of the self, as well as what people are like, and how to interact with them.

As children mature, parent-child relationships naturally change. Preschool and grade-school children are more capable, have their own preferences, and sometimes refuse or seek to compromise with parental expectations. This can lead to greater parent-child conflict, and how conflict is managed by parents further shapes the quality of parent-child relationships. In general, children develop greater competence and self-confidence when parents have high (but reasonable) expectations for children's behavior, communicate well with them, are warm and responsive, and use reasoning (rather than coercion) as preferred responses to children's misbehavior. This kind of parenting style has been described as **authoritative** (Baumrind, 2013). Authoritative parents are supportive and show interest in their kids' activities but are not overbearing and allow them to make constructive mistakes. By contrast, some less-constructive parent-child relationships result from authoritarian, uninvolved, or permissive parenting styles (see Table 1).

		Expectations/Control	
Warmth/ Responsiveness		Low	High
	Low	uninvolved	authoritarian
	High	permissive	authoritative

Table 1: Comparison of Four Parenting Styles

Parental roles in relation to their children change in other ways, too. Parents increasingly become mediators (or gatekeepers) of their children's involvement with peers and activities outside the family. Their communication and practice of values contributes to children's academic achievement, moral development, and activity preferences. As children reach adolescence, the parent-child relationship increasingly becomes one of "coregulation," in which both the parent(s) and the child recognizes the child's growing competence and autonomy, and together they rebalance authority relations. We often see evidence of this as parents start accommodating their teenage kids' sense of independence by allowing them to

get cars, jobs, attend parties, and stay out later.

Family relationships are significantly affected by conditions outside the home. For instance, the **Family Stress Model** describes how financial difficulties are associated with parents' depressed moods, which in turn lead to marital problems and poor parenting that contributes to poorer child adjustment (Conger, Conger, & Martin, 2010). Within the home, parental marital difficulty or divorce affects more than half the children growing up today in the United States. Divorce is typically associated with economic stresses for children and parents, the renegotiation of parent-child relationships (with one parent typically as primary custodian and the other assuming a visiting relationship), and many other significant adjustments for children. Divorce is often regarded by children as a sad turning point in their lives, although for most it is not associated with long-term problems of adjustment (Emery, 1999).

Peer Relationships

Parent-child relationships are not the only significant relationships in a child's life. Peer relationships are also important. Social interaction with another child who is similar in age, skills, and knowledge provokes the development of many social skills that are valuable for the rest of life (Bukowski, Buhrmester, & Underwood, 2011). In peer relationships, children learn how to initiate and maintain social interactions with other children. They learn skills for managing conflict, such as turn-taking, compromise, and bargaining. Play also involves the mutual, sometimes complex, coordination of goals, actions, and understanding. For example, as infants, children get their first encounter with sharing (of each other's toys); during pretend play as preschoolers they create narratives together, choose roles, and collaborate to act out their stories; and in primary school, they may join a sports team, learning to work together and support each other emotionally and strategically toward a common goal. Through these experiences, children develop friendships that provide additional sources of security and support to those provided



Peer relationships are particularly important for children. They can be supportive but also challenging. Peer rejection may lead to behavioral problems later in life. [Image: Twentyfour Students, <https://goo.gl/3IS2gV>, CC BY-SA 2.0, <https://goo.gl/jSSrcO>]

by their parents.

However, peer relationships can be challenging as well as supportive (Rubin, Coplan, Chen, Bowker, & McDonald, 2011). Being accepted by other children is an important source of affirmation and self-esteem, but peer rejection can foreshadow later behavior problems (especially when children are rejected due to aggressive behavior). With increasing age, children confront the challenges of bullying, peer victimization, and managing conformity pressures. Social comparison with peers is an important means by which children evaluate their skills, knowledge, and personal qualities, but it may cause them to feel that they do not measure up well against others. For example, a boy who is not athletic may feel unworthy of his football-playing peers and revert to shy behavior, isolating himself and avoiding conversation. Conversely, an athlete who doesn't "get" Shakespeare may feel embarrassed and avoid reading altogether. Also, with the approach of adolescence, peer relationships become focused on psychological intimacy, involving personal disclosure, vulnerability, and loyalty (or its betrayal)—which significantly affects a child's outlook on the world. Each of these aspects of peer relationships requires developing very different social and emotional skills than those that emerge in parent-child relationships. They also illustrate the many ways that peer relationships influence the growth of personality and self-concept.

Social Understanding

As we have seen, children's experience of relationships at home and the peer group contributes to an expanding repertoire of social and emotional skills and also to broadened social understanding. In these relationships, children develop expectations for specific people (leading, for example, to secure or insecure attachments to parents), understanding of how to interact with adults and peers, and developing self-concept based on how others respond to them. These relationships are also significant forums for emotional development.

Remarkably, young children begin developing social understanding very early in life. Before the end of the first year, infants are aware that other people have perceptions, feelings, and other mental states that affect their behavior, and which are different from the child's own mental states. This can be readily observed in a process called **social referencing**, in which an infant looks to the mother's face when confronted with an unfamiliar person or situation (Feinman, 1992). If the mother looks calm and reassuring, the infant responds positively as if the situation is safe. If the mother looks fearful or distressed, the infant is likely to respond with wariness or distress because the mother's expression signals danger. In a remarkably insightful manner, therefore, infants show an awareness that even though they are uncertain about the unfamiliar situation, their mother is not, and that by "reading" the emotion in her

face, infants can learn about whether the circumstance is safe or dangerous, and how to respond.

Although developmental scientists used to believe that infants are egocentric—that is, focused on their own perceptions and experience—they now realize that the opposite is true. Infants are aware at an early stage that people have different mental states, and this motivates them to try to figure out what others are feeling, intending, wanting, and thinking, and how these mental states affect their behavior. They are beginning, in other words, to develop a **theory of mind**, and although their understanding of mental states begins very simply, it rapidly expands (Wellman, 2011). For example, if an 18-month-old watches an adult try repeatedly to drop a necklace into a cup but inexplicably fail each time, they will immediately put the necklace into the cup themselves—thus completing what the adult intended, but failed, to do. In doing so, they reveal their awareness of the intentions underlying the adult's behavior (Meltzoff, 1995). Carefully designed experimental studies show that by late in the preschool years, young children understand that another's beliefs can be mistaken rather than correct, that memories can affect how you feel, and that one's emotions can be hidden from others (Wellman, 2011). Social understanding grows significantly as children's theory of mind develops.

How do these achievements in social understanding occur? One answer is that young children are remarkably sensitive observers of other people, making connections between their emotional expressions, words, and behavior to derive simple inferences about mental states (e.g., concluding, for example, that what Mommy is looking at is in her mind) (Gopnik, Meltzoff, & Kuhl, 2001). This is especially likely to occur in relationships with people whom the child knows well, consistent with the ideas of attachment theory discussed above. Growing language skills give young children words with which to represent these mental states (e.g., "mad," "wants") and talk about them with others. Thus in conversation with their parents about everyday experiences, children learn much about people's mental states from how adults talk about them ("Your sister was sad because she thought Daddy was coming home.") (Thompson, 2006b). Developing social understanding is, in other words, based on children's everyday interactions with others and their careful interpretations of what they see and hear. There are also some scientists who believe that infants are biologically prepared to perceive people in a special way, as organisms with an internal mental life, and this facilitates their interpretation of people's behavior with reference to those mental states (Leslie, 1994).

Personality

Parents look into the faces of their newborn infants and wonder, "What kind of person will



Although a child's temperament is partly determined by genetics, environmental influences also contribute to shaping personality. Positive personality development is supported by a "good fit" between a child's natural temperament, environment and experiences. [Image: Thomas Hawk, <https://goo.gl/2So40O>, CC BY-NC 2.0, <https://goo.gl/Filc2e>]

this child will become?" They scrutinize their baby's preferences, characteristics, and responses for clues of a developing personality. They are quite right to do so, because temperament is a foundation for personality growth. But **temperament** (defined as early-emerging differences in reactivity and self-regulation) is not the whole story. Although temperament is biologically based, it interacts with the influence of experience from the moment of birth (if not before) to shape personality (Rothbart, 2011). Temperamental dispositions are affected, for example, by the support level of parental care. More generally, personality is shaped by the **goodness of fit** between the child's temperamental qualities and characteristics of the environment (Chess & Thomas, 1999). For example, an adventurous child whose parents regularly take her on weekend

hiking and fishing trips would be a good "fit" to her lifestyle, supporting personality growth. Personality is the result, therefore, of the continuous interplay between biological disposition and experience, as is true for many other aspects of social and personality development.

Personality develops from temperament in other ways (Thompson, Winer, & Goodvin, 2010). As children mature biologically, temperamental characteristics emerge and change over time. A newborn is not capable of much self-control, but as brain-based capacities for self-control advance, temperamental changes in self-regulation become more apparent. For example, a newborn who cries frequently doesn't necessarily have a grumpy personality; over time, with sufficient parental support and increased sense of security, the child might be less likely to cry.

In addition, personality is made up of many other features besides temperament. Children's developing self-concept, their motivations to achieve or to socialize, their values and goals, their coping styles, their sense of responsibility and conscientiousness, and many other qualities are encompassed into personality. These qualities are influenced by biological dispositions, but even more by the child's experiences with others, particularly in close relationships, that guide the growth of individual characteristics.

Indeed, personality development begins with the biological foundations of temperament but becomes increasingly elaborated, extended, and refined over time. The newborn that parents gazed upon thus becomes an adult with a personality of depth and nuance.

Social and Emotional Competence

Social and personality development is built from the social, biological, and representational influences discussed above. These influences result in important developmental outcomes that matter to children, parents, and society: a young adult's capacity to engage in socially constructive actions (helping, caring, sharing with others), to curb hostile or aggressive impulses, to live according to meaningful moral values, to develop a healthy identity and sense of self, and to develop talents and achieve success in using them. These are some of the developmental outcomes that denote social and emotional competence.

These achievements of social and personality development derive from the interaction of many social, biological, and representational influences. Consider, for example, the development of conscience, which is an early foundation for moral development. **Conscience** consists of the cognitive, emotional, and social influences that cause young children to create and act consistently with internal standards of conduct (Kochanska, 2002). Conscience emerges from young children's experiences with parents, particularly in the development of a mutually responsive relationship that motivates young children to respond constructively to the parents' requests and expectations. Biologically based temperament is involved, as some children are temperamentally more capable of motivated self-regulation (a quality called **effortful control**) than are others, while some children are dispositionally more prone to the fear and anxiety that parental disapproval can evoke. Conscience development grows through a good fit between the child's temperamental qualities and how parents communicate and reinforce behavioral expectations. Moreover, as an illustration of the interaction of genes and experience, one research group found that young children with a particular gene allele (the 5-HTTLPR) were low on measures of conscience development when they had previously experienced unresponsive maternal care, but children with the same allele growing up with responsive care showed strong later performance on conscience measures (Kochanska, Kim, Barry, & Philibert, 2011).

Conscience development also expands as young children begin to represent moral values and think of themselves as moral beings. By the end of the preschool years, for example, young children develop a "moral self" by which they think of themselves as people who want to do the right thing, who feel badly after misbehaving, and who feel uncomfortable when others misbehave. In the development of conscience, young children become more socially and

emotionally competent in a manner that provides a foundation for later moral conduct (Thompson, 2012).

The development of gender and gender identity is likewise an interaction among social, biological, and representational influences (Ruble, Martin, & Berenbaum, 2006). Young children learn about gender from parents, peers, and others in society, and develop their own conceptions of the attributes associated with maleness or femaleness (called **gender schemas**). They also negotiate biological transitions (such as puberty) that cause their sense of themselves and their sexual identity to mature.

Each of these examples of the growth of social and emotional competence illustrates not only the interaction of social, biological, and representational influences, but also how their development unfolds over an extended period. Early influences are important, but not determinative, because the capabilities required for mature moral conduct, gender identity, and other outcomes continue to develop throughout childhood, adolescence, and even the adult years.



Social influences such as cultural norms impact children's interests, dress, style of speech and even life aspirations. [Image: Amanda Westmont, <https://goo.gl/ntS5qx>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Conclusion

As the preceding sentence suggests, social and personality development continues through adolescence and the adult years, and it is influenced by the same constellation of social, biological, and representational influences discussed for childhood. Changing social relationships and roles, biological maturation and (much later) decline, and how the individual represents experience and the self continue to form the bases for development throughout life. In this respect, when an adult looks forward rather than retrospectively to ask, “what kind of person am I becoming?”—a similarly fascinating, complex, multifaceted interaction of developmental processes lies ahead.

Outside Resources

Web: Center for the Developing Child, Harvard University

<http://developingchild.harvard.edu>

Web: Collaborative for Academic, Social, and Emotional Learning

<http://casel.org>

Discussion Questions

1. If parent-child relationships naturally change as the child matures, would you expect that the security of attachment might also change over time? What reasons would account for your expectation?
2. In what ways does a child's developing theory of mind resemble how scientists create, refine, and use theories in their work? In other words, would it be appropriate to think of children as informal scientists in their development of social understanding?
3. If there is a poor goodness of fit between a child's temperament and characteristics of parental care, what can be done to create a better match? Provide a specific example of how this might occur.
4. What are the contributions that parents offer to the development of social and emotional competence in children? Answer this question again with respect to peer contributions.

Vocabulary

Authoritative

A parenting style characterized by high (but reasonable) expectations for children's behavior, good communication, warmth and nurturance, and the use of reasoning (rather than coercion) as preferred responses to children's misbehavior.

Conscience

The cognitive, emotional, and social influences that cause young children to create and act consistently with internal standards of conduct.

Effortful control

A temperament quality that enables children to be more successful in motivated self-regulation.

Family Stress Model

A description of the negative effects of family financial difficulty on child adjustment through the effects of economic stress on parents' depressed mood, increased marital problems, and poor parenting.

Gender schemas

Organized beliefs and expectations about maleness and femaleness that guide children's thinking about gender.

Goodness of fit

The match or synchrony between a child's temperament and characteristics of parental care that contributes to positive or negative personality development. A good "fit" means that parents have accommodated to the child's temperamental attributes, and this contributes to positive personality growth and better adjustment.

Security of attachment

An infant's confidence in the sensitivity and responsiveness of a caregiver, especially when he or she is needed. Infants can be securely attached or insecurely attached.

Social referencing

The process by which one individual consults another's emotional expressions to determine how to evaluate and respond to circumstances that are ambiguous or uncertain.

Temperament

Early emerging differences in reactivity and self-regulation, which constitutes a foundation for personality development.

Theory of mind

Children's growing understanding of the mental states that affect people's behavior.

References

- Baumrind, D. (2013). Authoritative parenting revisited: History and current status. In R. E. Larzelere, A. Sheffield, & A. W. Harrist (Eds.), *Authoritative parenting: Synthesizing nurturance and discipline for optimal child development* (pp. 11–34). Washington, DC: American Psychological Association.
- Belsky, J., & Pasco Fearon, R. M. (2008). Precursors of attachment security. In J. Cassidy & P. R. Shaver (Eds.), *Handbook of attachment: Theory, research, and clinical applications* (2nd ed., pp. 295–316). New York, NY: Guilford.
- Bukowski, W. M., Buhrmester, D., & Underwood, M. K. (2011). Peer relations as a developmental context. In M. K. Underwood & L. H. Rosen (Eds.), *Social development* (pp. 153–179). New York, NY: Guilford.
- Cassidy, J. (2008). The nature of the child's ties. In J. Cassidy & P. R. Shaver (Eds.), *Handbook of attachment: Theory, research, and clinical applications* (2nd ed., pp. 3–22). New York, NY: Guilford.
- Chess, S., & Thomas, A. (1999). *Goodness of fit: Clinical applications from infancy through adult life*. New York, NY: Brunner-Mazel/Taylor & Francis.
- Conger, R. D., Conger, K. J., & Martin, M. J. (2010). Socioeconomic status, family processes, and individual development. *Journal of Marriage and Family*, 72, 685–704.
- Emery, R. E. (1999). *Marriage, divorce, and children's adjustment* (2nd ed.). Thousand Oaks, CA: Sage.
- Feinman, S. (Ed.) (1992). *Social referencing and the social construction of reality in infancy*. New York, NY: Plenum.
- Gopnik, A., Meltzoff, A. N., & Kuhl, P. K. (2001). *The scientist in the crib*. New York, NY: HarperCollins.
- Kochanska, G. (2002). Mutually responsive orientation between mothers and their young children: A context for the early development of conscience. *Current Directions in Psychological Science*, 11, 191–195.
- Kochanska, G., Kim, S., Barry, R. A., & Philibert, R. A. (2011). Children's genotypes interact with maternal responsive care in predicting children's competence: Diathesis-stress or differential susceptibility? *Development and Psychopathology*, 23, 605–616.
- Leslie, A. M. (1994). ToMM, ToBy, and agency: Core architecture and domain specificity in cognition and culture. In L. Hirschfeld & S. Gelman (Eds.), *Mapping the mind: Domain specificity in cognition and culture* (pp. 119–148). New York, NY: Cambridge University Press.

- Meltzoff, A. N. (1995). Understanding the intentions of others: Re-enactment of intended acts by 18-month-old children. *Developmental Psychology*, 31, 838-850.
- Rothbart, M. K. (2011). *Becoming who we are: Temperament and personality in development*. New York, NY: Guilford.
- Rubin, K. H., Coplan, R., Chen, X., Bowker, J., & McDonald, K. L. (2011). Peer relationships in childhood. In M. Bornstein & M. E. Lamb (Eds.), *Developmental science: An advanced textbook* (6th ed. pp. 519-570). New York, NY: Psychology Press/Taylor & Francis.
- Ruble, D. N., Martin, C., & Berenbaum, S. (2006). Gender development. In W. Damon & R. M. Lerner (Series Eds.) & N. Eisenberg (Vol. Ed.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (6th ed., pp. 858-932). New York, NY: Wiley.
- Solomon, J., & George, C. (2008). The measurement of attachment security and related constructs in infancy and early childhood. In J. Cassidy & P. R. Shaver (Eds.), *Handbook of attachment: Theory, research, and clinical applications* (2nd ed., pp. 383-416). New York, NY: Guilford.
- Thompson, R. A. (2012). Whither the preconventional child? Toward a life-span moral development theory. *Child Development Perspectives*, 6, 423-429.
- Thompson, R. A. (2008). Early attachment and later development: Familiar questions, new answers. In J. Cassidy & P. R. Shaver (Eds.), *Handbook of attachment: Theory, research, and clinical applications* (2nd ed., pp. 348-365). New York, NY: Guilford.
- Thompson, R. A. (2006a). Conversation and developing understanding: Introduction to the special issue. *Merrill-Palmer Quarterly*, 52, 1-16.
- Thompson, R. A. (2006b). The development of the person: Social understanding, relationships, self, conscience. In W. Damon & R. M. Lerner (Series Eds.) & N. Eisenberg (Vol. Ed.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (6th ed., pp. 24-98). New York, NY: Wiley.
- Thompson, R. A., Winer, A. C., & Goodvin, R. (2010). The individual child: Temperament, emotion, self, and personality. In M. Bornstein & M. E. Lamb (Eds.), *Developmental science: An advanced textbook* (6th ed., pp. 423-464). New York, NY: Psychology Press/Taylor & Francis.
- Wellman, H. M. (2011). Developing a theory of mind. In U. Goswami (Ed.), *Wiley-Blackwell handbook of childhood cognitive development* (2nd ed., pp. 258-284). New York, NY: Wiley-Blackwell

Adolescent Development

Jennifer Lansford

Adolescence is a period that begins with puberty and ends with the transition to adulthood (approximately ages 10–20). Physical changes associated with puberty are triggered by hormones. Cognitive changes include improvements in complex and abstract thought, as well as development that happens at different rates in distinct parts of the brain and increases adolescents' propensity for risky behavior because increases in sensation-seeking and reward motivation precede increases in cognitive control. Adolescents' relationships with parents go through a period of redefinition in which adolescents become more autonomous, and aspects of parenting, such as distal monitoring and psychological control, become more salient. Peer relationships are important sources of support and companionship during adolescence yet can also promote problem behaviors. Same-sex peer groups evolve into mixed-sex peer groups, and adolescents' romantic relationships tend to emerge from these groups. Identity formation occurs as adolescents explore and commit to different roles and ideological positions. Nationality, gender, ethnicity, socioeconomic status, religious background, sexual orientation, and genetic factors shape how adolescents behave and how others respond to them, and are sources of diversity in adolescence

Learning Objectives

- Describe major features of physical, cognitive, and social development during adolescence.
- Understand why adolescence is a period of heightened risk taking.
- Be able to explain sources of diversity in adolescent development.

Adolescence Defined



Adolescence is often characterized as a period of transformation, primarily, in terms of physical, cognitive, and social-relational change. [Image: Alex Proimos, <https://goo.gl/1jqpnI>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Adolescence is a developmental stage that has been defined as starting with puberty and ending with the transition to adulthood (approximately ages 10–20). Adolescence has evolved historically, with evidence indicating that this stage is lengthening as individuals start puberty earlier and transition to adulthood later than in the past. Puberty today begins, on average, at age 10–11 years for girls and 11–12 years for boys. This average age of onset has decreased gradually over time since the 19th century by 3–4 months per decade, which has been attributed to a range of factors including better nutrition, obesity, increased father absence, and other environmental factors (Steinberg, 2013). Completion of formal education, financial independence from parents,

marriage, and parenthood have all been markers of the end of adolescence and beginning of adulthood, and all of these transitions happen, on average, later now than in the past. In fact, the prolonging of adolescence has prompted the introduction of a new developmental period called *emerging adulthood* that captures these developmental changes out of adolescence and into adulthood, occurring from approximately ages 18 to 29 (Arnett, 2000).

This module will outline changes that occur during adolescence in three domains: physical, cognitive, and social. Within the social domain, changes in relationships with parents, peers, and romantic partners will be considered. Next, the module turns to adolescents' psychological and behavioral adjustment, including identity formation, aggression and antisocial behavior, anxiety and depression, and academic achievement. Finally, the module summarizes sources of diversity in adolescents' experiences and development.

Physical Changes

Physical changes of puberty mark the onset of adolescence (Lerner & Steinberg, 2009). For both boys and girls, these changes include a growth spurt in height, growth of pubic and underarm hair, and skin changes (e.g., pimples). Boys also experience growth in facial hair and a deepening of their voice. Girls experience breast development and begin menstruating.

These pubertal changes are driven by hormones, particularly an increase in testosterone for boys and estrogen for girls.

Cognitive Changes

Major changes in the structure and functioning of the brain occur during adolescence and result in cognitive and behavioral developments (Steinberg, 2008). Cognitive changes during adolescence include a shift from concrete to more abstract and complex thinking. Such changes are fostered by improvements during early adolescence in attention, memory, processing speed, and metacognition (ability to think about thinking and therefore make better use of strategies like mnemonic devices that can improve thinking). Early in adolescence, changes in the brain's dopaminergic system contribute to increases in adolescents' sensation-seeking and reward motivation. Later in adolescence, the brain's cognitive control centers in the prefrontal cortex develop, increasing adolescents' self-regulation and future orientation. The difference in timing of the development of these different regions of the brain contributes to more risk taking during middle adolescence because adolescents are motivated to seek thrills that sometimes come from risky behavior, such as reckless driving, smoking, or drinking, and have not yet developed the cognitive control to resist impulses or focus equally on the potential risks (Steinberg, 2008). One of the world's leading experts on adolescent development, Laurence Steinberg, likens this to engaging a powerful engine before the braking system is in place. The result is that adolescents are more prone to risky behaviors than are children or adults.



Dopamine is a neurotransmitter in the brain that produces feelings of pleasure. During adolescence, people tend to do whatever activities produce the most dopamine, without fully considering the consequences of such actions. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Social Changes

Parents

Although peers take on greater importance during adolescence, family relationships remain important too. One of the key changes during adolescence involves a renegotiation of parent–child relationships. As adolescents strive for more independence and autonomy during this time, different aspects of parenting become more salient. For example, parents’ distal supervision and monitoring become more important as adolescents spend more time away from parents and in the presence of peers. Parental monitoring encompasses a wide range of behaviors such as parents’ attempts to set rules and know their adolescents’ friends, activities, and whereabouts, in addition to adolescents’ willingness to disclose information to their parents (Stattin & Kerr, 2000). **Psychological control**, which involves manipulation and intrusion into adolescents’ emotional and cognitive world through invalidating adolescents’ feelings and pressuring them to think in particular ways (Barber, 1996), is another aspect of parenting that becomes more salient during adolescence and is related to more problematic adolescent adjustment.

Peers



Peer relationships are a big part of adolescent development. The influence of peers can be both positive and negative as adolescents experiment together with identity formation and new experiences. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

As children become adolescents, they usually begin spending more time with their peers and less time with their families, and these peer interactions are increasingly unsupervised by adults. Children’s notions of friendship often focus on shared activities, whereas adolescents’ notions of friendship increasingly focus on intimate exchanges of thoughts and feelings. During adolescence, peer groups evolve from primarily single-sex to mixed-sex. Adolescents within a peer group tend to be similar to one another in behavior and attitudes, which has been explained as being a function of **homophily** (adolescents who are similar to one another choose to spend time together in a “birds of a feather flock together” way) and influence (adolescents

who spend time together shape each other’s behavior and attitudes). One of the most widely studied aspects of adolescent peer influence is known as **deviant peer contagion** (Dishion & Tipsord, 2011), which is the process by which peers reinforce problem behavior by laughing or showing other signs of approval that then increase the likelihood of future problem

behavior.

Peers can serve both positive and negative functions during adolescence. Negative peer pressure can lead adolescents to make riskier decisions or engage in more problematic behavior than they would alone or in the presence of their family. For example, adolescents are much more likely to drink alcohol, use drugs, and commit crimes when they are with their friends than when they are alone or with their family. However, peers also serve as an important source of social support and companionship during adolescence, and adolescents with positive peer relationships are happier and better adjusted than those who are socially isolated or have conflictual peer relationships.

Crowds are an emerging level of peer relationships in adolescence. In contrast to friendships (which are reciprocal dyadic relationships) and cliques (which refer to groups of individuals who interact frequently), crowds are characterized more by shared reputations or images than actual interactions (Brown & Larson, 2009). These crowds reflect different prototypic identities (such as jocks or brains) and are often linked with adolescents' social status and peers' perceptions of their values or behaviors.

Romantic relationships

Adolescence is the developmental period during which romantic relationships typically first emerge. Initially, same-sex peer groups that were common during childhood expand into mixed-sex peer groups that are more characteristic of adolescence. Romantic relationships often form in the context of these mixed-sex peer groups (Connolly, Furman, & Konarski, 2000). Although romantic relationships during adolescence are often short-lived rather than long-term committed partnerships, their importance should not be minimized. Adolescents spend a great deal of time focused on romantic relationships, and their positive and negative emotions are more tied to romantic relationships (or lack thereof) than to friendships, family relationships, or school (Furman & Shaffer, 2003). Romantic relationships contribute to adolescents' identity formation, changes in family and peer relationships, and adolescents' emotional and behavioral adjustment.

Furthermore, romantic relationships are centrally connected to adolescents' emerging sexuality. Parents, policymakers, and researchers have devoted a great deal of attention to adolescents' sexuality, in large part because of concerns related to sexual intercourse, contraception, and preventing teen pregnancies. However, sexuality involves more than this narrow focus. For example, adolescence is often when individuals who are lesbian, gay, bisexual, or transgender come to perceive themselves as such (Russell, Clarke, & Clary, 2009).

Thus, romantic relationships are a domain in which adolescents experiment with new behaviors and identities.

Behavioral and Psychological Adjustment

Identity formation

Theories of adolescent development often focus on identity formation as a central issue. For example, in Erikson's (1968) classic theory of developmental stages, identity formation was highlighted as the primary indicator of successful development during adolescence (in contrast to role confusion, which would be an indicator of not successfully meeting the task of adolescence). Marcia (1966) described identity formation during adolescence as involving both decision points and commitments with respect to ideologies (e.g., religion, politics) and occupations. He described four identity statuses: foreclosure, identity diffusion, moratorium, and identity achievement. **Foreclosure** occurs when an individual commits to an identity without exploring options. **Identity diffusion** occurs when adolescents neither explore nor commit to any identities. **Moratorium** is a state in which adolescents are actively exploring options but have not yet made commitments. **Identity achievement** occurs when individuals have explored different options and then made identity commitments. Building on this work, other researchers have investigated more specific aspects of identity. For example, Phinney (1989) proposed a model of ethnic identity development that included stages of unexplored ethnic identity, ethnic identity search, and achieved ethnic identity.

Aggression and antisocial behavior

Several major theories of the development of antisocial behavior treat adolescence as an important period. Patterson's (1982) early versus late starter model of the development of aggressive and antisocial behavior distinguishes youths whose



Early, antisocial behavior leads to befriending others who also engage in antisocial behavior, which only perpetuates the downward cycle of aggression and wrongful acts. [Image: Philippe Put, <https://goo.gl/14H7HL>, CC BY 2.0, <https://goo.gl/BRvSA7>]

antisocial behavior begins during childhood (early starters) versus adolescence (late starters). According to the theory, early starters are at greater risk for long-term antisocial behavior that extends into adulthood than are late starters. Late starters who become antisocial during adolescence are theorized to experience poor parental monitoring and supervision, aspects of parenting that become more salient during adolescence. Poor monitoring and lack of supervision contribute to increasing involvement with deviant peers, which in turn promotes adolescents' own antisocial behavior. Late starters desist from antisocial behavior when changes in the environment make other options more appealing. Similarly, Moffitt's (1993) life-course persistent versus adolescent-limited model distinguishes between antisocial behavior that begins in childhood versus adolescence. Moffitt regards adolescent-limited antisocial behavior as resulting from a "maturity gap" between adolescents' dependence on and control by adults and their desire to demonstrate their freedom from adult constraint. However, as they continue to develop, and legitimate adult roles and privileges become available to them, there are fewer incentives to engage in antisocial behavior, leading to desistance in these antisocial behaviors.

Anxiety and depression

Developmental models of anxiety and depression also treat adolescence as an important period, especially in terms of the emergence of gender differences in prevalence rates that persist through adulthood (Rudolph, 2009). Starting in early adolescence, compared with males, females have rates of anxiety that are about twice as high and rates of depression that are 1.5 to 3 times as high (American Psychiatric Association, 2013). Although the rates vary across specific anxiety and depression diagnoses, rates for some disorders are markedly higher in adolescence than in childhood or adulthood. For example, prevalence rates for specific phobias are about 5% in children and 3%–5% in adults but 16% in adolescents. Anxiety and depression are particularly concerning because suicide is one of the leading causes of death during adolescence. Developmental models focus on interpersonal contexts in both childhood and adolescence that foster depression and anxiety (e.g., Rudolph, 2009). Family adversity, such as abuse and parental psychopathology, during childhood sets the stage for social and behavioral problems during adolescence. Adolescents with such problems generate stress in their relationships (e.g., by resolving conflict poorly and excessively seeking reassurance) and select into more maladaptive social contexts (e.g., "misery loves company" scenarios in which depressed youths select other depressed youths as friends and then frequently co-ruminate as they discuss their problems, exacerbating negative affect and stress). These processes are intensified for girls compared with boys because girls have more relationship-oriented goals related to intimacy and social approval, leaving them more vulnerable to disruption in these relationships. Anxiety and depression then exacerbate

problems in social relationships, which in turn contribute to the stability of anxiety and depression over time.

Academic achievement

Adolescents spend more waking time in school than in any other context (Eccles & Roeser, 2011). Academic achievement during adolescence is predicted by interpersonal (e.g., parental engagement in adolescents' education), intrapersonal (e.g., intrinsic motivation), and institutional (e.g., school quality) factors. Academic achievement is important in its own right as a marker of positive adjustment during adolescence but also because academic achievement sets the stage for future educational and occupational opportunities. The most serious consequence of school failure, particularly dropping out of school, is the high risk of unemployment or underemployment in adulthood that follows. High achievement can set the stage for college or future vocational training and opportunities.

Diversity



Although similar biological changes occur for all adolescents as they enter puberty, these changes can differ significantly depending on one's cultural, ethnic, and societal factors. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Adolescent development does not necessarily follow the same pathway for all individuals. Certain features of adolescence, particularly with respect to biological changes associated with puberty and cognitive changes associated with brain development, are relatively universal. But other features of adolescence depend largely on circumstances that are more environmentally variable. For example, adolescents growing up in one country might have different opportunities for risk taking than adolescents in a different country, and supports and sanctions for different behaviors in adolescence depend on laws and values that might be specific to where adolescents live. Likewise, different cultural norms regarding family and peer relationships shape adolescents' experiences

in these domains. For example, in some countries, adolescents' parents are expected to retain control over major decisions, whereas in other countries, adolescents are expected to begin

sharing in or taking control of decision making.

Even within the same country, adolescents' gender, ethnicity, immigrant status, religion, sexual orientation, socioeconomic status, and personality can shape both how adolescents behave and how others respond to them, creating diverse developmental contexts for different adolescents. For example, early puberty (that occurs before most other peers have experienced puberty) appears to be associated with worse outcomes for girls than boys, likely in part because girls who enter puberty early tend to associate with older boys, which in turn is associated with early sexual behavior and substance use. For adolescents who are ethnic or sexual minorities, discrimination sometimes presents a set of challenges that nonminorities do not face.

Finally, genetic variations contribute an additional source of diversity in adolescence. Current approaches emphasize gene X environment interactions, which often follow a **differential susceptibility** model (Belsky & Pluess, 2009). That is, particular genetic variations are considered riskier than others, but genetic variations also can make adolescents more or less susceptible to environmental factors. For example, the association between the CHRM2 genotype and adolescent externalizing behavior (aggression and delinquency) has been found in adolescents whose parents are low in monitoring behaviors (Dick et al., 2011). Thus, it is important to bear in mind that individual differences play an important role in adolescent development.

Conclusions

Adolescent development is characterized by biological, cognitive, and social changes. Social changes are particularly notable as adolescents become more autonomous from their parents, spend more time with peers, and begin exploring romantic relationships and sexuality. Adjustment during adolescence is reflected in identity formation, which often involves a period of exploration followed by commitments to particular identities. Adolescence is characterized by risky behavior, which is made more likely by changes in the brain in which reward-processing centers develop more rapidly than cognitive control systems, making adolescents more sensitive to rewards than to possible negative consequences. Despite these generalizations, factors such as country of residence, gender, ethnicity, and sexual orientation shape development in ways that lead to diversity of experiences across adolescence.

Outside Resources

Podcasts: Society for Research on Adolescence website with links to podcasts on a variety of topics, from autonomy-relatedness in adolescence, to the health ramifications of growing up in the United States.

<http://www.s-r-a.org/sra-news/podcasts>

Study: The National Longitudinal Study of Adolescent to Adult Health (Add Health) is a longitudinal study of a nationally representative sample of adolescents in grades 7-12 in the United States during the 1994-95 school year. Add Health combines data on respondents' social, economic, psychological and physical well-being with contextual data on the family, neighborhood, community, school, friendships, peer groups, and romantic relationships.

<http://www.cpc.unc.edu/projects/addhealth>

Video: This is a series of TED talks on topics from the mysterious workings of the adolescent brain, to videos about surviving anxiety in adolescence.

<http://tinyurl.com/lku4a3k>

Web: UNICEF website on adolescents around the world. UNICEF provides videos and other resources as part of an initiative to challenge common preconceptions about adolescence.

<http://www.unicef.org/adolescence/index.html>

Discussion Questions

1. What can parents do to promote their adolescents' positive adjustment?
2. In what ways do changes in brain development and cognition make adolescents particularly susceptible to peer influence?
3. How could interventions designed to prevent or reduce adolescents' problem behavior be developed to take advantage of what we know about adolescent development?
4. Reflecting on your own adolescence, provide examples of times when you think your experience was different from those of your peers as a function of something unique about you.
5. In what ways was your experience of adolescence different from your parents' experience of adolescence? How do you think adolescence may be different 20 years from now?

Vocabulary

Crowds

Adolescent peer groups characterized by shared reputations or images.

Deviant peer contagion

The spread of problem behaviors within groups of adolescents.

Differential susceptibility

Genetic factors that make individuals more or less responsive to environmental experiences.

Foreclosure

Individuals commit to an identity without exploration of options.

Homophily

Adolescents tend to associate with peers who are similar to themselves.

Identity achievement

Individuals have explored different options and then made commitments.

Identity diffusion

Adolescents neither explore nor commit to any roles or ideologies.

Moratorium

State in which adolescents are actively exploring options but have not yet made identity commitments.

Psychological control

Parents' manipulation of and intrusion into adolescents' emotional and cognitive world through invalidating adolescents' feelings and pressuring them to think in particular ways.

References

- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: American Psychiatric Publishing.
- Arnett, J. J. (2000). Emerging adulthood: A theory of development from the late teens through the twenties. *American Psychologist*, 55, 469–480.
- Barber, B. K. (1996). Parental psychological control: Revisiting a neglected construct. *Child Development*, 67, 3296–3319.
- Belsky, J., & Pluess, M. (2009). Beyond diathesis-stress: Differential susceptibility to environmental influences. *Psychological Bulletin*, 135, 885–908.
- Brown, B. B., & Larson, J. (2009). Peer relationships in adolescence. In R. M. Lerner & L. Steinberg (Eds.), *Handbook of adolescent psychology* (pp. 74–103). New York, NY: Wiley.
- Connolly, J., Furman, W., & Konarski, R. (2000). The role of peers in the emergence of heterosexual romantic relationships in adolescence. *Child Development*, 71, 1395–1408.
- Dick, D. M., Meyers, J. L., Latendresse, S. J., Creemers, H. E., Lansford, J. E., ... Huizink, A. C. (2011). CHRM2, parental monitoring, and adolescent externalizing behavior: Evidence for gene-environment interaction. *Psychological Science*, 22, 481–489.
- Dishion, T. J., & Tipsord, J. M. (2011). Peer contagion in child and adolescent social and emotional development. *Annual Review of Psychology*, 62, 189–214.
- Eccles, J. S., & Roeser, R. W. (2011). Schools as developmental contexts during adolescence. *Journal of Research on Adolescence*, 21, 225–241.
- Erikson, E. H. (1968). *Identity, youth, and crisis*. New York, NY: Norton.
- Furman, W., & Shaffer, L. (2003). The role of romantic relationships in adolescent development. In P. Florsheim (Ed.), *Adolescent romantic relations and sexual behavior: Theory, research, and practical implications* (pp. 3–22). Mahwah, NJ: Erlbaum.
- Lerner, R. M., & Steinberg, L. (Eds.). (2009). *Handbook of adolescent psychology*. New York, NY: Wiley.
- Marcia, J. E. (1966). Development and validation of ego identity status. *Journal of Personality and Social Psychology*, 3, 551–558.
- Moffitt, T. E. (1993). Adolescence-limited and life course persistent antisocial behavior: Developmental taxonomy. *Psychological Review*, 100, 674–701.
- Patterson, G. R. (1982). *Coercive family process*. Eugene, OR: Castalia Press.
- Phinney, J. (1989). Stages of ethnic identity in minority group adolescents. *Journal of Early*

Adolescence, 9, 34–49.

Rudolph, K. D. (2009). The interpersonal context of adolescent depression. In S. Nolen-Hoeksema & L. M. Hilt (Eds.), *Handbook of depression in adolescents* (pp. 377–418). New York, NY: Taylor and Francis.

Russell, S. T., Clarke, T. J., & Clary, J. (2009). Are teens “post-gay”? Contemporary adolescents’ sexual identity labels. *Journal of Youth and Adolescence*, 38, 884–890.

Stattin, H., & Kerr, M. (2000). Parental monitoring: A reinterpretation. *Child Development*, 71, 1072–1085.

Steinberg, L. (2013). *Adolescence* (10th ed.). New York, NY: McGraw-Hill.

Steinberg, L. (2008). A social neuroscience perspective on adolescent risk-taking. *Developmental Review*, 28, 78–106.

Attachment Through the Life Course

R. Chris Fraley

The purpose of this module is to provide a brief review of attachment theory—a theory designed to explain the significance of the close, emotional bonds that children develop with their caregivers and the implications of those bonds for understanding personality development. The module discusses the origins of the theory, research on individual differences in attachment security in infancy and childhood, and the role of attachment in adult relationships.

Learning Objectives

- Explain the way the attachment system works and its evolutionary significance.
- Identify three commonly studied attachment patterns and what is known about the development of those patterns.
- Describe what is known about the consequences of secure versus insecure attachment in adult relationships.

Introduction

Some of the most rewarding experiences in people's lives involve the development and maintenance of close relationships. For example, some of the greatest sources of joy involve falling in love, starting a family, being reunited with distant loved ones, and sharing experiences with close others. And, not surprisingly, some of the most painful experiences in people's lives involve the disruption of important social bonds, such as separation from a spouse, losing a

parent, or being abandoned by a loved one.

Why do close relationships play such a profound role in human experience? Attachment theory is one approach to understanding the nature of close relationships. In this module, we review the origins of the theory, the core theoretical principles, and some ways in which attachment influences human behavior, thoughts, and feelings across the life course.



Close relationships are the fabric of society, and are integral to the maintenance of our species.

[Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Attachment Theory: A Brief History and Core Concepts

Attachment theory was originally developed in the 1940s by John Bowlby, a British psychoanalyst who was attempting to understand the intense distress experienced by infants who had been separated from their parents. Bowlby (1969) observed that infants would go to extraordinary lengths to prevent separation from their parents or to reestablish proximity to a missing parent. For example, he noted that children who had been separated from their parents would often cry, call for their parents, refuse to eat or play, and stand at the door in desperate anticipation of their parents' return. At the time of Bowlby's initial writings, psychoanalytic writers held that these expressions were manifestations of immature defense mechanisms that were operating to repress emotional pain. However, Bowlby observed that such expressions are common to a wide variety of mammalian species and speculated that these responses to separation may serve an evolutionary function (see Focus Topic 1).

Focus Topic 1: Harlow's research on contact comfort

When Bowlby was originally developing his theory of attachment, there were alternative theoretical perspectives on why infants were emotionally attached to their primary caregivers (most often, their biological mothers). Bowlby and other theorists, for example, believed that there was something important about the responsiveness and contact provided by mothers. Other theorists, in contrast, argued that young infants feel emotionally connected to their mothers because mothers satisfy more basic needs, such as the need for food. That is, the child comes to feel emotionally connected to the mother because she is associated with the reduction of primary drives, such as hunger, rather than the reduction of drives that might be relational in nature.

In a classic set of studies, psychologist Harry Harlow placed young monkeys in cages that contained two artificial, surrogate “mothers” (Harlow, 1958). One of those surrogates was a simple wire contraption; the other was a wire contraption covered in cloth. Both of the surrogate mothers were equipped with a feeding tube so that Harlow and his colleagues had the option to allow the surrogate to deliver or not deliver milk. Harlow found that the young macaques spent a disproportionate amount of time with the cloth surrogate as opposed to the wire surrogate. Moreover, this was true even when the infants were fed by the wire surrogate rather than the cloth surrogate. This suggests that the strong emotional bond that infants form with their primary caregivers is rooted in something more than whether the caregiver provides food *per se*. Harlow’s research is now regarded as one of the first experimental demonstrations of the importance of “contact comfort” in the establishment of infant–caregiver bonds.

Drawing on evolutionary theory, Bowlby (1969) argued that these behaviors are adaptive responses to separation from a primary **attachment figure**—a caregiver who provides support, protection, and care. Because human infants, like other mammalian infants, cannot feed or protect themselves, they are dependent upon the care and protection of “older and wiser” adults for survival. Bowlby argued that, over the course of evolutionary history, infants who were able to maintain proximity to an attachment figure would be more likely to survive to a reproductive age.

According to Bowlby, a motivational system, what he called the **attachment behavioral system**, was gradually “designed” by natural selection to regulate proximity to an attachment figure. The attachment system functions much like a thermostat that continuously monitors the ambient temperature of a room, comparing that temperature against a desired state and

adjusting behavior (e.g., activating the furnace) accordingly. In the case of the attachment system, Bowlby argued that the system continuously monitors the accessibility of the primary attachment figure. If the child perceives the attachment figure to be nearby, accessible, and attentive, then the child feels loved, secure, and confident and, behaviorally, is likely to explore his or her environment, play with others, and be sociable. If, however, the child perceives the attachment figure to be inaccessible, the child experiences anxiety and, behaviorally, is likely to exhibit attachment behaviors ranging from simple visual searching on the low extreme to active searching, following, and vocal signaling on the other. These **attachment behaviors** continue either until the child is able to reestablish a desirable level of physical or psychological proximity to the attachment figure or until the child exhausts himself or herself or gives up, as may happen in the context of a prolonged separation or loss.

Individual Differences in Infant Attachment



Think of your earliest memory, does it involve just you, or does it include your loved ones, your family and caretakers? [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Although Bowlby believed that these basic dynamics captured the way the attachment system works in most children, he recognized that there are individual differences in the way children appraise the accessibility of the attachment figure and how they regulate their attachment behavior in response to threats. However, it was not until his colleague, Mary Ainsworth, began to systematically study infant–parent separations that a formal understanding of these individual differences emerged. Ainsworth and her students developed a technique called the **strange situation**—a laboratory task for studying infant–parent attachment (Ainsworth, Blehar, Waters, & Wall, 1978). In the strange situation, 12-month-old infants and their parents are brought to the

laboratory and, over a period of approximately 20 minutes, are systematically separated from and reunited with one another. In the strange situation, most children (about 60%) behave in the way implied by Bowlby's normative theory. Specifically, they become upset when the parent leaves the room, but, when he or she returns, they actively seek the parent and are easily comforted by him or her. Children who exhibit this pattern of behavior are often called

secure. Other children (about 20% or less) are ill at ease initially and, upon separation, become extremely distressed. Importantly, when reunited with their parents, these children have a difficult time being soothed and often exhibit conflicting behaviors that suggest they want to be comforted, but that they also want to “punish” the parent for leaving. These children are often called anxious-resistant. The third pattern of attachment that Ainsworth and her colleagues documented is often labeled avoidant. Avoidant children (about 20%) do not consistently behave as if they are stressed by the separation but, upon reunion, actively avoid seeking contact with their parent, sometimes turning their attention to play objects on the laboratory floor.

Ainsworth’s work was important for at least three reasons. First, she provided one of the first empirical demonstrations of how attachment behavior is organized in unfamiliar contexts. Second, she provided the first empirical taxonomy of individual differences in infant **attachment patterns**. According to her research, at least three types of children exist: those who are secure in their relationship with their parents, those who are anxious-resistant, and those who are anxious-avoidant. Finally, she demonstrated that these individual differences were correlated with infant–parent interactions in the home during the first year of life. Children who appear secure in the strange situation, for example, tend to have parents who are responsive to their needs. Children who appear insecure in the strange situation (i.e., anxious-resistant or avoidant) often have parents who are insensitive to their needs, or inconsistent or rejecting in the care they provide.

Antecedents of Attachment Patterns

In the years that have followed Ainsworth’s ground-breaking research, researchers have investigated a variety of factors that may help determine whether children develop secure or insecure relationships with their primary attachment figures. As mentioned above, one of the key determinants of attachment patterns is the history of sensitive and responsive interactions between the caregiver and the child. In short, when the child is uncertain or stressed, the ability of the caregiver to



Is attachment style multi-generational? How does one person’s childhood attachment style translate to the way they interact with their own children? [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

provide support to the child is critical for his or her psychological development. It is assumed that such supportive interactions help the child learn to regulate his or her emotions, give the child the confidence to explore the environment, and provide the child with a safe haven during stressful circumstances.

Evidence for the role of sensitive caregiving in shaping attachment patterns comes from longitudinal and experimental studies. For example, Grossmann, Grossmann, Spangler, Suess, and Unzner (1985) studied parent-child interactions in the homes of 54 families, up to three times during the first year of the child's life. At 12 months of age, infants and their mothers participated in the strange situation. Grossmann and her colleagues found that children who were classified as secure in the strange situation at 12 months of age were more likely than children classified as insecure to have mothers who provided responsive care to their children in the home environment.

Van den Boom (1994) developed an intervention that was designed to enhance maternal sensitive responsiveness. When the infants were 9 months of age, the mothers in the intervention group were rated as more responsive and attentive in their interaction with their infants compared to mothers in the control group. In addition, their infants were rated as more sociable, self-soothing, and more likely to explore the environment. At 12 months of age, children in the intervention group were more likely to be classified as secure than insecure in the strange situation.

Attachment Patterns and Child Outcomes

Attachment researchers have studied the association between children's attachment patterns and their adaptation over time. Researchers have learned, for example, that children who are classified as secure in the strange situation are more likely to have high functioning relationships with peers, to be evaluated favorably by teachers, and to persist with more diligence in challenging tasks. In contrast, insecure-avoidant children are more likely to be construed as "bullies" or to have a difficult time building and maintaining friendships (Weinfield, Sroufe, Egeland, & Carlson, 2008).

Attachment in Adulthood

Although Bowlby was primarily focused on understanding the nature of the infant-caregiver relationship, he believed that attachment characterized human experience across the life course. It was not until the mid-1980s, however, that researchers began to take seriously the possibility that attachment processes may be relevant to adulthood. Hazan and Shaver (1987)

were two of the first researchers to explore Bowlby's ideas in the context of romantic relationships. According to Hazan and Shaver, the emotional bond that develops between adult romantic partners is partly a function of the same motivational system—the attachment behavioral system—that gives rise to the emotional bond between infants and their caregivers. Hazan and Shaver noted that in both kinds of relationship, people (a) feel safe and secure when the other person is present; (b) turn to the other person during times of sickness, distress, or fear; (c) use the other person as a “secure base” from which to explore the world; and (d) speak to one another in a unique language, often called “motherese” or “baby talk.” (See Focus Topic 2)

Focus Topic 2: Attachment and social media

Social media websites and mobile communication services are coming to play an increasing role in people's lives. Many people use Facebook, for example, to keep in touch with family and friends, to update their loved ones regarding things going on in their lives, and to meet people who share similar interests. Moreover, modern cellular technology allows people to get in touch with their loved ones much easier than was possible a mere 20 years ago.

From an attachment perspective, these innovations in communications technology are important because they allow people to stay connected virtually to their attachment figures—regardless of the physical distance that might exist between them. Recent research has begun to examine how attachment processes play out in the use of social media. Oldmeadow, Quinn, and Kowert (2013), for example, studied a diverse sample of individuals and assessed their attachment security and their use of Facebook. Oldmeadow and colleagues found that the use of Facebook may serve attachment functions. For example, people were more likely to report using Facebook to connect with others when they were experiencing negative emotions. In addition, the researchers found that people who were more anxious in their attachment orientation were more likely to use Facebook frequently, but people who were more avoidant used Facebook less and were less open on the site.

On the basis of these parallels, Hazan and Shaver (1987) argued that adult romantic relationships, such as infant-caregiver relationships, are attachments. According to Hazan and Shaver, individuals gradually transfer attachment-related functions from parents to peers as they develop. Thus, although young children tend to use their parents as their primary attachment figures, as they reach adolescence and young adulthood, they come to rely more

upon close friends and/or romantic partners for basic attachment-related functions. Thus, although a young child may turn to his or her mother for comfort, support, and guidance when distressed, scared, or ill, young adults may be more likely to turn to their romantic partners for these purposes under similar situations.

Hazan and Shaver (1987) asked a diverse sample of adults to read the three paragraphs below and indicate which paragraph best characterized the way they think, feel, and behave in close relationships:

1. I am somewhat uncomfortable being close to others; I find it difficult to trust them completely, difficult to allow myself to depend on them. I am nervous when anyone gets too close, and often, others want me to be more intimate than I feel comfortable being.
2. I find it relatively easy to get close to others and am comfortable depending on them and having them depend on me. I don't worry about being abandoned or about someone getting too close to me.
3. I find that others are reluctant to get as close as I would like. I often worry that my partner doesn't really love me or won't want to stay with me. I want to get very close to my partner, and this sometimes scares people away.

Conceptually, these descriptions were designed to represent what Hazan and Shaver considered to be adult analogues of the kinds of attachment patterns Ainsworth described in the strange situation (avoidant, secure, and anxious, respectively). Hazan and Shaver (1987) found that the distribution of the three patterns was similar to that observed in infancy. In other words, about 60% of adults classified themselves as secure (paragraph B), about 20% described themselves as avoidant (paragraph A), and about 20% described themselves as anxious-resistant (paragraph C). Moreover, they found that people who described themselves as secure, for example, were more likely to report having had warm and trusting relationships with their parents when they were growing up. In addition, they were more likely to have positive views of romantic relationships. Based on these findings, Hazan and Shaver (1987) concluded that the same kinds of individual differences that exist in infant attachment also exist in adulthood.

Research on Attachment in Adulthood

Attachment theory has inspired a large amount of literature in social, personality, and clinical psychology. In the sections below, I provide a brief overview of some of the major research questions and what researchers have learned about attachment in adulthood.

Who Ends Up with Whom?



People who had relatively secure attachments as children go on to have more secure romantic attachments. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

than 83 heterosexual couples and found that, if the man was relatively secure, the woman was also likely to be secure.

When people are asked what kinds of psychological or behavioral qualities they are seeking in a romantic partner, a large majority of people indicate that they are seeking someone who is kind, caring, trustworthy, and understanding—the kinds of attributes that characterize a “secure” caregiver (Chappell & Davis, 1998). But we know that people do not always end up with others who meet their ideals. Are secure people more likely to end up with secure partners—and, vice versa, are insecure people more likely to end up with insecure partners? The majority of the research that has been conducted to date suggests that the answer is “yes.” Frazier, Byer, Fischer, Wright, and DeBord (1996), for example, studied the attachment patterns of more

One important question is whether these findings exist because (a) secure people are more likely to be attracted to other secure people, (b) secure people are likely to create security in their partners over time, or (c) some combination of these possibilities. Existing empirical research strongly supports the first alternative. For example, when people have the opportunity to interact with individuals who vary in security in a speed-dating context, they express a greater interest in those who are higher in security than those who are more insecure (McClure, Lydon, Baccus, & Baldwin, 2010). However, there is also some evidence that people’s attachment styles mutually shape one another in close relationships. For example, in a longitudinal study, Hudson, Fraley, Vicary, and Brumbaugh (2012) found that, if one person in a relationship experienced a change in security, his or her partner was likely to experience a change in the same direction.

Relationship Functioning

Research has consistently demonstrated that individuals who are relatively secure are more

likely than insecure individuals to have high functioning relationships—relationships that are more satisfying, more enduring, and less characterized by conflict. For example, Feeney and Noller (1992) found that insecure individuals were more likely than secure individuals to experience a breakup of their relationship. In addition, secure individuals are more likely to report satisfying relationships (e.g., Collins & Read, 1990) and are more likely to provide support to their partners when their partners were feeling distressed (Simpson, Rholes, & Nelligan, 1992).

Do Early Experiences Shape Adult Attachment?

The majority of research on this issue is retrospective—that is, it relies on adults' reports of what they recall about their childhood experiences. This kind of work suggests that secure adults are more likely to describe their early childhood experiences with their parents as being supportive, loving, and kind (Hazan & Shaver, 1987). A number of longitudinal studies are emerging that demonstrate prospective associations between early attachment experiences and adult attachment styles and/or interpersonal functioning in adulthood. For example, Fraley, Roisman, Booth-LaForce, Owen, and Holland (2013) found in a sample of more than 700 individuals studied from infancy to adulthood that maternal sensitivity across development prospectively predicted security at age 18. Simpson, Collins, Tran, and Haydon (2007) found that attachment security, assessed in infancy in the strange situation, predicted peer competence in grades 1 to 3, which, in turn, predicted the quality of friendship relationships at age 16, which, in turn, predicted the expression of positive and negative emotions in their adult romantic relationships at ages 20 to 23.

It is easy to come away from such findings with the mistaken assumption that early experiences “determine” later outcomes. To be clear: Attachment theorists assume that the relationship between early experiences and subsequent outcomes is probabilistic, not deterministic.



Sharing food, celebrations, and traditions are some of the ways we establish secure attachments with our loved ones from an early age. [Image: iwona_kellie, <https://goo.gl/B406LK>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Having supportive and responsive experiences with caregivers early in life is assumed to set the stage for positive social development. But that does not mean that attachment patterns are set in stone. In short, even if an individual has far from optimal experiences in early life, attachment theory suggests that it is possible for that individual to develop well-functioning adult relationships through a number of corrective experiences—including relationships with siblings, other family members, teachers, and close friends. Security is best viewed as a culmination of a person's attachment history rather than a reflection of his or her early experiences alone. Those early experiences are considered important not because they determine a person's fate, but because they provide the foundation for subsequent experiences.

Outside Resources

Hazan, C., & Shaver, P. (1987). Romantic love conceptualized as an attachment process. *Journal of Personality and Social Psychology*, 52, 511-524. Retrieved from: <http://www2.psych.ubc.ca/~schaller/Psyc591Readings/HazanShaver1987.pdf>

Hofer, M. A. (2006). Psychobiological roots of early attachment. *Current Directions in Psychological Science*, 15, 84-88. doi:10.1111/j.0963-7214.2006.00412.x
<http://cdp.sagepub.com/content/15/2/84.short>

Strange Situation Video

<https://www.youtube.com/watch?v=QTsewNrHUHU>

Survey: Learn more about your attachment patterns via this online survey

<http://www.yourpersonality.net/relstructures/>

Video on Harry Harlow's Research with Rhesus Monkeys

<https://www.youtube.com/watch?v=OrNBEhzjg8I>

Discussion Questions

1. What kind of relationship did you have with your parents or primary caregivers when you were young? Do you think that had any bearing on the way you related to others (e.g., friends, relationship partners) as you grew older?
2. There is variation across cultures in the extent to which people value independence. Do you think this might have implications for the development of attachment patterns?
3. As parents age, it is not uncommon for them to have to depend on their adult children. Do you think that people's history of experiences in their relationships with their parents might shape people's willingness to provide care for their aging parents? In other words, are secure adults more likely to provide responsive care to their aging parents?
4. Some people, despite reporting insecure relationships with their parents, report secure, well-functioning relationships with their spouses. What kinds of experiences do you think might enable someone to develop a secure relationship with their partners despite having an insecure relationship with other central figures in their lives?
5. Most attachment research on adults focuses on attachment to peers (e.g., romantic

partners). What other kinds of things may serve as attachment figures? Do you think siblings, pets, or gods can serve as attachment figures?

Vocabulary

Attachment behavioral system

A motivational system selected over the course of evolution to maintain proximity between a young child and his or her primary attachment figure.

Attachment behaviors

Behaviors and signals that attract the attention of a primary attachment figure and function to prevent separation from that individual or to reestablish proximity to that individual (e.g., crying, clinging).

Attachment figure

Someone who functions as the primary safe haven and secure base for an individual. In childhood, an individual's attachment figure is often a parent. In adulthood, an individual's attachment figure is often a romantic partner.

Attachment patterns

(also called "attachment styles" or "attachment orientations") Individual differences in how securely (vs. insecurely) people think, feel, and behave in attachment relationships.

Strange situation

A laboratory task that involves briefly separating and reuniting infants and their primary caregivers as a way of studying individual differences in attachment behavior.

References

- Ainsworth, M. D. S., Blehar, M. C., Waters, E., & Wall, S. (1978). *Patterns of attachment*. Hillsdale, NJ: Erlbaum.
- Bowlby, J. (1969). *Attachment and loss: Vol. 1. Attachment*. New York, NY: Basic Books
- Chappell, K. D., & Davis, K. E. (1998). Attachment, partner choice, and perception of romantic partners: An experimental test of the attachment-security hypothesis. *Personal Relationships*, 5, 327–342.
- Collins, N., & Read, S. (1990). Adult attachment, working models and relationship quality in dating couples. *Journal of Personality and Social Psychology*, 58, 644–663.
- Feeney, J. A., & Noller, P. (1992). Attachment style and romantic love: Relationship dissolution. *Australian Journal of Psychology*, 44, 69–74.
- Fraley, R. C., Roisman, G. I., Booth-LaForce, C., Owen, M. T., & Holland, A. S. (2013). Interpersonal and genetic origins of adult attachment styles: A longitudinal study from infancy to early adulthood. *Journal of Personality and Social Psychology*, 104, 8817–838.
- Frazier, P. A., Byer, A. L., Fischer, A. R., Wright, D. M., & DeBord, K. A. (1996). Adult attachment style and partner choice: Correlational and experimental findings. *Personal Relationships*, 3, 117–136.
- Grossmann, K., Grossmann, K. E., Spangler, G., Suess, G., & Unzner, L. (1985). Maternal sensitivity and newborns orientation responses as related to quality of attachment in northern Germany. *Monographs of the Society for Research in Child Development*, 50(1-2), 233–256.
- Harlow, H. F. (1958). The nature of love. *American Psychologist*, 13, 673–685.
- Hazan, C., & Shaver, P. R. (1987). Romantic love conceptualized as an attachment process. *Journal of Personality and Social Psychology*, 52, 511–524.
- Hudson, N. W., Fraley, R. C., Vicary, A. M., & Brumbaugh, C. C. (2012). *Attachment coregulation: A longitudinal investigation of the coordination in romantic partners' attachment styles*. Manuscript under review.
- McClure, M. J., Lydon, J. E., Baccus, J., & Baldwin, M. W. (2010). A signal detection analysis of the anxiously attached at speed-dating: Being unpopular is only the first part of the problem. *Personality and Social Psychology Bulletin*, 36, 1024–1036.
- Oldmeadow, J. A., Quinn, S., & Kowert, R. (2013). Attachment style, social skills, and Facebook use amongst adults. *Computers in Human Behavior*, 28, 1142–1149.
- Simpson, J. A., Collins, W. A., Tran, S., & Haydon, K. C. (2007). Attachment and the experience

and expression of emotions in adult romantic relationships: A developmental perspective. *Journal of Personality and Social Psychology*, 92, 355–367.

Simpson, J. A., Rholes, W. S., & Nelligan, J. S. (1992). Support seeking and support giving within couples in an anxiety-provoking situation. *Journal of Personality and Social Psychology*, 62, 434–446.

Weinfield, N. S., Sroufe, L. A., Egeland, B., Carlson, E. A. (2008). Individual differences in infant-caregiver attachment: Conceptual and empirical aspects of security. In J. Cassidy & P. R. Shaver (Eds.), *Handbook of attachment: Theory, research, and clinical applications* (2nd ed., pp. 78–101). New York, NY: Guilford Press.

van den Boom, D. C. (1994). The influence of temperament and mothering on attachment and exploration: An experimental manipulation of sensitive responsiveness among lower-class mothers with irritable infants. *Child Development*, 65, 1457–1477.

Emerging Adulthood

Jeffrey Jensen Arnett

Emerging adulthood has been proposed as a new life stage between adolescence and young adulthood, lasting roughly from ages 18 to 25. Five features make emerging adulthood distinctive: identity explorations, instability, self-focus, feeling in-between adolescence and adulthood, and a sense of broad possibilities for the future. Emerging adulthood is found mainly in industrialized countries, where most young people obtain tertiary education and median ages of entering marriage and parenthood are around 30. There are variations in emerging adulthood within industrialized countries. It lasts longest in Europe, and in Asian industrialized countries, the self-focused freedom of emerging adulthood is balanced by obligations to parents and by conservative views of sexuality. In non-industrialized countries, although today emerging adulthood exists only among the middle-class elite, it can be expected to grow in the 21st century as these countries become more affluent.

Learning Objectives

- Explain where, when, and why a new life stage of emerging adulthood appeared over the past half-century.
- Identify the five features that distinguish emerging adulthood from other life stages.
- Describe the variations in emerging adulthood in countries around the world.

Introduction

Think for a moment about the lives of your grandparents and great-grandparents when they

were in their twenties. How do their lives at that age compare to your life? If they were like most other people of their time, their lives were quite different than yours. What happened to change the twenties so much between their time and our own? And how should we understand the 18–25 age period today?



In industrialized countries young people just out of high school and into their 20's are spending more time experimenting with potential directions for their lives. This new way of transitioning into adulthood is different enough from generations past that it is considered a new developmental phase - Emerging Adulthood. [Image: City Year, <https://goo.gl/1ZGKWw>, CC BY-NC-ND 2.0, <https://goo.gl/62XJAl>]

teenage adolescence led quickly and directly to stable adult roles in love and work by their late teens or early twenties. These roles would form the structure of their adult lives for decades to come.

Now all that has changed. A higher proportion of young people than ever before—about 70% in the United States—pursue education and training beyond secondary school (National Center for Education Statistics, 2012). The early twenties are not a time of entering stable adult work but a time of immense job instability: In the United States, the average number of job changes from ages 20 to 29 is seven. The median age of entering marriage in the United States is now 27 for women and 29 for men (U.S. Bureau of the Census, 2011). Consequently, a new stage of the life span, **emerging adulthood**, has been created, lasting from the late teens through the mid-twenties, roughly ages 18 to 25.

The theory of emerging adulthood proposes that a new life stage has arisen between adolescence and young adulthood over the past half-century in industrialized countries. Fifty years ago, most young people in these countries had entered stable adult roles in love and work by their late teens or early twenties. Relatively few people pursued education or training beyond secondary school, and, consequently, most young men were full-time workers by the end of their teens. Relatively few women worked in occupations outside the home, and the median marriage age for women in the United States and in most other industrialized countries in 1960 was around 20 (Arnett & Taber, 1994; Douglass, 2005). The median marriage age for men was around 22, and married couples usually had their first child about one year after their wedding day. All told, for most young people half a century ago, their

The Five Features of Emerging Adulthood

Five characteristics distinguish emerging adulthood from other life stages (Arnett, 2004). Emerging adulthood is:

1. the age of identity explorations;
2. the age of instability;
3. the self-focused age;
4. the age of feeling in-between; and
5. the age of possibilities.

Perhaps the most distinctive characteristic of emerging adulthood is that it is the age of identity explorations. That is, it is an age when people explore various possibilities in love and work as they move toward making enduring choices. Through trying out these different possibilities, they develop a more definite identity, including an understanding of who they are, what their capabilities and limitations are, what their beliefs and values are, and how they fit into the society around them. Erik Erikson (1950), who was the first to develop the idea of identity, proposed that it is mainly an issue in adolescence; but that was more than 50 years ago, and today it is mainly in emerging adulthood that identity explorations take place (Côté, 2006).



The years of emerging adulthood are often times of identity exploration through work, fashion, music, education, and other venues. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

The explorations of emerging adulthood also make it the age of instability. As emerging adults explore different possibilities in love and work, their lives are often unstable. A good illustration of this instability is their frequent moves from one residence to another. Rates of residential change in American society are much higher at ages 18 to 29 than at any other period of life (Arnett, 2004). This reflects the explorations going on in emerging adults' lives. Some move out of their parents' household for the first time in their late teens to attend a residential

college, whereas others move out simply to be independent (Goldscheider & Goldscheider, 1999). They may move again when they drop out of college or when they graduate. They may move to cohabit with a romantic partner, and then move out when the relationship ends. Some move to another part of the country or the world to study or work. For nearly half of American emerging adults, residential change includes moving back in with their parents at least once (Goldscheider & Goldscheider, 1999). In some countries, such as in southern Europe, emerging adults remain in their parents' home rather than move out; nevertheless, they may still experience instability in education, work, and love relationships (Douglass, 2005, 2007).

Emerging adulthood is also a self-focused age. Most American emerging adults move out of their parents' home at age 18 or 19 and do not marry or have their first child until at least their late twenties (Arnett, 2004). Even in countries where emerging adults remain in their parents' home through their early twenties, as in southern Europe and in Asian countries such as Japan, they establish a more independent lifestyle than they had as adolescents (Rosenberger, 2007). Emerging adulthood is a time between adolescents' reliance on parents and adults' long-term commitments in love and work, and during these years, emerging adults focus on themselves as they develop the knowledge, skills, and self-understanding they will need for adult life. In the course of emerging adulthood, they learn to make independent decisions about everything from what to have for dinner to whether or not to get married.

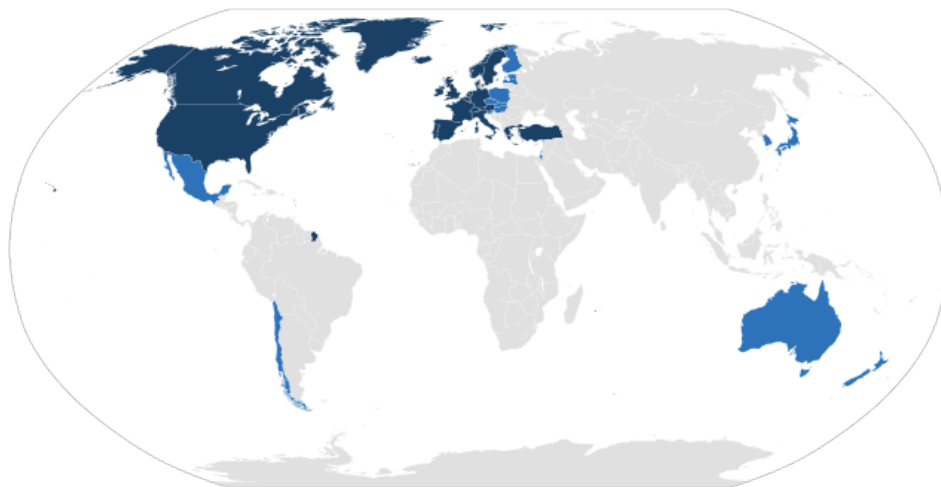
Another distinctive feature of emerging adulthood is that it is an age of feeling in-between, not adolescent but not fully adult, either. When asked, "Do you feel that you have reached adulthood?" the majority of emerging adults respond neither yes nor no but with the ambiguous "in some ways yes, in some ways no" (Arnett, 2003, 2012). It is only when people reach their late twenties and early thirties that a clear majority feels adult. Most emerging adults have the subjective feeling of being in a transitional period of life, on the way to adulthood but not there yet. This "in-between" feeling in emerging adulthood has been found in a wide range of countries, including Argentina (Facio & Micocci, 2003), Austria (Sirsch, Dreher, Mayr, & Willinger, 2009), Israel (Mayseless & Scharf, 2003), the Czech Republic (Macek, Bejček, & Vaníčková, 2007), and China (Nelson & Chen, 2007).

Finally, emerging adulthood is the age of possibilities, when many different futures remain possible, and when little about a person's direction in life has been decided for certain. It tends to be an age of high hopes and great expectations, in part because few of their dreams have been tested in the fires of real life. In one national survey of 18- to 24-year-olds in the United States, nearly all—89%—agreed with the statement, "I am confident that one day I will get to where I want to be in life" (Arnett & Schwab, 2012). This optimism in emerging adulthood has been found in other countries as well (Nelson & Chen, 2007).

International Variations

The five features proposed in the theory of emerging adulthood originally were based on research involving about 300 Americans between ages 18 and 29 from various ethnic groups, social classes, and geographical regions (Arnett, 2004). To what extent does the theory of emerging adulthood apply internationally?

The answer to this question depends greatly on what part of the world is considered. Demographers make a useful distinction between the non-industrialized countries that comprise the majority of the world's population and the industrialized countries that are part of the Organization for Economic Co-operation and Development (OECD), including the United States, Canada, western Europe, Japan, South Korea, Australia, and New Zealand. The current population of **OECD countries** (also called **industrialized countries**) is 1.2 billion, about 18% of the total world population (UNDP, 2011). The rest of the human population resides in **non-industrialized countries**, which have much lower median incomes; much lower median educational attainment; and much higher incidence of illness, disease, and early death. Let us consider emerging adulthood in OECD countries first, then in non-industrialized countries.



Map of OECD countries. Darker shaded countries are original members. [Image: Parastscilveks, <https://goo.gl/MLvm0Y>, CC BY-SA 2.0, <https://goo.gl/eH69he>]

EA in OECD Countries: The Advantages of Affluence

The same demographic changes as described above for the United States have taken place

in other OECD countries as well. This is true of participation in postsecondary education as well as median ages for entering marriage and parenthood (UNdata, 2010). However, there is also substantial variability in how emerging adulthood is experienced across OECD countries. Europe is the region where emerging adulthood is longest and most leisurely. The median ages for entering marriage and parenthood are near 30 in most European countries (Douglass, 2007). Europe today is the location of the most affluent, generous, and egalitarian societies in the world—in fact, in human history (Arnett, 2007). Governments pay for **tertiary education**, assist young people in finding jobs, and provide generous unemployment benefits for those who cannot find work. In northern Europe, many governments also provide housing support. Emerging adults in European societies make the most of these advantages, gradually making their way to adulthood during their twenties while enjoying travel and leisure with friends.

The lives of Asian emerging adults in industrialized countries such as Japan and South Korea are in some ways similar to the lives of emerging adults in Europe and in some ways strikingly different. Like European emerging adults, Asian emerging adults tend to enter marriage and parenthood around age 30 (Arnett, 2011). Like European emerging adults, Asian emerging adults in Japan and South Korea enjoy the benefits of living in affluent societies with generous social welfare systems that provide support for them in making the transition to adulthood—for example, free university education and substantial unemployment benefits.

However, in other ways, the experience of emerging adulthood in Asian OECD countries is markedly different than in Europe. Europe has a long history of **individualism**, and today's emerging adults carry that legacy with them in their focus on self-development and leisure during emerging adulthood. In contrast, Asian cultures have a shared cultural history emphasizing **collectivism** and family obligations. Although Asian cultures have become more individualistic in recent decades as a consequence of globalization, the legacy of collectivism persists in the lives of emerging adults. They pursue identity explorations and self-development during emerging adulthood, like their American and European counterparts, but within narrower boundaries set by their sense of obligations to others, especially their parents (Phinney & Baldelomar, 2011). For example, in their views of the most important criteria for becoming an adult, emerging adults in the United States and Europe consistently rank *financial independence* among the most important markers of adulthood. In contrast, emerging adults with an Asian cultural background especially emphasize becoming *capable of supporting parents financially* as among the most important criteria (Arnett, 2003; Nelson, Badger, & Wu, 2004). This sense of family obligation may curtail their identity explorations in emerging adulthood to some extent, as they pay more heed to their parents' wishes about what they should study, what job they should take, and where they should live than emerging adults do in the West (Rosenberger, 2007).

Another notable contrast between Western and Asian emerging adults is in their sexuality. In the West, premarital sex is normative by the late teens, more than a decade before most people enter marriage. In the United States and Canada, and in northern and eastern Europe, cohabitation is also normative; most people have at least one cohabiting partnership before marriage. In southern Europe, cohabiting is still taboo, but premarital sex is tolerated in emerging adulthood. In contrast, both premarital sex and cohabitation remain rare and forbidden throughout Asia. Even dating is discouraged until the late twenties, when it would be a prelude to a serious relationship leading to marriage. In cross-cultural comparisons, about three fourths of emerging adults in the United States and Europe report having had premarital sexual relations by age 20, versus less than one fifth in Japan and South Korea (Hatfield and Rapson, 2006).

EA in Non-Industrialized Countries: Low But Rising

Emerging adulthood is well established as a normative life stage in the industrialized countries described thus far, but it is still growing in non-industrialized countries. Demographically, in non-industrialized countries as in OECD countries, the median ages for entering marriage and parenthood have been rising in recent decades, and an increasing proportion of young

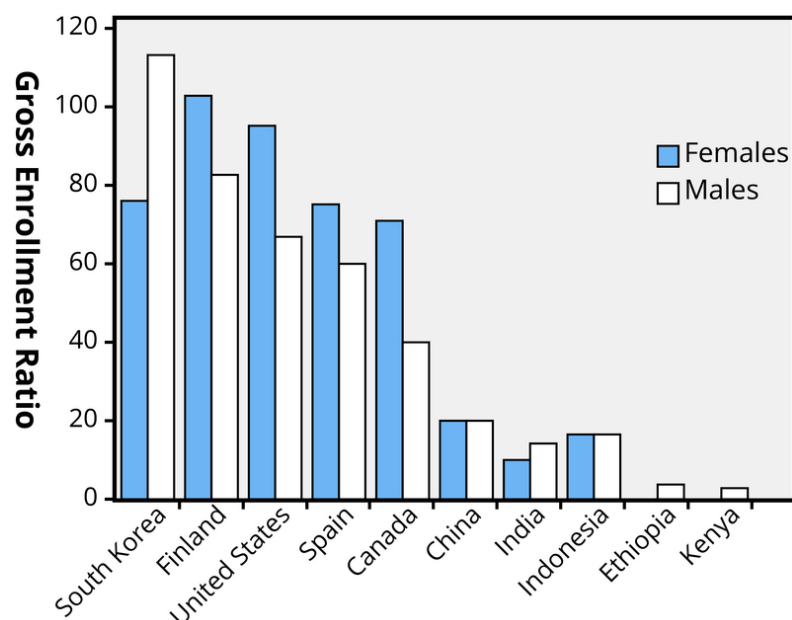


Figure 1: Gross tertiary enrollment, selected countries, 2007. Source: UNdata (2010). Note. Gross enrollment ratio is the total enrollment in a specific level of education, regardless of age, expressed as a percentage of the eligible official school-age population corresponding to the same level of education in a given school year. For the tertiary level, the population used is that of the five-year age group following the end of secondary schooling.

people have obtained post-secondary education. Nevertheless, currently it is only a minority of young people in non-industrialized countries who experience anything resembling emerging adulthood. The majority of the population still marries around age 20 and has long finished education by the late teens. As you can see in Figure 1, rates of enrollment in tertiary education are much lower in non-industrialized countries (represented by the five countries on the right) than in OECD countries (represented by the five countries on the left).

For young people in non-industrialized countries, emerging adulthood exists only for the wealthier segment of society, mainly the urban middle class, whereas the rural and urban poor—the majority of the population—have no emerging adulthood and may even have no adolescence because they enter adult-like work at an early age and also begin marriage and parenthood relatively early. What Saraswathi and Larson (2002) observed about adolescence applies to emerging adulthood as well: “In many ways, the lives of middle-class youth in India, South East Asia, and Europe have more in common with each other than they do with those of poor youth in their own countries.” However, as globalization proceeds, and economic development along with it, the proportion of young people who experience emerging adulthood will increase as the middle class expands. By the end of the 21st century, emerging adulthood is likely to be normative worldwide.

Conclusion

The new life stage of emerging adulthood has spread rapidly in the past half-century and is continuing to spread. Now that the transition to adulthood is later than in the past, is this change positive or negative for emerging adults and their societies? Certainly there are some negatives. It means that young people are dependent on their parents for longer than in the past, and they take longer to become full contributing members of their societies. A substantial proportion of them have trouble sorting through the opportunities available to them and struggle with anxiety and depression, even though most are optimistic. However, there are advantages to having this new life stage as well. By



College and other educational opportunities are important for emerging adults to help transition successfully to the next stages of their lives. [Image: Jirka Matousek, <https://goo.gl/WliY5W>, CC BY 2.0, <https://goo.gl/BRvSA7>]

waiting until at least their late twenties to take on the full range of adult responsibilities, emerging adults are able to focus on obtaining enough education and training to prepare themselves for the demands of today's information- and technology-based economy. Also, it seems likely that if young people make crucial decisions about love and work in their late twenties or early thirties rather than their late teens and early twenties, their judgment will be more mature and they will have a better chance of making choices that will work out well for them in the long run.

What can societies do to enhance the likelihood that emerging adults will make a successful transition to adulthood? One important step would be to expand the opportunities for obtaining tertiary education. The tertiary education systems of OECD countries were constructed at a time when the economy was much different, and they have not expanded at the rate needed to serve all the emerging adults who need such education. Furthermore, in some countries, such as the United States, the cost of tertiary education has risen steeply and is often unaffordable to many young people. In non-industrialized countries, tertiary education systems are even smaller and less able to accommodate their emerging adults. Across the world, societies would be wise to strive to make it possible for every emerging adult to receive tertiary education, free of charge. There could be no better investment for preparing young people for the economy of the future.

Outside Resources

Article: "Average Age of First-Time Moms Keeps Climbing In The U.S" - This NPR story was released in January of 2016 and discusses the rising age of first time pregnancies in US women. The rising average age is reflective of emerging adulthood.

<http://www.npr.org/sections/health-shots/2016/01/14/462816458/average-age-of-first-time-moms-keeps-climbing-in-the-u-s>

Article: "Emerging Adulthood: A Theory of Development from the Late Teens Through the Early Twenties." - The author of the module, Dr. Arnett wrote this American Psychologist article. It summarizes the theory of Emerging Adulthood.

http://jeffre yarnett.com/articles/ARNETT_Emerging_Adulthood_theory.pdf

Article: "Why are so many people in their 20s taking so long to grow up?" - This article presents an interesting perspective that discusses the changing lifestyle of US individuals in their 20's.

<http://www.nytimes.com/2010/08/22/magazine/22Adulthood-t.html?pagewanted=all&r=0>

Video: "Jeffrey Jensen Arnett: Emerging Adulthood" - This video shows an interview with the author of the module, Jeffrey Jensen Arnett. Dr. Arnett talks about his book "Emerging Adulthood" as well as emerging adulthood as a lifespan.

https://www.youtube.com/watch?v=Y_f8DmU-gQQ

Web: Jeffrey Jensen Arnett website

<http://www.jeffreyarnett.com>

Web: Society for the Study of Emerging Adulthood. SSEA is "a multidisciplinary, international organization with a focus on theory and research related to emerging adulthood, which includes the age range of approximately 18 through 29 years. The website includes information on topics, events, and publications pertaining to emerging adults from diverse backgrounds, cultures, and countries."

<http://www.ssea.org>

Discussion Questions

1. What kind of variations in emerging adulthood would you predict within your country? Would there be social class differences? Gender differences? Ethnic differences?

2. Looking at Figure 1, what contrasts do you observe between OECD countries and non-industrialized countries? Between males and females? What economic and cultural differences might explain these contrasts?
3. Do you agree or disagree with the author's prediction that emerging adulthood is likely to become a life stage experienced worldwide in the decades to come? What factors are likely to determine whether this turns out to be true?

Vocabulary

Collectivism

Belief system that emphasizes the duties and obligations that each person has toward others.

Emerging adulthood

A new life stage extending from approximately ages 18 to 25, during which the foundation of an adult life is gradually constructed in love and work. Primary features include identity explorations, instability, focus on self-development, feeling incompletely adult, and a broad sense of possibilities.

Individualism

Belief system that exalts freedom, independence, and individual choice as high values.

Industrialized countries

The economically advanced countries of the world, in which most of the world's wealth is concentrated.

Non-industrialized countries

The less economically advanced countries that comprise the majority of the world's population. Most are currently developing at a rapid rate.

OECD countries

Members of the Organization for Economic Co-operation and Development, comprised of the world's wealthiest countries.

Tertiary education

Education or training beyond secondary school, usually taking place in a college, university, or vocational training program.

References

- Arnett, J. J. (2012). New horizons in emerging and young adulthood. In A. Booth & N. Crouter (Eds.), *Early adulthood in a family context* (pp. 231–244). New York, NY: Springer.
- Arnett, J. J. (2011). Emerging adulthood(s): The cultural psychology of a new life stage. In L.A. Jensen (Ed.), *Bridging cultural and developmental psychology: New syntheses in theory, research, and policy* (pp. 255–275). New York, NY: Oxford University Press.
- Arnett, J. J. (2004). *Emerging adulthood: The winding road from late teens through the twenties*. Oxford, England: Oxford University Press.
- Arnett, J. J. (2003). Conceptions of the transition to adulthood among emerging adults in American ethnic groups. *New Directions for Child and Adolescent Development*, 100, 63–75.
- Arnett, J. J., & Taber, S. (1994). Adolescence terminable and interminable: When does adolescence end? *Journal of Youth & Adolescence*, 23, 517–537.
- Arnett, J. J. & Schwab, J. (2012). *The Clark University poll of emerging adults: Thriving, struggling, & hopeful*. Worcester, MA: Clark University.
- Arnett, J. J. (2007). The long and leisurely route: Coming of age in Europe today. *Current History*, 106, 130–136.
- Côté, J. (2006). Emerging adulthood as an institutionalized moratorium: Risks and benefits to identity formation. In J. J. Arnett & J. L. Tanner (Eds.), *Emerging adults in America: Coming of age in the 21st century* (pp. 85–116). Washington, DC: American Psychological Association Press.
- Douglass, C. B. (2007). From duty to desire: Emerging adulthood in Europe and its consequences. *Child Development Perspectives*, 1, 101–108.
- Douglass, C. B. (2007). From duty to desire: Emerging adulthood in Europe and its consequences. *Child Development Perspectives*, 1, 101–108.
- Douglass, C. B. (2005). *Barren states: The population "implosion" in Europe*. New York, NY: Berg.
- Erikson, E. H. (1950). *Childhood and society*. New York, NY: Norton.
- Facio, A., & Micocci, F. (2003). Emerging adulthood in Argentina. *New Directions in Child and Adolescent Development*, 100, 21–31.
- Goldscheider, F., & Goldscheider, C. (1999). *The changing transition to adulthood: Leaving and returning home*. Thousand Oaks, CA: Sage.
- Hatfield, E., & Rapson, R. L. (2006). *Love and sex: Cross-cultural perspectives*. New York, NY: University Press of America.

- Macek, P., Bejček, J., & Vaníčková, J. (2007). Contemporary Czech emerging adults: Generation growing up in the period of social changes. *Journal of Adolescent Research, 22*, 444–475.
- Mayseless, O., & Scharf, M. (2003). What does it mean to be an adult? The Israeli experience. *New Directions in Child and Adolescent Development, 100*, 5–20.
- National Center for Education Statistics (NCES) (2012). *The condition of education, 2012*. Washington, DC: U.S. Department of Education. Retrieved from <http://www.nces.gov>
- Nelson, L. J., & Chen, X. (2007) Emerging adulthood in China: The role of social and cultural factors. *Child Development Perspectives, 1*, 86–91.
- Nelson, L. J., Badger, S., & Wu, B. (2004). The influence of culture in emerging adulthood: Perspectives of Chinese college students. *International Journal of Behavioral Development, 28*, 26–36.
- Phinney, J. S. & Baldelomar, O. A. (2011). Identity development in multiple cultural contexts. In L. A. Jensen (Ed.), *Bridging cultural and developmental psychology: New syntheses in theory, research and policy* (pp. 161-186). New York, NY: Oxford University Press.
- Rosenberger, N. (2007). Rethinking emerging adulthood in Japan: Perspectives from long-term single women. *Child Development Perspectives, 1*, 92–95.
- Saraswathi, T. S., & Larson, R. (2002). Adolescence in global perspective: An agenda for social policy. In B. B. Brown, R. Larson, & T. S. Saraswathi, (Eds.), *The world's youth: Adolescence in eight regions of the globe* (pp. 344–362). New York, NY: Cambridge University Press.
- Sirsch, U., Dreher, E., Mayr, E., & Willinger, U. (2009). What does it take to be an adult in Austria? Views of adulthood in Austrian adolescents, emerging adults, and adults. *Journal of Adolescent Research, 24*, 275–292.
- U.S. Bureau of the Census (2011). *Statistical abstract of the United States*. Washington, DC: Author.
- UNdata (2010). Gross enrollment ratio in tertiary education. United Nations Statistics Division. Retrieved November 5, 2010, from <http://data.un.org/Data.aspx?d=GenderStat&f=inID:68>
- United Nations Development Programme (UNDP) (2011). *Human development report*. New York, NY: Oxford University Press.

36

The Developing Parent

Marissa L. Diener

This module focuses on parenthood as a developmental task of adulthood. Parents take on new roles as their children develop, transforming their identity as a parent as the developmental demands of their children change. The main influences on parenting, parent characteristics, child characteristics, and contextual factors, are described.

Learning Objectives

- Identify and describe the stages of parenthood.
- Identify and describe the influences on parenting.

The Development of Parents

Think back to an emotional event you experienced as a child. How did your parents react to you? Did your parents get frustrated or criticize you, or did they act patiently and provide support and guidance? Did your parents provide lots of rules for you or let you make decisions on your own? Why do you think your parents behaved the way they did?

Psychologists have attempted to answer these questions about the influences on parents and understand why parents behave the way they do. Because parents are critical to a child's development, a great deal of research has been focused on the impact that parents have on children. Less is known, however, about the development of parents themselves and the impact of children on parents. Nonetheless, parenting is a major role in an adult's life. Parenthood is often considered a normative developmental task of adulthood. Cross-cultural

studies show that adolescents around the world plan to have children. In fact, most men and women in the United States will become parents by the age of 40 years (Martinez, Daniels, & Chandra, 2012). People have children for many reasons, including emotional reasons (e.g., the emotional bond with children and the gratification the parent-child relationship brings), economic and utilitarian reasons (e.g., children provide help in the family and support in old age), and social-normative reasons (e.g., adults are expected to have children; children provide status) (Nauck, 2007).

Parenthood is undergoing changes in the United States and elsewhere in the world. Children are less likely to be living with both parents, and women in the United States have fewer children than they did previously. The average fertility rate of women in the United States was about seven children in the early 1900s and has remained relatively stable at 2.1 since the 1970s (Hamilton, Martin, & Ventura, 2011; Martinez, Daniels, & Chandra, 2012). Not only are parents having fewer children, the context of parenthood has also changed. Parenting outside of marriage has increased dramatically among most socioeconomic, racial, and ethnic groups, although college-educated women are substantially more likely to be married at the birth of a child than are mothers with less education (Dye, 2010). Parenting is occurring outside of marriage for many reasons, both economic and social. People are having children at older ages, too. Despite the fact that young people are more often delaying childbearing, most 18- to 29-year-olds want to have children and say that being a good parent is one of the most important things in life (Wang & Taylor, 2011).



Parenthood has a huge impact on a person's identity, emotions, daily behaviors, and many other aspects of their lives. [Image: Kim881231, CC0 Public Domain, <https://goo.gl/m25gce>]

Galinsky (1987) was one of the first to emphasize the development of parents themselves, how they respond to their children's development, and how they grow as parents. Parenthood is an experience that transforms one's identity as parents take on new roles. Children's growth and development force parents to change their roles. They must develop new skills and abilities in response to children's development. Galinsky identified six stages of parenthood that focus on different tasks and goals (see Table 2).

	1960	2012
Average number of children (fertility rate)	3.6 ⁶	2.1 ²
Percent of births to unmarried women	5% ¹	41% ³
Median age at first marriage for women	20.8 years ⁵	26.5 years ²
Percent of adults ages 18 to 29 married	59% ⁴	20% ⁴

Table 1. Demographic Changes in Parenthood in the United States

1. Ventura & Bachrach, 1999

2. Martinez, Daniels, & Chandra, 2012

3. Hamilton, Martin, & Ventura, 2012

4. Cohn, Passel, Wang, & Livingston, 2011

5. U.S. Decennial Census (1890-2000).

6. Wetzell, J. R. (1990).

1. The Image-Making Stage

As prospective parents think about and form images about their roles as parents and what parenthood will bring, and prepare for the changes an infant will bring, they enter the **image-making stage**. Future parents develop their ideas about what it will be like to be a parent and the type of parent they want to be. Individuals may evaluate their relationships with their own parents as a model of their roles as parents.

2. The Nurturing Stage

The second stage, the **nurturing stage**, occurs at the birth of the baby. A parent's main goal during this stage is to develop an attachment relationship to their baby. Parents must adapt their romantic relationships, their relationships with their other children, and with their own parents to include the new infant. Some parents feel attached to the baby immediately, but for other parents, this occurs more gradually. Parents may have imagined their infant in specific ways, but they now have to reconcile those images with their actual baby. In incorporating their relationship with their child into their other relationships, parents often have to reshape their conceptions of themselves and their identity. Parenting responsibilities are the most demanding during infancy because infants are completely dependent on caregiving.

3. The Authority Stage

	Age of Child	Main Tasks and Goals
Stage 1: The Image-Making Stage	Planning for a child; Pregnancy	Consider what it means to be a parent and plan for changes to accommodate a child.
Stage 2: The Nurturing Stage	Infancy	Develop an attachment relationship with child and adapt to the new baby
Stage 3: The Authority Stage	Toddler and preschool	Parents create rules and figure out how to effectively guide their children's behavior.
Stage 4: The Interpretive Stage	Middle childhood	Parents help their children interpret their experiences with the social world beyond the family.
Stage 5: The Interdependent Stage	Adolescence	Parents renegotiate their relationship with their adolescent children to allow for shared power in decision-making.
Stage 6: The Departure Stage	Early adulthood	Parents evaluate their successes and failures as parents.

Table 2. Galinsky's Stages of Parenthood

The **authority stage** occurs when children are 2 years old until about 4 or 5 years old. In this stage, parents make decisions about how much authority to exert over their children's behavior. Parents must establish rules to guide their child's behavior and development. They have to decide how strictly they should enforce rules and what to do when rules are broken.

4. The Interpretive Stage

The **interpretive stage** occurs when children enter school (preschool or kindergarten) to the beginning of adolescence. Parents interpret their children's experiences as children are increasingly exposed to the world outside the family. Parents answer their children's questions, provide explanations, and determine what behaviors and values to teach. They decide what experiences to provide their children, in terms of schooling, neighborhood, and extracurricular activities. By this time, parents have experience in the parenting role and often reflect on their strengths and weaknesses as parents, review their images of parenthood, and determine how realistic they have been. Parents have to negotiate how involved to be with their children, when to step in, and when to encourage children to make choices independently.

5. The Interdependent Stage

Parents of teenagers are in the interdependent stage. They must redefine their authority and renegotiate their relationship with their adolescent as the children increasingly make decisions independent of parental control and authority. On the other hand, parents do not permit their adolescent children to have complete autonomy over their decision-making and behavior, and thus adolescents and parents must adapt their relationship to allow for greater negotiation and discussion about rules and limits.

6. The Departure Stage

During the departure stage of parenting, parents evaluate the entire experience of parenting. They prepare for their child's departure, redefine their identity as the parent of an adult child, and assess their parenting accomplishments and failures. This stage forms a transition to a new era in parents' lives. This stage usually spans a long time period from when the oldest child moves away (and often returns) until the youngest child leaves. The parenting role must be redefined as a less central role in a parent's identity.

Despite the interest in the development of parents among lay people and helping professionals, little research has examined developmental changes in parents' experience and behaviors over time. Thus, it is not clear whether these theoretical stages are generalizable to parents of different races, ages, and religions, nor do we have empirical data on the factors that influence individual differences in these stages. On a practical note, how-to books and websites geared toward parental development should be evaluated with caution, as not all advice provided is supported by research.



When a child achieves a new level of independence and leaves the home it marks another turning point in the identity of a parent. [Image: State Farm, <https://goo.gl/Npw2fb>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Influences on Parenting

Parenting is a complex process in which parents and children influence one another. There are many reasons that parents behave the way they do. The multiple influences on parenting are still being explored. Proposed influences on parental behavior include 1) parent characteristics, 2) child characteristics, and 3) contextual and sociocultural characteristics (Belsky, 1984; Demick, 1999) (see Figure 1).

Parent Characteristics

Parents bring unique traits and qualities to the parenting relationship that affect their decisions as parents. These characteristics include the age of the parent, gender, beliefs, personality, developmental history, knowledge about parenting and child development, and mental and physical health. Parents' personalities affect parenting behaviors. Mothers and fathers who are more agreeable, conscientious, and outgoing are warmer and provide more structure to their children. Parents who are more agreeable, less anxious, and less negative also support their children's autonomy more than parents who are anxious and less agreeable (Prinz, Stams, Dekovic, Reijntjes, & Belsky, 2009). Parents who have these personality traits appear to be better able to respond to their children positively and provide a more consistent, structured environment for their children.

Parents' developmental histories, or their experiences as children, also affect their parenting strategies. Parents may learn parenting practices from their own parents. Fathers whose own parents provided monitoring, consistent and age-appropriate discipline, and warmth were more likely to provide this constructive parenting to their own children (Kerr, Capaldi, Pears, & Owen, 2009). Patterns of negative parenting and ineffective discipline also appear from one generation to the next. However, parents who are dissatisfied with their own parents' approach may be more likely to change their parenting methods with their own children.

Child Characteristics

Parenting is bidirectional. Not only do parents affect their children, children influence their parents. Child characteristics, such as gender, birth order, temperament, and health status, affect parenting behaviors and roles. For example, an infant with an easy temperament may enable parents to feel more effective, as they are easily able to soothe the child and elicit smiling and cooing. On the other hand, a cranky or fussy infant elicits fewer positive reactions from his or her parents and may result in parents feeling less effective in the parenting role.

(Eisenberg et al., 2008). Over time, parents of more difficult children may become more punitive and less patient with their children (Clark, Kochanska, & Ready, 2000; Eisenberg et al., 1999; Kiff, Lengua, & Zalewski, 2011). Parents who have a fussy, difficult child are less satisfied with their marriages and have greater challenges in balancing work and family roles (Hyde, Else-Quest, & Goldsmith, 2004). Thus, child temperament is one of the child characteristics that influences how parents behave with their children.



A child with a difficult temperament can have a significant impact on a parent. [Image: Harald Groven, <https://goo.gl/cwemLg>, CC BY-SA 2.0, <https://goo.gl/eH69he>]

Another child characteristic is the gender of the child. Parents respond differently to boys and girls. Parents often assign different household chores to their sons and daughters. Girls are more often responsible for caring for younger siblings and household chores, whereas boys are more likely to be asked to perform chores outside the home, such as mowing the lawn (Grusec, Goodnow, & Cohen, 1996). Parents also talk differently with their sons and daughters, providing more scientific explanations to their sons and using more emotion words with their daughters (Crowley, Callanan, Tenenbaum, & Allen, 2001).

Contextual Factors and Sociocultural Characteristics

The parent-child relationship does not occur in isolation. Sociocultural characteristics, including economic hardship, religion, politics, neighborhoods, schools, and social support, also influence parenting. Parents who experience economic hardship are more easily frustrated, depressed, and sad, and these emotional characteristics affect their parenting skills (Conger & Conger, 2002). Culture also influences parenting behaviors in fundamental ways. Although promoting the development of skills necessary to function effectively in one's community is a universal goal of parenting, the specific skills necessary vary widely from culture to culture. Thus, parents have different goals for their children that partially depend on their culture (Tamis-LeMonda et al., 2008). For example, parents vary in how much they emphasize goals for independence and individual achievements, and goals involving maintaining harmonious relationships and being embedded in a strong network of social relationships. These differences in parental goals are influenced by culture and by immigration status. Other important contextual characteristics, such as the neighborhood, school, and

social networks, also affect parenting, even though these settings don't always include both the child and the parent (Bronfenbrenner, 1989). For example, Latina mothers who perceived their neighborhood as more dangerous showed less warmth with their children, perhaps because of the greater stress associated with living a threatening environment (Gonzales et al., 2011). Many contextual factors influence parenting.

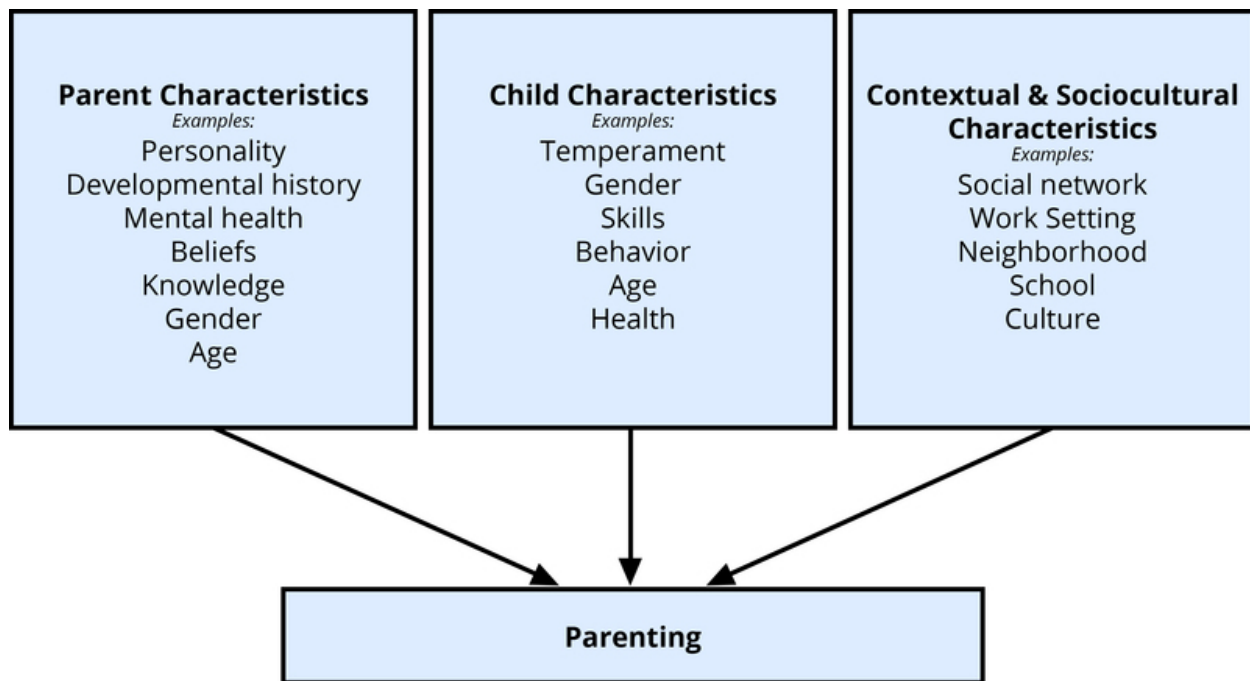


Figure 1. Influences on Parenting

Conclusion

Many factors influence parenting decisions and behaviors. These factors include characteristics of the parent, such as gender and personality, as well as characteristics of the child, such as age. The context is also important. The interaction among all these factors creates many different patterns of parenting behavior. Furthermore, parenting influences not just a child's development, but also the development of the parent. As parents are faced with new challenges, they change their parenting strategies and construct new aspects of their identity. The goals and tasks of parents change over time as their children develop.

Outside Resources

Article: "Is a Child's Behavior Always a Reflection of His Parents?" - This article is written by Dr. Peggy Drexler and discusses the notion that child behavior is not always a reflection of parenting.

http://www.huffingtonpost.com/peggy-drexler/is-a-childs-behavior-alwa_b_1886367.html

Article: "Parent behavior toward first and second children" - This journal article describes how parents behave differently and the same with their first and second born children. This is an interesting read to learn more about parenting behavior and how it changes based on a child characteristic, birth order.

<http://psycnet.apa.org/psycinfo/1954-08594-001>

Org: American Psychological Association (APA), Parenting - Parenting is a psychology topic explored by APA. They state that, "Parenting practices around the world share three major goals: ensuring children's health and safety, preparing children for life as productive adults and transmitting cultural values. A high-quality parent-child relationship is critical for healthy development." This webpage links to articles to support these goals.

<http://www.apa.org/topics/parenting/>

Org: Society for the Research in Child Development (SRCD) - SRCD works to coordinate and integrate research in human development. It aims to assist in the dissemination of research findings and in this way can be a great resource to teachers and students.

<http://www.srcd.org>

Web: American Psychological Association- Information and Resources on Parenting

<http://www.apa.org/topics/parenting/index.aspx>

Web: NPR, Parenting - National Public Radio presented interesting stories on many topics related to child development. The page linked here has many stories on parenting.

<http://www.npr.org/tags/126952921/parenting>

Web: PBS Parents: Child Development - PBS has some interesting resources for parents including articles, games and products. This is a good resource for students looking for some friendly and less psychology based sources that they can read or share with their own families.

<http://www.pbs.org/parents/child-development/>

Discussion Questions

1. Reflect on the way you were raised. Consider the parenting behaviors (e.g., rules, discipline strategies, warmth, and support) used in your household when you were a child. Why do you think your parents behaved this way? How do these factors fit with the influences on parenting described here? Provide specific examples of multiple influences on parenting.
2. Think about different parents and grandparents you know. Do the challenges they face as parents differ based on the age of their children? Do your observations fit with Galinsky's stages of parenting?
3. What type of parent do you envision yourself becoming? If you are a parent, how do you parent your child/children? How do you think this is similar to or different than the way you were raised? What influences exist in your life that will make you parent differently from your own parents?

Vocabulary

Authority stage

Stage from approximately 2 years to age 4 or 5 when parents create rules and figure out how to effectively guide their children's behavior.

Bidirectional

The idea that parents influence their children, but their children also influence the parents; the direction of influence goes both ways, from parent to child, and from child to parent.

Departure stage

Stage at which parents prepare for a child to depart and evaluate their successes and failures as parents.

Image-making stage

Stage during pregnancy when parents consider what it means to be a parent and plan for changes to accommodate a child.

Interdependent stage

Stage during teenage years when parents renegotiate their relationship with their adolescent children to allow for shared power in decision-making.

Interpretive stage

Stage from age 4 or 5 to the start of adolescence when parents help their children interpret their experiences with the social world beyond the family.

Nurturing stage

Stage from birth to around 18-24 months in which parents develop an attachment relationship with child and adapt to the new baby.

Temperament

A child's innate personality; biologically based personality, including qualities such as activity level, emotional reactivity, sociability, mood, and soothability.

References

- Belsky, J. (1984). The determinants of parenting: A process model. *Child Development, 55*, 83–96.
- Bronfenbrenner, U. (1989). Ecological systems theory. In R. Vasta (Ed.), *Annals of Child Development, Vol. 6* (pp. 187–251). Greenwich, CT: JAI Press.
- Clark, L. A., Kochanska, G., & Ready, R. (2000). Mothers' personality and its interaction with child temperament as predictors of parenting behavior. *Journal of Personality and Social Psychology, 79*, 274–285.
- Cohn, D., Passel, J., Wang, W., & Livingston, G. (2011). Barely Half of U.S. Adults are Married – A Record Low. *Social & Demographic Trends*. Washington, DC: Pew Research Center.
- Conger, R. D., & Conger, K. J. (2002). Resilience in Midwestern families: Selected findings from the first decade of a prospective longitudinal study. *Journal of Marriage and Family, 64*, 361–373.
- Crowley, K., Callanan, M. A., Tenenbaum, H. R., & Allen, E. (2001). Parents explain more often to boys than to girls during shared scientific thinking. *Psychological Science, 12*, 258–261.
- Demick, J. (1999). Parental development: Problem, theory, method, and practice. In R. L. Mosher, D. J. Youngman, & J. M. Day (Eds.), *Human Development Across the Life Span: Educational and Psychological Applications* (pp. 177–199). Westport, CT: Praeger.
- Dye, J. L. (2010). Fertility of American women: 2008. Current Population Reports P20-563. Retrieved on May 18, 2012 from www.census.gov/prod/2010pubs/p20-563.pdf.
- Eisenberg, N., Fabes, R. A., Shepard, S. A., Guthrie, I.K., Murphy, B.C., & Reiser, M. (1999). Parental reactions to children's negative emotions: Longitudinal relations to quality of children's social functioning. *Child Development, 70*, 513–534.
- Eisenberg, N., Hofer, C., Spinrad, T., Gershoff, E., Valiente, C., Losoya, S. L., Zhou, Q., Cumberland, A., Liew, J., Reiser, M., & Maxon, E. (2008). Understanding parent-adolescent conflict discussions: Concurrent and across-time prediction from youths' dispositions and parenting. *Monographs of the Society for Research in Child Development, 73*, (Serial No. 290, No. 2), 1–160.
- Galinsky, E. (1987). *The Six Stages of Parenthood*. Cambridge, MA: Perseus Books.
- Gonzales, N. A., Coxe, S., Roosa, M. W., White, R. M. B., Knight, G. P., Zeiders, K. H., & Saenz, D. (2011). Economic hardship, neighborhood context, and parenting: Prospective effects on Mexican-American adolescent's mental health. *American Journal of Community Psychology, 47*, 98–113. doi: 10.1007/s10464-010-9366-1

- Grusec, J. E., Goodnow, J. J., & Cohen, L. (1996). Household work and the development of concern for others. *Developmental Psychology, 32*, 999–1007.
- Hamilton, B. E., Martin, J. A., & Ventura, S. J. (2011). Births: Preliminary data for 2010. *National Vital Statistics Reports, 60*(2). Hyattsville, MD: U.S. Department of Health and Human Services.
- Hyde, J. S., Else-Quest, N. M., & Goldsmith, H. H. (2004). Children's temperament and behavior problems predict their employed mothers' work functioning. *Child Development, 75*, 580–594.
- Kerr, D. C. R., Capaldi, D. M., Pears, K. C., & Owen, L. D. (2009). A prospective three generational study of fathers' constructive parenting: Influences from family of origin, adolescent adjustment, and offspring temperament. *Developmental Psychology, 45*, 1257–1275.
- Kiff, C. J., Lengua, L. J., & Zalewski, M. (2011). Nature and nurturing: Parenting in the context of child temperament. *Clinical Child and Family Psychology Review, 14*, 251–301. doi: 10.1007/s10567-011-0093-4
- Martinez, G., Daniels, K., & Chandra, A. (2012). Fertility of men and women aged 15-44 years in the United States: National Survey of Family Growth, 2006-2010. *National Health Statistics Reports, 51*(April). Hyattsville, MD: U.S., Department of Health and Human Services.
- Nauck, B. (2007). Value of children and the framing of fertility: Results from a cross-cultural comparative survey in 10 societies. *European Sociological Review, 23*, 615–629.
- Prinz, P., Stams, G. J., Dekovic, M., Reijntjes, A. H., & Belsky, J. (2009). The relations between parents' Big Five personality factors and parenting: A meta-analytic review. *Journal of Personality and Social Psychology, 97*, 351–362.
- Tamis-LeMonda, C. S., Way, N., Hughes, D., Yoshikawa, H., Kalman, R. K., & Niwa, E. Y. (2008). Parents' goals for children: The dynamic coexistence of individualism and collectivism in cultures and individuals. *Social Development, 17*, 183–209. doi: 10.1111/j.1467.9507.2007.00419.x
- U.S. Decennial Census (1890-2000). <http://www.census.gov>
- Ventura, S. J., & Bachrach, C. A. (1999). Nonmarital childbearing in the United States, 1940-1999. *National Vital Statistics, 48*, No. 16. Hyattsville, MD: National Center for Health Statistics.
- Wang, W., & Taylor, P. (2011). *For Millennials, parenthood trumps marriage*. Washington, DC: Pew Research Center.
- Wetzel, J. R. (1990). *American Families: 75 Years of change*. Washington, DC: Bureau of Labor Statistics.

Aging

Tara Queen & Jacqui Smith

Traditionally, research on aging described only the lives of people over age 65 and the very old. Contemporary theories and research recognizes that biogenetic and psychological processes of aging are complex and lifelong. Functioning in each period of life is influenced by what happened earlier and, in turn, affects subsequent change. We all age in specific social and historical contexts. Together, these multiple influences on aging make it difficult to define when middle-age or old age begins. This module describes central concepts and research about adult development and aging. We consider contemporary questions about cognitive aging and changes in personality, self-related beliefs, social relationships, and subjective well-being. These four aspects of psychosocial aging are related to health and longevity

Learning Objectives

- Explain research approaches to studying aging.
- Describe cognitive, psychosocial, and physical changes that occur with age.
- Provide examples of how age-related changes in these domains are observed in the context of everyday life.

Introduction

We are currently living in an aging society (Rowe, 2009). Indeed, by 2030 when the last of the Baby Boomers reach age 65, the U.S. older population will be double that of 2010. Furthermore, because of increases in average life expectancy, each new generation can expect to live longer



Due to positive health trends the population of older adults is increasing steadily. Understanding the psychology of aging will be more important than ever to support this group and help them thrive. [Photo: Woody Hibbard, <https://goo.gl/VP7pfz>, CC BY 2.0, <https://goo.gl/JD0cjj>]

than their parents' generation and certainly longer than their grandparents' generation. As a consequence, it is time for individuals of all ages to rethink their personal life plans and consider prospects for a long life. When is the best time to start a family? Will the education gained up to age 20 be sufficient to cope with future technological advances and marketplace needs? What is the right balance between work, family, and leisure throughout life? What's the best age to retire? How can I age successfully and enjoy life to the fullest when I'm 80 or 90? In this module we will discuss several different domains of psychological research on aging that will help answer these important questions.

Overview: Life Span and Life

Course Perspectives on Aging

Just as young adults differ from one another, older adults are also not all the same. In each decade of adulthood, we observe substantial **heterogeneity** in cognitive functioning, personality, social relationships, lifestyle, beliefs, and satisfaction with life. This heterogeneity reflects differences in rates of biogenetic and psychological aging and the sociocultural contexts and history of people's lives (Bronfenbrenner, 1979; Fingerman, Berg, Smith, & Antonucci, 2011). Theories of aging describe how these multiple factors interact and change over time. They describe why functioning differs on average between young, middle-aged, young-old, and very old adults and why there is heterogeneity within these age groups. **Life course theories**, for example, highlight the effects of social expectations and the normative timing of life events and social roles (e.g., becoming a parent, retirement). They also consider the lifelong cumulative effects of membership in specific **cohorts** (generations) and sociocultural subgroups (e.g., race, gender, socioeconomic status) and exposure to historical events (e.g., war, revolution, natural disasters; Elder, Johnson, & Crosnoe, 2003; Settersten, 2005). **Life span theories** complement the life-course perspective with a greater focus on processes within the individual (e.g., the aging brain). This approach emphasizes the patterning of lifelong **intra- and inter-individual differences** in the shape (gain, maintenance,

loss), level, and rate of change (Baltes, 1987, 1997). Both life course and life span researchers generally rely on longitudinal studies to examine hypotheses about different patterns of aging associated with the effects of biogenetic, life history, social, and personal factors. Cross-sectional studies provide information about age-group differences, but these are confounded with cohort, time of study, and historical effects.

Cognitive Aging

Researchers have identified areas of both losses and gains in cognition in older age. Cognitive ability and intelligence are often measured using standardized tests and validated measures. The psychometric approach has identified two categories of intelligence that show different rates of change across the life span (Schaie & Willis, 1996). Fluid intelligence refers to information processing abilities, such as logical reasoning, remembering lists, spatial ability, and reaction time. Crystallized intelligence encompasses abilities that draw upon experience and knowledge. Measures of crystallized intelligence include vocabulary tests, solving number problems, and understanding texts.

With age, systematic declines are observed on cognitive tasks requiring self-initiated, effortful processing, without the aid of supportive memory cues (Park, 2000). Older adults tend to perform poorer than young adults on memory tasks that involve recall of information, where individuals must retrieve information they learned previously without the help of a list of possible choices. For example, older adults may have more difficulty recalling facts such as names or contextual details about where or when something happened (Craik, 2000). What might explain these deficits as we age? As we age, working memory, or our ability to simultaneously store and use information, becomes less efficient (Craik & Bialystok, 2006). The ability to process information quickly also decreases with age. This slowing of processing speed may explain age differences on many different cognitive



There are many stereotypes of older adults. They are sometimes seen as slow because of changes in cognitive processing speed. They are though, on average, excellent at drawing on personal experience and knowledge. And they tend to outperform young adults when it comes to social and emotional challenges. [Image: Alex E. Proimos, <https://goo.gl/20SbW8>, CC BY-NC 2.0, <https://goo.gl/Fllc2e>]

tasks (Salthouse, 2004). Some researchers have argued that inhibitory functioning, or the ability to focus on certain information while suppressing attention to less pertinent information, declines with age and may explain age differences in performance on cognitive tasks (Hasher & Zacks, 1988). Finally, it is well established that our hearing and vision decline as we age. Longitudinal research has proposed that deficits in sensory functioning explain age differences in a variety of cognitive abilities (Baltes & Lindenberger, 1997).

Fewer age differences are observed when memory cues are available, such as for recognition memory tasks, or when individuals can draw upon acquired knowledge or experience. For example, older adults often perform as well if not better than young adults on tests of word knowledge or vocabulary. With age often comes expertise, and research has pointed to areas where aging experts perform as well or better than younger individuals. For example, older typists were found to compensate for age-related declines in speed by looking farther ahead at printed text (Salthouse, 1984). Compared to younger players, older chess experts are able to focus on a smaller set of possible moves, leading to greater cognitive efficiency (Charness, 1981). Accrued knowledge of everyday tasks, such as grocery prices, can help older adults to make better decisions than young adults (Tentori, Osheron, Hasher, & May, 2001).

How do changes or maintenance of cognitive ability affect older adults' everyday lives? Researchers have studied cognition in the context of several different everyday activities. One example is driving. Although older adults often have more years of driving experience, cognitive declines related to reaction time or attentional processes may pose limitations under certain circumstances (Park & Gutchess, 2000). Research on interpersonal problem solving suggested that older adults use more effective strategies than younger adults to navigate through social and emotional problems (Blanchard-Fields, 2007). In the context of work, researchers rarely find that older individuals perform poorer on the job (Park & Gutchess, 2000). Similar to everyday problem solving, older workers may develop more efficient strategies and rely on expertise to compensate for cognitive decline.

Personality and Self-Related Processes

Research on adult personality examines normative age-related increases and decreases in the expression of the so-called "Big Five" traits—extraversion, neuroticism, conscientiousness, agreeableness, and openness to new experience. Does personality change throughout adulthood? Previously the answer was no, but contemporary research shows that although some people's personalities are relatively stable over time, others' are not (Lucas & Donnellan, 2011; Roberts & Mroczek, 2008). Longitudinal studies reveal average changes during adulthood in the expression of some traits (e.g., neuroticism and openness decrease with age

and conscientiousness increases) and individual differences in these patterns due to idiosyncratic life events (e.g., divorce, illness). Longitudinal research also suggests that adult personality traits, such as conscientiousness, predict important life outcomes including job success, health, and longevity (Friedman, Tucker, Tomlinson-Keasey, Schwartz, Wingard, & Criqui, 1993; Roberts, Kuncel, Shiner, Caspi, & Goldberg, 2007).

In contrast to the relative stability of personality traits, theories about the aging self-propose changes in self-related knowledge, beliefs, and **autobiographical narratives**. Responses to questions such as "Tell me something about yourself. Who are you?" "What are your hopes for the future?" provide insight into the characteristics and life themes that an individual considers uniquely distinguish him or herself from others. These self-descriptions enhance self-esteem and guide behavior (Markus & Nurius, 1986; McAdams, 2006). Theory suggests that as we age, themes that were relatively unimportant in young and middle adulthood gain in salience (e.g., generativity, health) and that people view themselves as improving over time (Ross & Wilson, 2003). Reorganizing personal life narratives and self-descriptions are the major tasks of midlife and young-old age due to transformations in professional and family roles and obligations. In advanced old age, self-descriptions are often characterized by a life review and reflections about having lived a long life. Birren and Schroots (2006), for example, found the process of life review in late life helped individuals confront and cope with the challenges of old age.



There is a difference between physical age and subjective age as summarized in the saying "You are only as old as you feel." [Image: Emar, CC BY-NC-SA 2.0, <https://goo.gl/HEXbAA>]

One aspect of the self that particularly interests life span and life course psychologists is the individual's perception and evaluation of their own aging and identification with an age group. **Subjective age** is a multidimensional construct that indicates how old (or young) a person feels and into which age group a person categorizes him- or herself. After early adulthood, most people say that they feel younger than their chronological age and the gap between subjective age and actual age generally increases. On average, after age 40 people report feeling 20% younger than their actual age (e.g., Rubin & Berntsen, 2006). Asking people how satisfied they are with their own aging

assesses an evaluative component of age identity. Whereas some aspects of age identity are positively valued (e.g., acquiring seniority in a profession or becoming a grandparent), others may be less valued, depending on societal context. Perceived physical age (i.e., the age one looks in a mirror) is one aspect that requires considerable self-related adaptation in social and cultural contexts that value young bodies. Feeling younger and being satisfied with one's own aging are expressions of positive self-perceptions of aging. They reflect the operation of self-related processes that enhance well-being. Levy (2009) found that older individuals who are able to adapt to and accept changes in their appearance and physical capacity in a positive way report higher well-being, have better health, and live longer.

Social Relationships

Social ties to family, friends, mentors, and peers are primary resources of information, support, and comfort. Individuals develop and age together with family and friends and interact with others in the community. Across the life course, social ties are accumulated, lost, and transformed. Already in early life, there are multiple sources of heterogeneity in the characteristics of each person's social network of relationships (e.g., size, composition, and quality). Life course and life span theories and research about age-related patterns in social relationships focus on understanding changes in the processes underlying social connections. Antonucci's Convoy Model of Social Relations (2001; Kahn & Antonucci, 1980), for example, suggests that the social connections that people accumulate are held together by exchanges in social support (e.g., tangible and emotional). The frequency, types, and reciprocity of the exchanges change with age and in response to need, and in turn, these exchanges impact the health and well-being of the givers and receivers in the convoy. In many relationships, it is not the actual objective exchange of support that is critical but instead the perception that support is available if needed (Uchino, 2009). Carstensen's Socioemotional Selectivity Theory (1993; Carstensen, Isaacowitz, & Charles, 1999) focuses on changes in motivation for actively seeking social contact with others. She proposes that with increasing age our motivational goals change from information gathering to emotion regulation. To optimize the experience of positive affect, older adults actively restrict their social life to prioritize time spent with emotionally close significant others. In line with this, older marriages are found to be characterized by enhanced positive and reduced negative interactions and older partners show more affectionate behavior during conflict discussions than do middle-aged partners (Carstensen, Gottman, & Levenson, 1995). Research showing that older adults have smaller networks compared to young adults and tend to avoid negative interactions also supports this theory. Similar selective processes are also observed when time horizons for interactions with close partners shrink temporarily for young adults (e.g., impending geographical

separations).

Much research focuses on the associations between specific effects of long-term social relationships and health in later life. Older married individuals who receive positive social and emotional support from their partner generally report better health than their unmarried peers (Antonucci, 2001; Umberson, Williams, Powers, Liu, & Needham, 2006; Waite & Gallagher, 2000). Despite the overall positive health effects of being married in old age (compared with being widowed, divorced, or single), living as a couple can have a "dark side" if the relationship is strained or if one partner is the primary caregiver. The consequences of positive and negative aspects of relationships are complex (Birditt & Antonucci, 2008; Rook, 1998; Uchino, 2009). For example, in some circumstances, criticism from a partner may be perceived as valid and useful feedback whereas in others it is considered unwarranted and hurtful. In long-term relationships, habitual negative exchanges might have diminished effects. Parent-child and sibling relationships are often the most long-term and emotion-laden social ties. Across the life span, the parent-child tie, for example, is characterized by a paradox of solidarity, conflict, and ambivalence (Fingerman, Chen, Hay, Cichy, & Lefkowitz, 2006).

Emotion and Well-being

As we get older, the likelihood of losing loved ones or experiencing declines in health increases. Does the experience of such losses result in decreases in well-being in older adulthood? Researchers have found that well-being differs across the life span and that the patterns of these differences depend on how well-being is measured.

Measures of **global subjective well-being** assess individuals' overall perceptions of their lives. This can include questions about life satisfaction or judgments of whether individuals are currently living the best life possible. What factors may contribute to how people respond to these questions? Age, health, personality, social support, and life experiences have been shown to influence judgments of global well-being. It is important to note that predictors of well-being may change as we age. What is important to life satisfaction in young adulthood can be different in later adulthood (George, 2010). Early research on well-being argued that life events such as marriage or divorce can temporarily influence well-being, but people quickly adapt and return to a neutral baseline (called the hedonic treadmill; Diener, Lucas, & Scollon, 2006). More recent research suggests otherwise. Using longitudinal data, researchers have examined well-being prior to, during, and after major life events such as widowhood, marriage, and unemployment (Lucas, 2007). Different life events influence well-being in different ways, and individuals do not often adapt back to baseline levels of well-being. The influence of events, such as unemployment, may have a lasting negative influence on well-being as people

age. Research suggests that global well-being is highest in early and later adulthood and lowest in midlife (Stone, Schwartz, Broderick, & Deaton, 2010).

Hedonic well-being refers to the emotional component of well-being and includes measures of positive (e.g., happiness, contentment) and negative affect (e.g., stress, sadness). The pattern of positive affect across the adult life span is similar to that of global well-being, with experiences of positive emotions such as happiness and enjoyment being highest in young and older adulthood. Experiences of negative affect, particularly stress and anger, tend to decrease with age. Experiences of sadness are lowest in early and later adulthood compared to midlife (Stone et al., 2010). Other research finds that older adults report more positive and less negative affect than middle age and younger adults (Magai, 2008; Mroczek, 2001). It should be noted that both global well-being and positive affect tend to taper off during late older adulthood and these declines may be accounted for by increases in health-related losses during these years (Charles & Carstensen, 2010).

Psychological well-being aims to evaluate the positive aspects of psychosocial development, as opposed to factors of ill-being, such as depression or anxiety. Ryff's model of psychological well-being proposes six core dimensions of positive well-being. Older adults tend to report higher environmental mastery (feelings of competence and control in managing everyday life) and autonomy (independence), lower personal growth and purpose in life, and similar levels

of positive relations with others as younger individuals (Ryff, 1995). Links between health and interpersonal flourishing, or having high-quality connections with others, may be important in understanding how to optimize quality of life in old age (Ryff & Singer, 2000).

Successful Aging and Longevity

Increases in **average life expectancy** in the 20th century and evidence from twin studies that suggests that genes account for only 25% of the variance in human life spans have opened new questions about implications for individuals and society (Christensen, Doblhammer, Rau, &



Physical activity is one of the pillars of successful aging. [Image: William Murphy, <https://goo.gl/Khsbsb>, CC BY-SA 2.0, <https://goo.gl/jSSrcO>]

Vaupel, 2009). What environmental and behavioral factors contribute to a healthy long life? Is it possible to intervene to slow processes of aging or to minimize cognitive decline, prevent dementia, and ensure life quality at the end of life (Fratiglioni, Paillard-Borg, & Winblad, 2004; Hertzog, Kramer, Wilson, & Lindenberger, 2009; Lang, Baltes, & Wagner, 2007)? Should interventions focus on late life, midlife, or indeed begin in early life? Suggestions that pathological change (e.g., dementia) is not an inevitable component of aging and that pathology could at least be delayed until the very end of life led to theories about successful aging and proposals about targets for intervention. Rowe and Kahn (1997) defined three criteria of successful aging: (a) the relative avoidance of disease, disability, and risk factors like high blood pressure, smoking, or obesity; (b) the maintenance of high physical and cognitive functioning; and (c) active engagement in social and productive activities. Although such definitions of successful aging are value-laden, research and behavioral interventions have subsequently been guided by this model. For example, research has suggested that age-related declines in cognitive functioning across the adult life span may be slowed through physical exercise and lifestyle interventions (Kramer & Erickson, 2007). It is recognized, however, that societal and environmental factors also play a role and that there is much room for social change and technical innovation to accommodate the needs of the Baby Boomers and later generations as they age in the next decades.

Outside Resources

Web: Columbia Aging Society

<http://www.agingsocietynetwork.org/>

Web: Columbia International Longevity Center

<http://www.mailman.columbia.edu/academic-departments/centers/columbia-aging/international-longevity-center-knowledge-transfer>

Web: National Institute on Aging

<http://www.nia.nih.gov/>

Web: Stanford Center Longevity

<http://longevity3.stanford.edu/>

Discussion Questions

1. How do age stereotypes and intergenerational social interactions shape quality of life in older adults? What are the implications of the research of Levy and others?
2. Researchers suggest that there is both stability and change in Big Five personality traits after age 30. What is stable? What changes?
3. Describe the Social Convoy Model of Antonucci. What are the implications of this model for older adults?
4. Memory declines during adulthood. Is this statement correct? What does research show?
5. Is dementia inevitable in old age? What factors are currently thought to be protective?
6. What are the components of successful aging described by Rowe and Kahn (1998) and others? What outcomes are used to evaluate successful aging?

Vocabulary

Age identity

How old or young people feel compared to their chronological age; after early adulthood, most people feel younger than their chronological age.

Autobiographical narratives

A qualitative research method used to understand characteristics and life themes that an individual considers to uniquely distinguish him- or herself from others.

Average life expectancy

Mean number of years that 50% of people in a specific birth cohort are expected to survive. This is typically calculated from birth but is also sometimes re-calculated for people who have already reached a particular age (e.g., 65).

Cohort

Group of people typically born in the same year or historical period, who share common experiences over time; sometimes called a generation (e.g., Baby Boom Generation).

Convoy Model of Social Relations

Theory that proposes that the frequency, types, and reciprocity of social exchanges change with age. These social exchanges impact the health and well-being of the givers and receivers in the convoy.

Cross-sectional studies

Research method that provides information about age group differences; age differences are confounded with cohort differences and effects related to history and time of study.

Crystallized intelligence

Type of intellectual ability that relies on the application of knowledge, experience, and learned information.

Fluid intelligence

Type of intelligence that relies on the ability to use information processing resources to reason logically and solve novel problems.

Global subjective well-being

Individuals' perceptions of and satisfaction with their lives as a whole.

Hedonic well-being

Component of well-being that refers to emotional experiences, often including measures of positive (e.g., happiness, contentment) and negative affect (e.g., stress, sadness).

Heterogeneity

Inter-individual and subgroup differences in level and rate of change over time.

Inhibitory functioning

Ability to focus on a subset of information while suppressing attention to less relevant information.

Intra- and inter-individual differences

Different patterns of development observed within an individual (intra-) or between individuals (inter-).

Life course theories

Theory of development that highlights the effects of social expectations of age-related life events and social roles; additionally considers the lifelong cumulative effects of membership in specific cohorts and sociocultural subgroups and exposure to historical events.

Life span theories

Theory of development that emphasizes the patterning of lifelong within- and between-person differences in the shape, level, and rate of change trajectories.

Longitudinal studies

Research method that collects information from individuals at multiple time points over time, allowing researchers to track cohort differences in age-related change to determine cumulative effects of different life experiences.

Processing speed

The time it takes individuals to perform cognitive operations (e.g., process information, react to a signal, switch attention from one task to another, find a specific target object in a complex picture).

Psychometric approach

Approach to studying intelligence that examines performance on tests of intellectual functioning.

Recall

Type of memory task where individuals are asked to remember previously learned information without the help of external cues.

Recognition

Type of memory task where individuals are asked to remember previously learned information with the assistance of cues.

Self-perceptions of aging

An individual's perceptions of their own aging process; positive perceptions of aging have been shown to be associated with greater longevity and health.

Social network

Network of people with whom an individual is closely connected; social networks provide emotional, informational, and material support and offer opportunities for social engagement.

Socioemotional Selectivity Theory

Theory proposed to explain the reduction of social partners in older adulthood; posits that older adults focus on meeting emotional over information-gathering goals, and adaptively select social partners who meet this need.

Subjective age

A multidimensional construct that indicates how old (or young) a person feels and into which age group a person categorizes him- or herself

Successful aging

Includes three components: avoiding disease, maintaining high levels of cognitive and physical functioning, and having an actively engaged lifestyle.

Working memory

Memory system that allows for information to be simultaneously stored and utilized or manipulated.

References

- Antonucci, T. C. (2001). Social relations: An examination of social networks, social support and sense of control. In J.E. Birren & K. W. Schaie (Eds.), *Handbook of the psychology of aging* (5th ed., pp. 427–453). New York: Academic Press.
- Baltes, P. B. (1987). Theoretical propositions of lifespan developmental psychology: On the dynamics between growth and decline. *Developmental Psychology*, 23, 611–626.
- Baltes, P. B. & Lindenberger, U. (1997). Emergence of powerful connection between sensory and cognitive functions across the adult life span: A new window to the study of cognitive aging? *Psychology and Aging*, 12, 12–21.
- Birditt, K., & Antonucci, T. C. (2008). Life sustaining irritations? Relationship quality and mortality in the context of chronic illness. *Social Science & Medicine*, 67, 1291.
- Birren, J. E. & Schroots, J. J. F. (2006). Autobiographical memory and the narrative self over the life span. In J. E. Birren and K. Warner Schaie (Eds.) *Handbook of the psychology of aging* (6th ed. pp. 477–499). Burlington, MA: Elsevier Academic Press.
- Blanchard-Fields, F. (2007). Everyday problem solving and emotion: An adult development perspective. *Current Directions in Psychological Science*, 16, 26–31.
- Bronfenbrenner, U. (1979). *The ecology of human development*. Cambridge, MA: Harvard University Press.
- Carstensen, L. L. (1993). Motivation for social contact across the life span: A theory of socioemotional selectivity. In J. E. Jacobs (Ed.), *Nebraska Symposium on Motivation*, 1992: *Developmental perspectives on motivation* (pp. 209–254). Lincoln, NE: University of Nebraska Press.
- Carstensen, L. L., Gottman, J. M., & Levensen, R. W. (1995). Emotional behavior in long-term marriage. *Psychology and Aging*, 10, 140–149.
- Carstensen, L. L., Isaacowitz, D. M., & Charles, S. T. (1999). Taking time seriously: A theory of socioemotional selectivity. *American Psychologist*, 54, 165–181.
- Charles, S. T., & Carstensen, L. L. (2010). Social and emotional aging. *Annual review of psychology*, 61, 383–409.
- Charness, N. (1981). Search in chess: Age and skill differences. *Journal of Experimental Psychology: Human Perception and Performance*, 7, 467.
- Christensen, K., Doblhammer, G., Rau, R., Vaupel, J. W. (2009). Ageing populations: the challenges ahead. *Lancet*, 374, 1196–208.
- Craik, F. I., & Bialystok, E. (2006). Cognition through the lifespan: mechanisms of change. *Trends*

- in *Cognitive Sciences*, 10, 131–138.
- Craik, F. I. M. (2000). Age-related changes in human memory. In D. C. Park & N. Schwarz (Eds.), *Cognitive Aging: A Primer* (pp. 75–92). New York: Psychology Press.
- Diener, E., Lucas, R. E., & Scollon, C. N. (2006). Beyond the hedonic treadmill: revising the adaptation theory of well-being. *American Psychologist*, 61, 305.
- Elder, G. H., Johnson, M. K., & Crosnoe, R. (2003). The emergence and development of life course theory. In J. T. Mortimer & M. J. Shanahan (Eds.), *Handbook of the Life Course*, 3-19. New York: Kluwer Academic/Plenum Publishers.
- Fingerman, K. L., Berg, C. A., Smith, J., & Antonucci, T.C. (Eds.) (2011). *Handbook of Life-Span Development*. New York: Springer
- Fingerman, K. L., Chen, P. C., Hay, E., Cichy, K. E., & Lefkowitz, E. S. (2006). Ambivalent reactions in the parent and offspring relationship. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 61, 152-160.
- Fratiglioni, L., Paillard-Borg, S., & Winblad, B. (2004). An active and socially integrated lifestyle in late life might protect against dementia. *Lancet Neurology*, 3, 343–353
- Friedman, H. S., Tucker, J. S., Tomlinson-Keasey, C., Schwartz, J. E., Wingard, D. L., & Criqui, M. H. (1993). Does childhood personality predict longevity? *Journal of Personality and Social Psychology*, 65, 176–185. doi:10.1037/0022-3514.65.1.176
- George, L. K. (2010). Still happy after all these years: Research frontiers on subjective well-being in later life. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 65, 331.
- Hasher, L. & Zacks, R. T. (1988). Working memory, comprehension, and aging: A review and a new view. In G.H. Bower (Ed.), *The Psychology of Learning and Motivation*, (Vol. 22, pp. 193–225). San Diego, CA: Academic Press.
- Hertzog, C., Kramer, A. F., Wilson, R. S., & Lindenberger, U. (2009). Enrichment effects on adult cognitive development: Can the functional capacity of older adults be preserved and enhanced? *Psychological Science in the Public Interest*, 9, 1–65
- Kahn, R. L., & Antonucci, T. C. (1980). Convoys over the life course: Attachment, roles, and social support. In P. B. Baltes & O. G. Brim (Eds.), *Life-span development and behavior* (pp. 253–286). New York: Academic Press.
- Kramer, A. F. & Erickson, K. I. (2007). Capitalizing on cortical plasticity: The influence of physical activity on cognition and brain function. *Trends in Cognitive Sciences*, 11, 342–348.
- Lang, F. R., Baltes, P. B., & Wagner, G. G. (2007). Desired lifetime and end-of-life desires across adulthood From 20 to 90: A dual-source information model. *Journal of Gerontology*:

- Psychological Sciences*, 62B, 268–276.
- Levy, B. (2009). Stereotype Embodiment: A Psychosocial Approach to Aging. *Current Directions in Psychological Science*, 18(6), 332–336
- Lucas, R. E. (2007). Adaptation and the Set-Point Model of Subjective Well-Being Does Happiness Change After Major Life Events?. *Current Directions in Psychological Science*, 16, 75–79.
- Lucas, R. E. & Donnellan, A. B. (2011). Personality development across the life span: Longitudinal analyses with a national sample from Germany. *Journal of Personality and Social Psychology*, 101, 847–861
- Magai, C. (2008). Long-lived emotions. In M. Lewis, J. M. Haviland-Jones, & L. Feldman Barrett (Eds.), *Handbook of Emotions*, 376–392. New York: Guilford.
- Markus, H., & Nurius, P. (1986). Possible selves. *American Psychologist*, 41, 954–969.
- McAdams, D. P. (2006). The redemptive self: Generativity and the stories Americans live by. *Research in Human Development*, 3(2-3), 81–100.
- Mroczek, D. K. (2001). Age and emotion in adulthood. *Current Directions in Psychological Science*, 10(3), 87–90.
- Park, D. C. (2000). The basic mechanisms accounting for age-related decline in cognitive function. In D.C. Park & N. Schwarz (Eds.), *Cognitive Aging: A Primer* (pp. 3–21). New York: Psychology Press.
- Park, D. C. & Gutchess, A. H. (2000). Cognitive aging and everyday life. In D.C. Park & N. Schwarz (Eds.), *Cognitive Aging: A Primer* (pp. 217–232). New York: Psychology Press.
- Roberts, B. W., & Mroczek, D. K. (2008). Personality trait change in adulthood. *Current Directions in Psychological Science*, 17, 31–35.
- Roberts, B. W., Kuncel, N., Shiner, R., N., Caspi, A., & Goldberg, L. R. (2007). The power of personality: The comparative validity of personality traits, socioeconomic status, and cognitive ability for predicting important life outcomes. *Perspectives on Psychological Science*, 2(4), 313–345. doi:10.1111/j.1745-6916.2007.00047.
- Rook, K. S. (1998). Investigating the positive and negative sides of personal relationships: Through a lens darkly? In W. R. Cupach & B. H. Spitzberg (Eds.), *The dark side of close relationships* (pp. 369–393). Mahwah, NJ: Lawrence Erlbaum.
- Ross, M., & Wilson, A. E. (2003). Autobiographical memory and conceptions of self: Getting better all the time. *Current Directions in Psychological Science*, 12, 66–69.
- Rowe, J. W. (2009). Facts and fictions about an aging America. *Contexts*, 8, 16–21.
- Rowe, J. W. & Kahn, R. L. (1997). Successful aging. *The Gerontologist*, 37(4), 433–440.

- Rubin, D., & Berntsen, D. (2006). People over 40 feel 20% younger than their age: Subjective age across the lifespan. *Psychonomic Bulletin & Review*, 13, 776–780
- Ryff, C. D. (1995). Psychological well-being in adult life. *Current Directions in Psychological Science*, 4(4), 99–104
- Ryff, C. D., & Singer, B. (2000). Interpersonal flourishing: A positive health agenda for the new millennium. *Personality and Social Psychology Review*, 4, 30–44.
- Salthouse, T. A. (2004). What and when of cognitive aging. *Current Directions in Psychological Science*, 13, 140–144.
- Salthouse, T. A. (1984). Effects of age and skill in typing. *Journal of Experimental Psychology: General*, 113, 345.
- Schaie, K. W. & Willis, S. L. (1996). Psychometric intelligence and aging. In F. Blanchard-Fields & T.M. Hess (Eds.), *Perspectives on Cognitive Change in Adulthood and Aging* (pp. 293–322). New York: McGraw Hill.
- Settersten, R. A., Jr. (2005). Toward a stronger partnership between lifecourse sociology and life-span psychology. *Research in Human Development*, 2(1–2), 25–41.
- Stone, A. A., Schwartz, J. E., Broderick, J. E., & Deaton, A. (2010). A snapshot of the age distribution of psychological well-being in the United States. *Proceedings of the National Academy of Sciences*, 107, 9985–9990.
- Tentori, K., Osherson, D., Hasher, L., & May, C. (2001). Wisdom and aging: Irrational preferences in college students but not older adults. *Cognition*, 81, B87–B96.
- Uchino, B. N. (2009). What a lifespan approach might tell us about why distinct measures of social support have differential links to physical health. *Journal of Social and Personal Relationships*, 26(1), 53–62. DOI: 10.1177/0265407509105521
- Umberson, D., Williams, K., Powers, D. A., Liu, H., & Needham, B. (2006). You make me sick: Marital quality and health over the life course. *Journal of Health and Social Behavior*, 47, 1–16.
- Waite, L. J. & Gallagher, M. (2000). *The case for marriage: Why married people are happier, healthier, and better off financially*. New York: Doubleday.